COMPUTER FUNDAMENTALS

A Complete Beginner-to-Advanced Practical Guide

Learn Computer Basics, Hardware, Software, Windows, Internet, and Essential Digital Skills



AUTHOR

ZOHA SATTAR

Computer Teacher | Web Developer | IT Professional

FIRST EDITION

ENGLISH LANGUAGE

ABOUT THE AUTHOR



ZOHA SATTAR

Computer Teacher | Web Developer | IT Professional

Zoha Sattar is a computer teacher, web developer, and IT professional with an interest in computer education and practical technology skills.

With experience in teaching Computer Science and working on web development projects, the author aims to make computer concepts easier to understand for beginners and students.

This book has been prepared as a learning resource for readers who want to understand computers, explore their essential components, learn Windows operations, and develop practical digital skills.

PURPOSE OF THIS BOOK

The purpose of Computer Fundamentals is to provide readers with clear explanations of important computer concepts in an organized and accessible format. It is intended to support independent learning, classroom instruction, and practical computer education.

WHO SHOULD READ THIS BOOK?

- Students beginning their computer education.
- Beginners who want to develop computer skills.
- Teachers preparing introductory computer lessons.
- Office workers seeking to improve their digital knowledge.
- Anyone interested in understanding computers and their everyday uses.

The author hopes this book will help readers build a strong foundation in computer knowledge and use technology with greater confidence.

BOOK TITLE

Computer Fundamentals — A Complete Beginner-to-Advanced Practical Guide

TABLE OF CONTENTS

Chapter 1 — Introduction to Computers ..... Topics 1–15

Chapter 2 — Types and Classification of Computers ..... Topics 16–30

Chapter 3 — Computer System Components ..... Topics 31–45

Chapter 4 — Computer Hardware in Detail ..... Topics 46–57

Chapter 5 — Memory and Storage Fundamentals ..... Topics 58–80

Chapter 6 — Input Devices ..... Topics 81–91

Chapter 7 — Output Devices ..... Topics 92–100

Chapter 8 — Operating Systems Fundamentals ..... Topics 101–110

Chapter 9 — Windows Desktop and Basic Navigation ..... Topics 111–120

Chapter 10 — Windows Desktop Personalization ..... Topics 121–135

Chapter 11 — Start Menu and Search ..... Topics 136–150

Chapter 12 — Taskbar and Application Management ..... Topics 151–170

Chapter 13 — Working with Windows and Applications ..... Topics 171–190

Chapter 14 — Keyboard, Mouse, and Touchpad ..... Topics 191–210

Chapter 15 — File and Folder Management ..... Topics 211–230

Chapter 16 — File Explorer ..... Topics 231–255

Chapter 17 — Windows Settings and Control Panel ..... Topics 256–280

Chapter 18 — Windows Accounts and Sign-In ..... Topics 281–295

Chapter 19 — Software, Apps, and Programs ..... Topics 296–315

Chapter 20 — Internet, Browsers, and Online Work ..... Topics 316–340

Chapter 21 — Computer Networking and Connectivity ..... Topics 341–355

Chapter 22 — Email and Digital Communication ..... Topics 356–370

Chapter 23 — Messaging, Video Conferencing, and Digital Collaboration Topics 371–380

Chapter 24 — Audio, Video, and Digital Media ..... Topics 381–390

Chapter 25 — Printing, Scanning, PDF Documents, and Projectors .. Topics 391–400

Chapter 1: Introduction to Computers

Introduction

Computers are an essential part of modern life. They are used in education, business, healthcare, communication, science, entertainment, and many other fields. Understanding computers helps people use technology effectively, solve problems, and develop valuable digital skills.

This book introduces computer fundamentals and gradually progresses towards practical computer skills. It is designed for students, beginners, and anyone who wants to improve their computer knowledge.

Topic 1: What is a Computer?

A computer is an electronic device that accepts data, processes it according to instructions, stores it, and produces meaningful results called information.

A computer performs four basic operations:

- **Input:** Receives data through devices such as a keyboard or mouse.
- **Processing:** Performs calculations and operations on the data.
- **Output:** Presents results through devices such as a monitor or printer.
- **Storage:** Saves data and information for future use.

Example: When a student enters marks into a computer, it can calculate the total and percentage and display the results.

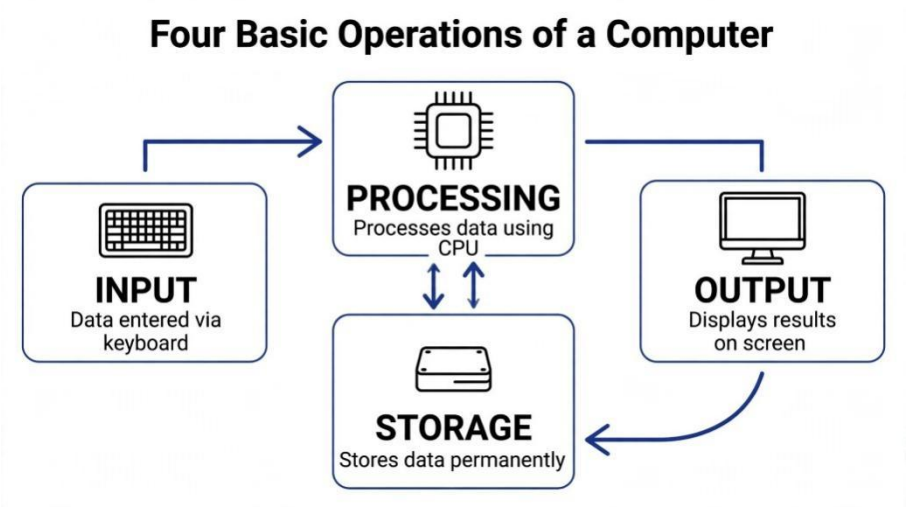


Figure 1.1: Basic Components of a Computer System

Topic 2: History and Evolution of Computers

The history of computers describes how computing devices developed from simple calculating tools into powerful electronic systems.

Early tools, such as the abacus, helped people perform arithmetic. Mechanical calculators later automated calculations. During the twentieth century, electronic computers introduced much faster data processing. Modern computers use advanced processors, large storage capacities, and sophisticated software. Computer development has made technology smaller, faster, more affordable, and more accessible.

Example: Large early computers occupied entire rooms, while modern laptops can perform many of the same types of tasks in a portable form.

Topic 3: Generations of Computers

Computer generations are commonly used to explain major stages in the development of electronic computers. Each generation introduced important technological improvements.

1. **First Generation:** Used vacuum tubes and consumed considerable electricity.
2. **Second Generation:** Used transistors, making computers smaller and more reliable.
3. **Third Generation:** Used integrated circuits containing multiple electronic components.
4. **Fourth Generation:** Used microprocessors, supporting personal computers and compact systems.

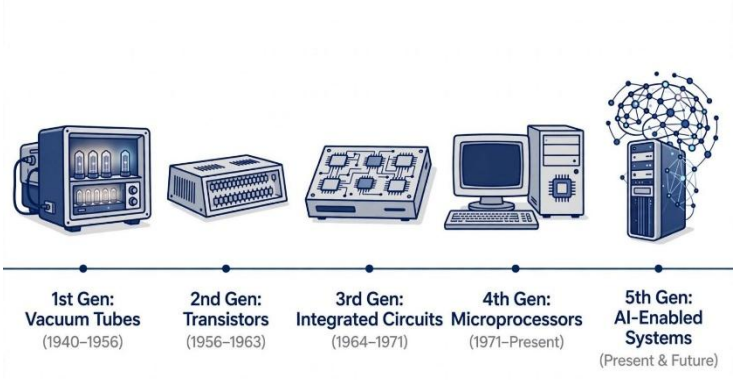


Figure 1.2: Five Generations of Computers

5. **Fifth Generation:** Commonly associated with artificial intelligence, advanced processing, and intelligent computing systems.

These categories are simplified historical classifications; modern computers combine technologies developed across several generations.

Topic 4: Characteristics of Computers

Computers have several characteristics that make them useful for performing different tasks.

- **Speed:** They process large amounts of data quickly.
- **Accuracy:** They can produce accurate results when the data and instructions are correct.
- **Storage:** They can store large amounts of information.
- **Automation:** They can perform programmed tasks with limited human intervention.
- **Consistency:** They can repeat operations without becoming tired.
- **Versatility:** They can perform many different types of tasks.

A computer's output still depends on the quality of its input, instructions, hardware, and software.

Topic 5: Limitations of Computers

Although computers are powerful tools, they have limitations.

- They require suitable instructions or software to perform tasks.
- Incorrect input can produce incorrect results.
- They do not possess human judgment or understanding in the same way people do.
- They depend on electricity or another suitable power source.
- They may experience hardware failures, software errors, or security threats.
- They cannot independently guarantee that information is true or reliable.

Example: A computer may calculate a student's percentage correctly but cannot automatically know whether the entered marks belong to the correct student.

Topic 6: Importance of Computers in Daily Life

Computers help people perform everyday activities efficiently. They are used to prepare documents, communicate with others, manage finances, study online, organize information, and access digital services.
Computing devices include desktop computers, laptops, tablets, and smartphones. Many everyday appliances also contain small embedded computers.
Computers can save time and improve productivity, but users should protect their personal information and use technology responsibly.

Topic 7: Uses of Computers in Education

Computers support teaching, learning, assessment, and educational administration.
Common uses include:

- Preparing notes, assignments, and presentations.
- Attending online classes and accessing learning materials.
- Conducting research using reliable digital sources.
- Practising programming and other technical skills.
- Maintaining student records and examination results.
- Preparing reports and managing school information.

Example: A teacher can use a spreadsheet to calculate students' marks and a word processor to prepare examination papers.

Topic 8: Uses of Computers in Business

Businesses use computers to manage operations, communicate with customers, process transactions, and maintain records.
Common applications include accounting, inventory management, payroll, employee records, customer support, marketing, and online sales.
Business software can reduce repetitive work and help organizations analyse information. However, businesses must protect customer data and maintain reliable backups.
Example: An online store can use a computer system to display products, record orders, calculate bills, and update inventory.

Topic 9: Uses of Computers in Hospitals
Hospitals and healthcare facilities use computers to support administration, medical services, and communication.
They can help manage patient records, appointments, laboratory results, billing, medical images, and hospital inventory. Specialized systems also support medical research and the operation of certain diagnostic devices.
Healthcare professionals must verify important information and protect the confidentiality of patient records.
Example: A hospital information system can help authorized staff find a patient's appointment details and recorded test results.

Topic 10: Uses of Computers in Banks
Banks use computer systems to manage accounts, process transactions, maintain financial records, and provide digital banking services.
Computers support services such as:

- Account management and balance enquiries.
- Electronic fund transfers.
- ATM operations.
- Online and mobile banking.
- Transaction monitoring and fraud detection.
- Preparation of statements and financial reports.

Banking systems require strong security, accurate records, and controlled access because they handle sensitive financial information.

Topic 11: Uses of Computers in Science and Research
Scientists and researchers use computers to collect, organize, process, and analyse data. Computers also support simulations, mathematical calculations, laboratory equipment, and the preparation of research reports.
High-performance computers can process complex scientific problems that would take much longer using ordinary computing systems.
Example: Researchers can analyse weather observations using computer models to help prepare forecasts.
Computers assist scientific work, but research conclusions still require suitable methods, evidence, and expert evaluation.

Topic 12: Uses of Computers in Communication
Computers make it possible to exchange information across local and global networks.
Common communication tools include email, instant messaging, video conferencing, online discussion platforms, and collaborative applications.
These tools allow people to share documents, attend remote meetings, and work together from different locations.
Example: A student can send an assignment to a teacher by email instead of delivering a printed copy.
Users should verify recipients, avoid suspicious links, and protect confidential information when communicating online.

Topic 13: Uses of Computers in Entertainment
Computers support many forms of digital entertainment, including games, music, films, animation, photography, and video editing.
They allow users to create, edit, store, and enjoy digital media. Streaming platforms and online gaming services also use computer systems and networks to deliver content.
Example: A user can edit a video on a laptop and share the finished project through an online platform.
Responsible use includes balancing screen time, respecting copyright, and choosing age-appropriate content.

Topic 14: Advantages and Disadvantages of Computers
Computers offer many benefits, but they can also create challenges when used carelessly.
Advantages

- Fast processing and calculations.

- Efficient storage and retrieval of information.
- Improved communication and collaboration.
- Support for education, research, and creativity.
- Automation of repetitive tasks.

Disadvantages

- Security and privacy risks.
- Dependence on electricity and technology.
- Possible health and comfort problems from poor working habits.
- Costs of equipment, maintenance, and upgrades.
- Risk of inaccurate results from incorrect data or instructions.

Using computers effectively requires technical knowledge, responsible behaviour, and appropriate security practices.

Topic 15: Human Intelligence and Computer Processing

Human intelligence and computer processing are different, although they can complement each other. Humans can use experience, emotions, social understanding, judgment, and creativity in complex situations. Computers can process data rapidly, repeat calculations consistently, and follow instructions at scale. Artificial intelligence can perform tasks that resemble aspects of human problem-solving, but its abilities and reliability depend on the system, data, and task.

Example: A computer can organize examination results and calculate averages, while a teacher can consider a student's circumstances when providing educational guidance.

Chapter 2: Types and Classification of Computers

Topic 16: Introduction to Types and Classification of Computers
Computers can be classified according to their size, processing capability, purpose, and the type of data they handle.
This classification helps users understand why different computing systems are designed for different needs.
Common categories include supercomputers, mainframes, personal computers, workstations, servers, embedded computers, and mobile devices. Computers can also be classified as analog, digital, or hybrid according to how they process data.
Some categories overlap because a computer can belong to more than one classification.

Topic 17: Supercomputers
A supercomputer is a high-performance computing system designed to perform extremely complex calculations and process large amounts of data.
Supercomputers use many processing units working together to solve demanding problems. They are typically found in specialized research centres, scientific institutions, and national computing facilities.
Common applications include climate modelling, scientific simulations, advanced physics, and certain types of medical and engineering research.
Example: Researchers can use a supercomputer to simulate complex weather patterns.
Supercomputers are expensive to build and operate and require specialized facilities and technical expertise.

Topic 18: Mainframe Computers
A mainframe computer is a powerful system designed to process large volumes of transactions and support many users or applications reliably.
Mainframes are often used by banks, insurance companies, government departments, and large organizations that manage critical data.
Their strengths include high transaction capacity, reliability, security controls, and support for continuous business operations.
Example: A large bank may use a mainframe to process account transactions and maintain financial records.
Mainframes differ from supercomputers in their typical purpose: mainframes emphasize large-scale transaction processing and dependable service, while supercomputers emphasize extremely demanding calculations.
Topic 19: Minicomputers
Minicomputers were historically medium-sized multi-user systems positioned between mainframes and smaller computers. They allowed organizations and departments to share computing resources.
Such systems were used for industrial control, scientific work, and departmental data processing.
The term is now largely historical. Many tasks once handled by minicomputers are performed by modern servers and other networked systems.
Example: A department could historically use a minicomputer to support several users accessing shared applications.
Understanding minicomputers helps explain the evolution of multi-user computing systems.

Topic 20: Microcomputers
A microcomputer is a computer that uses a microprocessor as its central processing unit. The category includes many devices designed for individual use.
Common examples include desktop computers, laptops, and many single-board computers. Microcomputers are used for education, office work, programming, entertainment, and internet access.
Their capabilities vary according to processor performance, memory, storage, and other components.
Example: A student can use a laptop to write assignments, attend online classes, learn programming, and prepare presentations.
Microcomputers have made computing widely accessible by combining useful processing power with relatively compact designs.

Topic 21: Workstations and Servers

A **workstation** is a computer designed for professional tasks that may require higher processing power, memory, or graphics performance than a typical personal computer.
Workstations are commonly used for engineering, scientific analysis, 3D modelling, software development, and video production.
A **server** is a computer system that provides services, resources, or data to other computers called clients over a network.
Servers can host websites, store files, manage databases, and support applications.
Example: An engineer may use a workstation to design a building, while a web server hosts the website of an engineering company.

Topic 22: Embedded Computers
An embedded computer is a computing system built into a larger device to perform specific functions. It usually combines hardware and software designed for a particular task.
Embedded computers are found in household appliances, vehicles, medical equipment, industrial machines, and smart devices.
They may operate independently or communicate with other systems. Many are designed to work reliably with limited memory and processing resources.
Example: A washing machine uses an embedded computer to control its washing cycles, water levels, and timing.

Topic 23: Analog Computers
An analog computer processes information represented by continuously varying physical quantities, such as voltage, temperature, pressure, or rotational movement.
Analog computing systems can model physical processes and have been used in specialized scientific, engineering, and control applications.
Unlike conventional digital computers, which represent information using discrete values, analog computers work with continuous signals.
Example: A traditional analog computing system may use electrical voltages to model the behaviour of a physical system.
Analog computers are different from ordinary digital computers, although modern systems can combine analog and digital components.

Topic 24: Digital Computers
A digital computer processes information using discrete values, usually represented internally through binary digits: 0 and 1.
Most modern computers, including desktops, laptops, smartphones, and servers, are digital computers.
They perform operations according to instructions provided by software and can process text, numbers, images, audio, and video.
Example: A laptop uses digital processing to calculate percentages, display documents, and play videos.
Digital computers are widely used because they can perform many different tasks with speed, accuracy, and flexibility.

Topic 25: Hybrid Computers
A hybrid computer combines analog and digital computing capabilities to handle different forms of information.
Analog components may measure or process continuous physical signals, while digital components perform calculations, make logical decisions, and store results.
Hybrid systems are useful in specialized scientific, industrial, and medical applications.
Example: A monitoring system may measure a continuously changing physical signal and use digital processing to analyse and display the measurements.
Hybrid computing does not necessarily mean that every device containing analog and digital circuits is classified as a hybrid computer. The term is generally used for systems that combine these computing approaches for a specific purpose.

Topic 26: General-Purpose and Special-Purpose Computers
Computers can be classified according to whether they are designed for many tasks or a particular function.
General-purpose computers can run different programs and perform a wide variety of activities.
Examples include desktop computers, laptops, and smartphones.
Special-purpose computers are designed primarily for a particular task or a limited set of related tasks.
Examples include dedicated industrial controllers and computing systems built into certain appliances.
A general-purpose computer can often be configured for a specialized job, while a special-purpose system may have a narrower range of functions.

Topic 27: Classification by Size and Processing Power
Computers vary in physical size, processing capability, storage capacity, and the number of users they can support.
Common categories include:

- **Supercomputers:** Designed for extremely demanding calculations and simulations.
- **Mainframes:** Designed for large-scale transaction processing and reliable services.
- **Servers:** Provide network resources and services to other systems.
- **Workstations:** Support demanding individual professional tasks.
- **Personal computers:** Support everyday computing and productivity.
- **Embedded computers:** Perform dedicated functions within larger devices.

These categories are not always mutually exclusive. For example, a server may be physically compact but still provide powerful services to many users.

Topic 28: Personal Computers and Enterprise Computer Systems
A **personal computer (PC)** is designed primarily for individual use. It may be used for education, office work, communication, programming, and entertainment.
An **enterprise computer system** supports the operations of an organization. It may include servers, databases, employee computers, networking equipment, and business applications.
Personal computers generally focus on individual productivity, while enterprise systems often emphasize shared resources, centralized administration, security, reliability, and organizational requirements.
Example: An employee uses a PC to prepare a report, while the organization's enterprise system stores the report and controls who can access it.

Topic 29: Mobile Computers
Mobile computers are portable computing devices designed to be carried and used in different locations.
Common examples include laptops, tablets, and handheld computing devices. Smartphones also provide many mobile computing functions.
Mobile computers often include wireless connectivity, rechargeable batteries, portable storage, and built-in input or output components.
They support activities such as online learning, communication, document editing, navigation, and business operations.
Example: A student can use a laptop at home and continue working on the same assignment at school.

Topic 30: Wearable Computers
Wearable computers are electronic computing devices designed to be worn on the body or integrated into clothing and accessories.
Examples include smartwatches, fitness trackers, and specialized smart glasses.
Depending on the device, they may collect sensor data, display notifications, track activity, support communication, or provide navigation information.
Wearable devices have limited physical space and battery capacity, so their computing capabilities are usually optimized for particular functions.
Users should also consider privacy because some wearables collect personal or activity-related data.

Chapter 3: Computer System Components

Topic 31: Introduction to Computer System Components
A computer system consists of hardware, software, data, and the people who use or manage it. These elements work together to perform computing tasks.
Hardware refers to the physical components, while **software** provides the instructions that tell the hardware what to do.
Data is the raw material processed by the system, and information is the meaningful result produced from that processing.

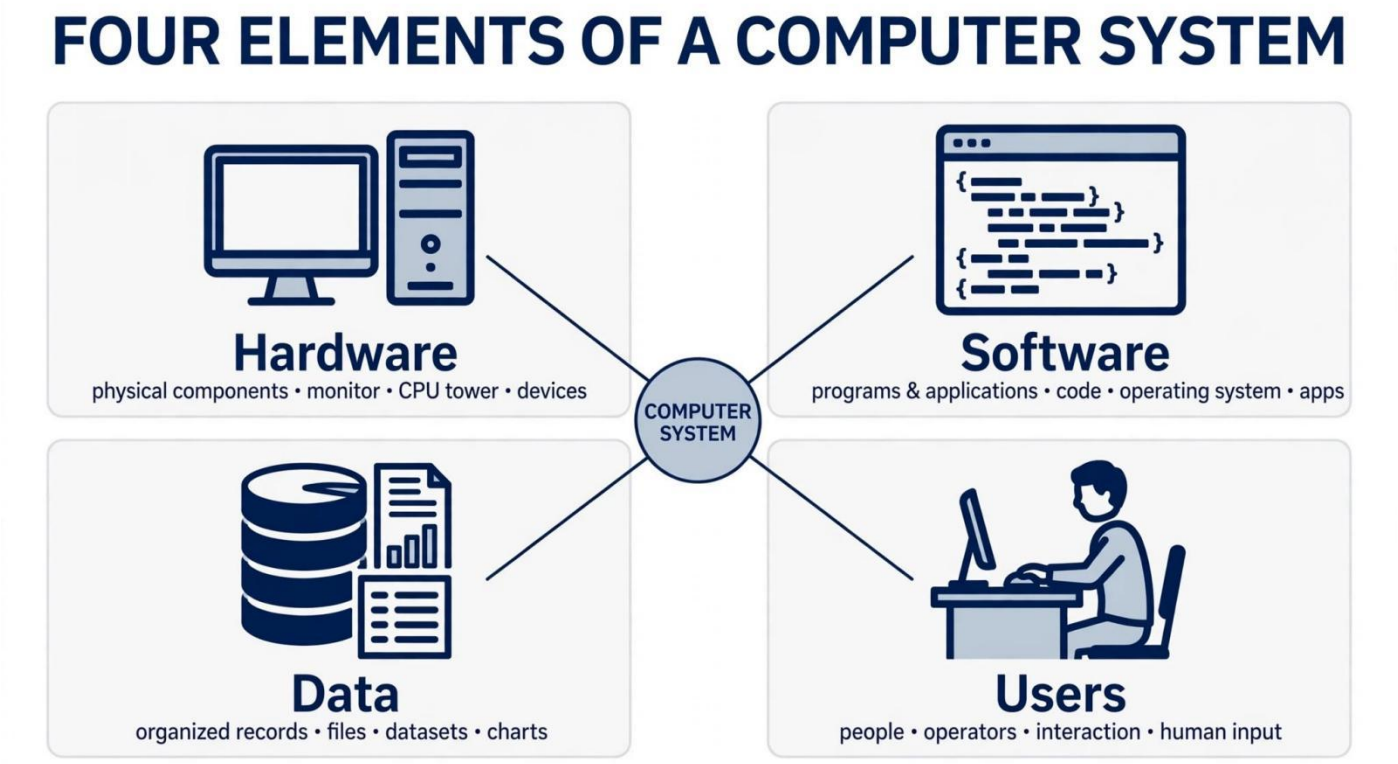


Figure 3.1: Main Elements of a Computer System

Topic 32: Hardware and Software
Hardware and software are two essential parts of a computer system.
Hardware includes the physical components that can be touched, such as the keyboard, monitor, motherboard, processor, and storage drive.
Software consists of programs and instructions that tell a computer how to perform tasks. Examples include operating systems, web browsers, word processors, and games.
Hardware and software depend on each other. Hardware provides the physical resources, while software directs those resources to perform useful operations.
Example: A word processor is software, while the keyboard used to type a document is hardware.

Topic 33: Data and Information
Data consists of raw facts, figures, symbols, or observations that can be collected and processed.
Information is data that has been processed or organized to make it meaningful and useful.
The same data may produce different information depending on how it is organized and interpreted.
Example: The marks 70, 80, and 90 are data. Their calculated average of 80 is information that summarizes the results.
Accurate information depends on accurate data and suitable processing.

Topic 34: Input Devices
An input device is a hardware component used to enter data or instructions into a computer.
Common input devices include:

- **Keyboard:** Enters text, numbers, and commands.
- **Mouse:** Controls the pointer and selects items.
- **Scanner:** Converts physical documents or images into digital form.

- **Microphone:** Captures sound.
- **Webcam:** Captures images and video.
- **Touchscreen:** Accepts input through touch.

Different devices support different forms of input. A computer may use several input devices at the same time.
Example: A student uses a keyboard to type an assignment and a mouse to select formatting options.

Topic 35: Processing Devices
Processing devices perform operations on data according to instructions.
The **Central Processing Unit (CPU)** executes program instructions and performs arithmetic and logical operations. Other components, such as the Graphics Processing Unit (GPU), specialize in particular types of processing.
The CPU works with memory and other system components to execute programs. Different processing devices are designed for different workloads.
Example: When a user opens a program, the CPU processes its instructions while other components provide the required data and display the results.

Topic 36: Output Devices
An output device presents the results of computer processing in a form that users can see, hear, or otherwise use.
Common output devices include:

- **Monitor:** Displays text, images, and video.
- **Printer:** Produces printed documents or images.
- **Speakers:** Produce sound.
- **Headphones:** Provide personal audio output.
- **Projector:** Displays computer content on a larger surface.

Some devices perform both input and output functions. For example, a touchscreen displays information and accepts touch input.
Example: A teacher displays a presentation on a projector and prints worksheets using a printer.

Topic 37: Storage Devices and Their Role
Storage devices preserve data, programs, and files so they can be used later.
Common storage devices include hard disk drives (HDDs), solid-state drives (SSDs), USB flash drives, memory cards, and optical discs.
Storage is important because most data must remain available after a computer is switched off. Storage devices differ in capacity, speed, cost, durability, and portability.
Example: A student saves an assignment on an SSD and keeps an additional copy on an external drive.
Storage is not the same as RAM. RAM temporarily holds data and instructions being used, while storage devices retain files for longer periods.

Topic 38: The Motherboard
The motherboard is the main circuit board of a computer. It connects important components and provides pathways for communication between them.
A motherboard commonly contains or provides connections for:

- CPU socket.
- Memory slots.
- Storage interfaces.
- Expansion slots.
- Power connectors.
- Firmware components.
- Input/output ports and internal connectors.

The motherboard's design determines which processors, memory types, and other components it can support.
Example: When a compatible memory module is installed in a motherboard's memory slot, the system can use it as RAM.

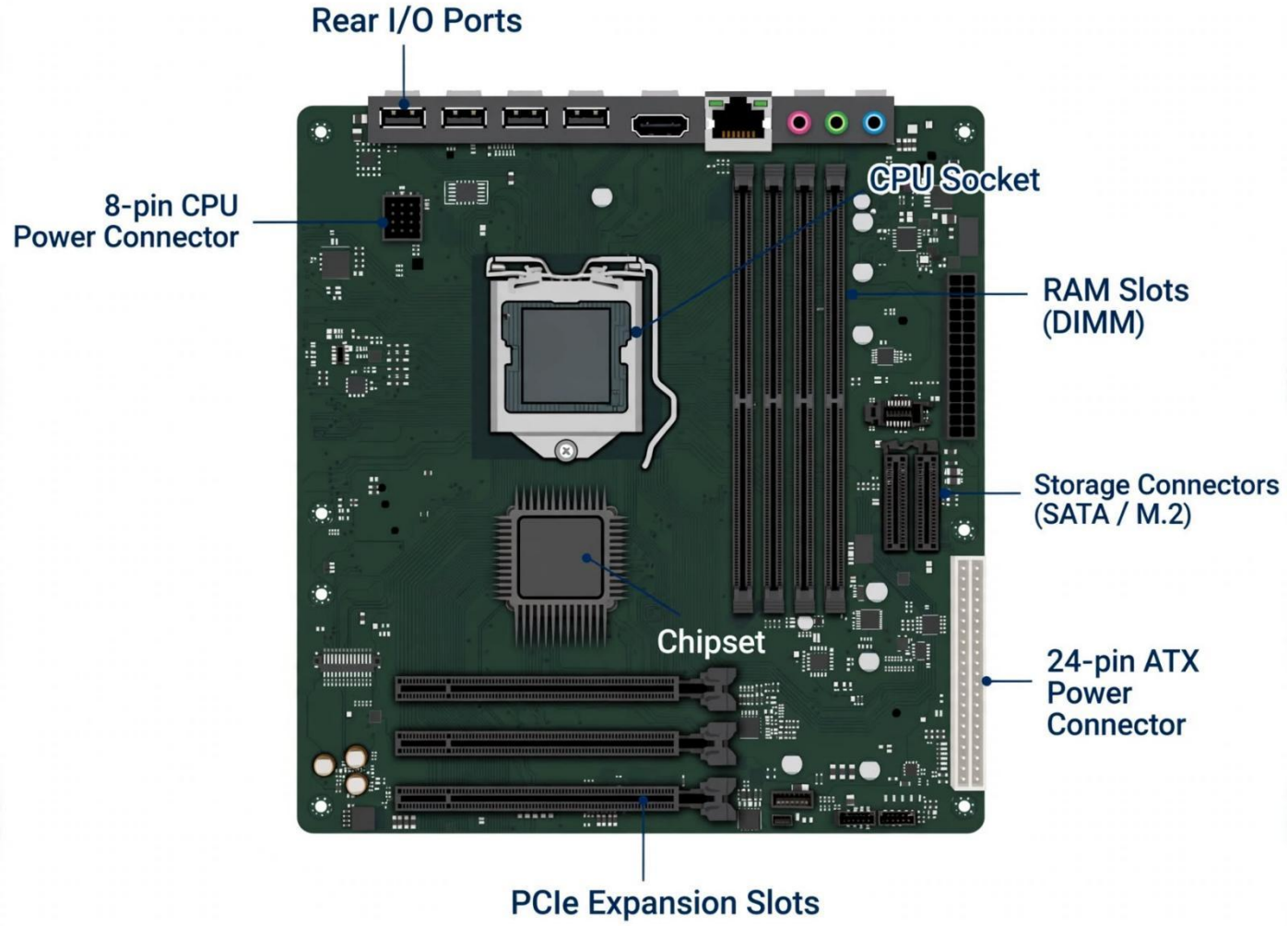


Figure 3.2: Main Components of a Motherboard

Topic 39: The Central Processing Unit (CPU)

The Central Processing Unit (CPU) is the main processor responsible for executing program instructions and coordinating many computer operations.

Its important functions include performing arithmetic and logical operations, processing instructions, and controlling the execution of programs.

Common CPU components include:

- **Arithmetic Logic Unit (ALU):** Performs arithmetic and logical operations.
- **Control Unit (CU):** Coordinates instruction execution and directs operations.
- **Registers:** Small, fast storage locations used for data and instructions during processing.

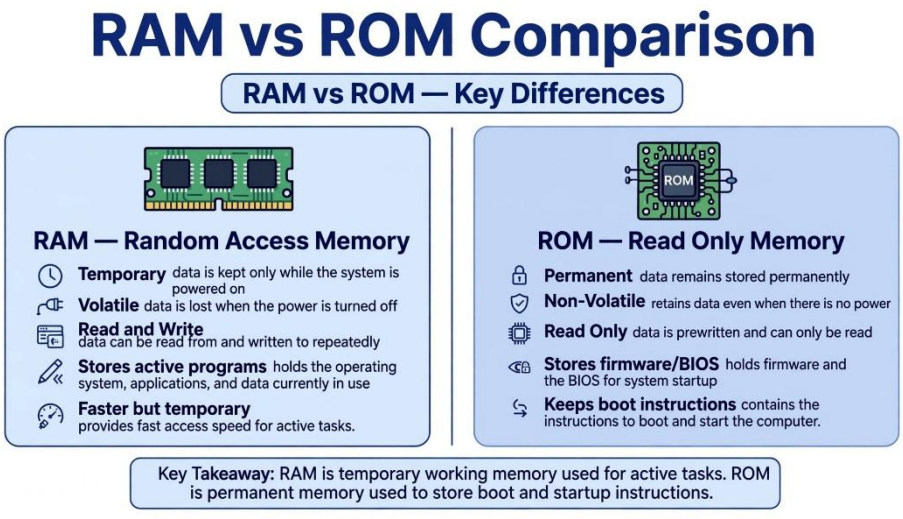
Modern CPUs also include additional components and optimizations to improve performance.

CPU performance depends on several factors, including architecture, core count, clock speed, cache, and the workload being performed.

Topic 40: Random Access Memory (RAM)

Random Access Memory (RAM) is the computer's main working memory. It temporarily stores data and instructions that the CPU needs while programs are running.

RAM allows the computer to access active data much faster than it generally can from long-term storage. It is usually volatile, meaning its contents are lost when power is removed.



More RAM can help a computer handle more applications and larger workloads, although the benefit depends on the task and the rest of the system.
Example: When a user opens a web browser and a word processor, RAM holds data needed by those running applications.
RAM is different from an SSD or HDD, which stores files for long-term use.

Topic 41: Read-Only Memory (ROM)
Read-Only Memory (ROM) is a type of memory used to store instructions that are needed for specific functions, such as starting or initializing a computer.
Unlike ordinary RAM, ROM is non-volatile, meaning it retains its contents when power is switched off.
Modern computers commonly use flash memory for firmware, which can be updated when necessary.
ROM and related non-volatile firmware storage help a computer begin operating before the operating system loads.
Example: Firmware stored on a motherboard helps initialize hardware during startup.

Topic 42: Power Supply Unit (PSU)
The Power Supply Unit (PSU) converts electrical power from a wall outlet into the regulated DC power required by a desktop computer's internal components.
A PSU supplies power to components such as the motherboard, processor, storage drives, and graphics card.
Its capacity and connector types must be suitable for the computer's hardware.
An unsuitable or poor-quality power supply may cause instability or damage.
Safety note: Never open a PSU yourself. It may contain components that retain dangerous electrical energy even after it has been disconnected.

Topic 43: Cooling Systems in Computers
Computer cooling systems remove excess heat generated by components during operation. Effective cooling helps maintain performance, reliability, and safe operating temperatures.
Common cooling methods include:

- **Heat sinks:** Transfer heat away from components.
- **Fans:** Move air through the computer case.
- **Liquid cooling:** Circulates liquid to transfer heat away from components.
- **Thermal interface material:** Helps transfer heat between a component and its cooler.

Dust buildup, blocked vents, and failed fans can reduce cooling effectiveness.
Example: A CPU cooler transfers heat away from the processor while a fan helps carry that heat away.

Topic 44: Computer Case and Its Components
A computer case, also called a chassis, houses and protects the internal components of a desktop computer. It provides mounting points for the motherboard, power supply, storage drives, and other components. It also helps organize cables and manage airflow.
Computer cases vary in size and design. A case must provide enough space for the selected motherboard, graphics card, cooling system, and other hardware.
Example: Before installing a large graphics card, a user should check that the case has sufficient clearance.

Topic 45: Expansion Cards and Slots
An expansion card is a circuit board installed in a compatible expansion slot to add or improve a computer's capabilities.
Common examples include:

- **Graphics card:** Handles graphics processing.
- **Network interface card:** Provides wired or additional network connectivity.
- **Sound card:** Provides specialized audio capabilities.
- **Storage controller card:** Adds or expands storage connections.

Many modern motherboards already include these functions, so separate expansion cards are needed only when additional capabilities are required.
Compatibility depends on the slot type, available space, power requirements, and operating-system support.

Chapter 4: Computer Hardware in Detail

Topic 46: Introduction to Computer Hardware
Computer hardware refers to the physical parts of a computer system that can be seen or physically handled. Hardware includes internal components, external devices, connectors, and accessories. Each component performs a specific function and works with other parts of the system.
Hardware can be grouped into input, processing, output, storage, and communication components. Understanding hardware helps users select suitable equipment, maintain their computers, and identify possible technical problems.

Topic 47: Internal and External Computer Hardware
Computer hardware can be divided into internal and external components according to where the components are installed or connected.
Internal hardware is installed inside the computer case or integrated into the computer. Examples include the motherboard, CPU, RAM, internal storage drive, and PSU.
External hardware consists of devices connected from outside the computer. Examples include keyboards, mice, monitors, printers, webcams, and external drives.
Laptops integrate many components into a compact body, so not every internal component is separately replaceable.
Example: A desktop's RAM module is internal hardware, while a USB keyboard is external hardware.

Topic 48: Ports, Connectors, and Interfaces
A port is a connection point through which a computer communicates with an external device or another system. A connector is the physical plug or socket used to establish a connection, while an interface defines how devices communicate.
Common examples include:

- **USB:** Connects keyboards, storage devices, and many accessories.
- **HDMI:** Carries digital video and audio to compatible displays.
- **Ethernet:** Provides wired network connectivity.
- **Audio connectors:** Connect microphones, headphones, and speakers.
- **DisplayPort:** Carries digital video and, where supported, audio.

The shape of a connector alone does not guarantee that every feature or speed is supported. Compatibility depends on the port, cable, device, and relevant standards.

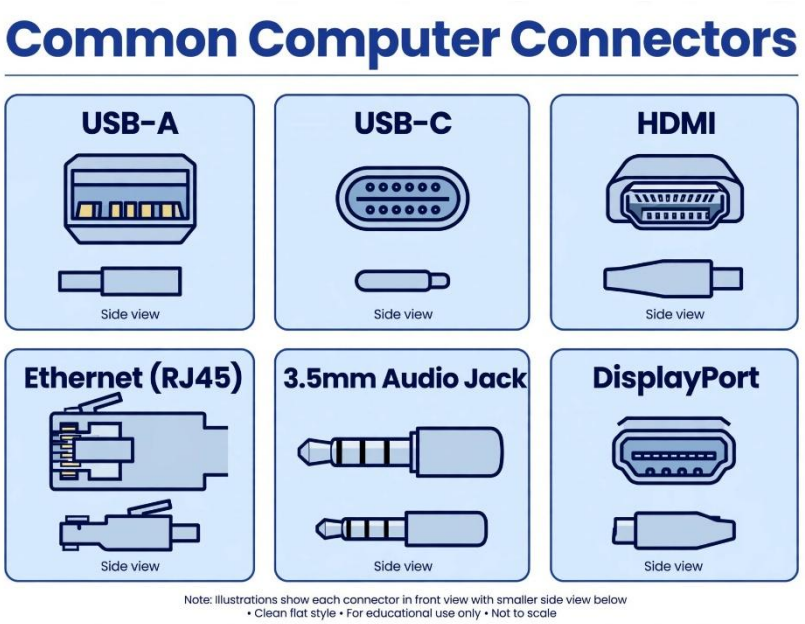


Figure 4.1: Common Computer Ports and Connectors

Topic 49: Plug-and-Play Devices and Device Drivers
Plug and Play (PnP) is a technology that allows an operating system to detect and configure compatible hardware with limited manual setup.
A **device driver** is software that enables the operating system to communicate with and control a particular hardware device.
Many devices work automatically because the required driver is already included with the operating system or installed through an update. Some devices require a separate driver from the manufacturer.
Example: When a compatible USB mouse is connected, the operating system usually detects it and makes it available for use.
If a device does not work correctly, users should check its connection, compatibility, driver status, and manufacturer instructions.

Topic 50: Computer Performance and Its Factors
Computer performance describes how effectively and quickly a computer completes tasks.
Important factors include:

- **CPU:** Influences instruction processing and calculations.
- **RAM:** Affects how much active data can be handled efficiently.
- **Storage:** Influences file access and application loading times.
- **GPU:** Accelerates graphics and certain parallel computing tasks.
- **Cooling:** Helps prevent excessive temperatures that can reduce performance.
- **Software:** Efficient programs and a well-maintained operating system can improve responsiveness.

Network speed can also affect online tasks, even when the computer itself is performing well.
No single component determines all aspects of performance. The best configuration depends on the intended workload.

Topic 51: Basic Computer Troubleshooting
Troubleshooting is the process of identifying the cause of a problem and applying an appropriate solution.
A systematic approach can prevent unnecessary changes and help users solve common issues.
Basic troubleshooting steps:

1. Identify the problem and note any error messages.
2. Check power, cables, connections, and visible warning indicators.
3. Restart the affected application or device if appropriate.
4. Review relevant settings and recent changes.
5. Use built-in diagnostic tools or trusted support documentation.
6. Test whether the problem has been resolved.
7. Seek qualified technical support if the issue continues.

Example: If a monitor shows no picture, check its power, input selection, cable connection, and whether the computer is running.

Avoid opening electrical equipment or attempting repairs beyond your training.
Topic 52: Safe Handling of Computer Hardware
Safe handling protects computer components from physical damage, electrical hazards, and electrostatic discharge.
Follow these precautions:

- Shut down the computer and disconnect power before installing or removing internal components.
- Follow the manufacturer's instructions.
- Handle circuit boards by their edges.
- Use appropriate electrostatic-discharge precautions.
- Keep liquids and food away from electronic equipment.
- Never force a connector into a port.
- Do not open power supplies or attempt unsafe electrical repairs.

Example: Before installing a RAM module, disconnect power and follow the motherboard manufacturer's instructions for handling and installation.
When in doubt, ask a qualified technician for help.

Topic 53: Hardware Maintenance and Preventive Care
Preventive maintenance helps keep computer hardware clean, reliable, and in good working condition.
Recommended practices include:

- Keep air vents clear and the computer in a well-ventilated location.
- Clean external surfaces using suitable methods recommended by the manufacturer.
- Inspect cables and connectors for visible damage.
- Monitor unusual heat, noise, or recurring errors.
- Keep important files backed up.
- Arrange professional internal cleaning when necessary.

Always power down and disconnect equipment before cleaning it in ways that could expose electrical parts.
Never spray liquid directly onto a computer.

Regular maintenance can reduce avoidable problems, but it cannot prevent every hardware failure.

Topic 54: Choosing Computer Hardware for Different Tasks
Choosing hardware requires matching the system's capabilities to the work it needs to perform.
Consider these factors:

- **Study and office work:** A suitable processor, sufficient RAM, and an SSD are useful for everyday applications.
- **Programming:** Memory, processor capability, storage, and compatibility with development tools matter.
- **Graphic design:** A suitable display, adequate RAM, and graphics capabilities may be important.
- **Video editing:** A capable CPU, suitable GPU, sufficient memory, and fast storage can improve workflow.
- **Gaming:** Graphics performance, processor capability, memory, display specifications, and cooling all matter.

Budget, upgrade options, reliability, warranty, and software requirements should also be considered.
The most expensive hardware is not always the best choice. Select components according to actual needs.

Topic 55: Introduction to Hardware Troubleshooting Tools
Hardware troubleshooting tools help technicians inspect, test, and diagnose computer equipment.
Common tools include:

- **Screwdriver set:** Used for compatible computer screws.
- **Flashlight:** Helps inspect small components and connections.
- **Multimeter:** Measures electrical quantities when used by someone with suitable training.
- **Diagnostic software:** Checks selected hardware conditions, such as memory status or drive health.
- **Compressed-air products designed for electronics:** May help remove dust when used according to safety instructions.
- **Electrostatic-discharge protection:** Helps reduce the risk of static damage to sensitive components.

Different problems require different tools. Always follow safety instructions and avoid electrical measurements or repairs beyond your skill level.

Topic 56: Computer Hardware Upgrades
A hardware upgrade replaces or adds a component to improve a computer's capabilities, compatibility, or useful life.
Common upgrades include increasing RAM, replacing an HDD with an SSD, installing a compatible graphics card, or upgrading a supported processor.
Before upgrading, check:

1. Whether the component is compatible with the motherboard and system.
2. Whether the required slot, connector, and physical space are available.
3. Whether power and cooling requirements can be met.
4. Whether the operating system supports the upgrade.
5. Whether the upgrade offers a worthwhile benefit for the intended task.

Example: Replacing a traditional HDD with a compatible SSD can significantly improve startup and application-loading times.
Some laptops have soldered memory or storage, so their upgrade options may be limited.

Topic 57: Hardware Compatibility and System Requirements
Hardware compatibility means that components and devices can work together correctly within a computer system.
System requirements describe the hardware and software needed to run an operating system or application.
Important considerations include processor architecture, RAM capacity, available storage, graphics support, connection standards, power requirements, and operating-system compatibility.
Minimum requirements may allow an application to run, while recommended requirements generally provide a better experience.
Example: Before installing a new application, check that the computer meets its supported operating-system version, memory, storage, and processor requirements.

Checking compatibility before purchasing hardware can prevent wasted money and installation problems.

Chapter 5: Memory and Storage Fundamentals

Topic 58: Introduction to Computer Memory
Computer memory refers to components that hold data and instructions for processing. Different types of memory serve different purposes.
Primary memory, such as RAM, provides fast access to information currently needed by the processor. Cache memory keeps frequently needed data close to the CPU. Non-volatile memory and firmware storage retain information without continuous power.
The term *memory* is sometimes used broadly, but it is useful to distinguish working memory from long-term storage.

Understanding these differences helps users learn how computers run programs and preserve files.

Topic 59: Primary Memory and Secondary Storage
Primary memory provides the processor with fast access to data and instructions needed for current operations. RAM and CPU cache are important examples.
Secondary storage retains programs and files for longer periods. Examples include SSDs, HDDs, USB flash drives, and memory cards.
The main differences are:

- **Purpose:** Primary memory supports active processing; secondary storage retains files and programs.
- **Speed:** Primary memory is generally faster than secondary storage.
- **Capacity:** Secondary storage usually offers more space for a given cost.
- **Persistence:** RAM is generally volatile, while common secondary storage devices are non-volatile.

Example: A document may be stored on an SSD, loaded into RAM when opened, and then saved back to the SSD after editing.

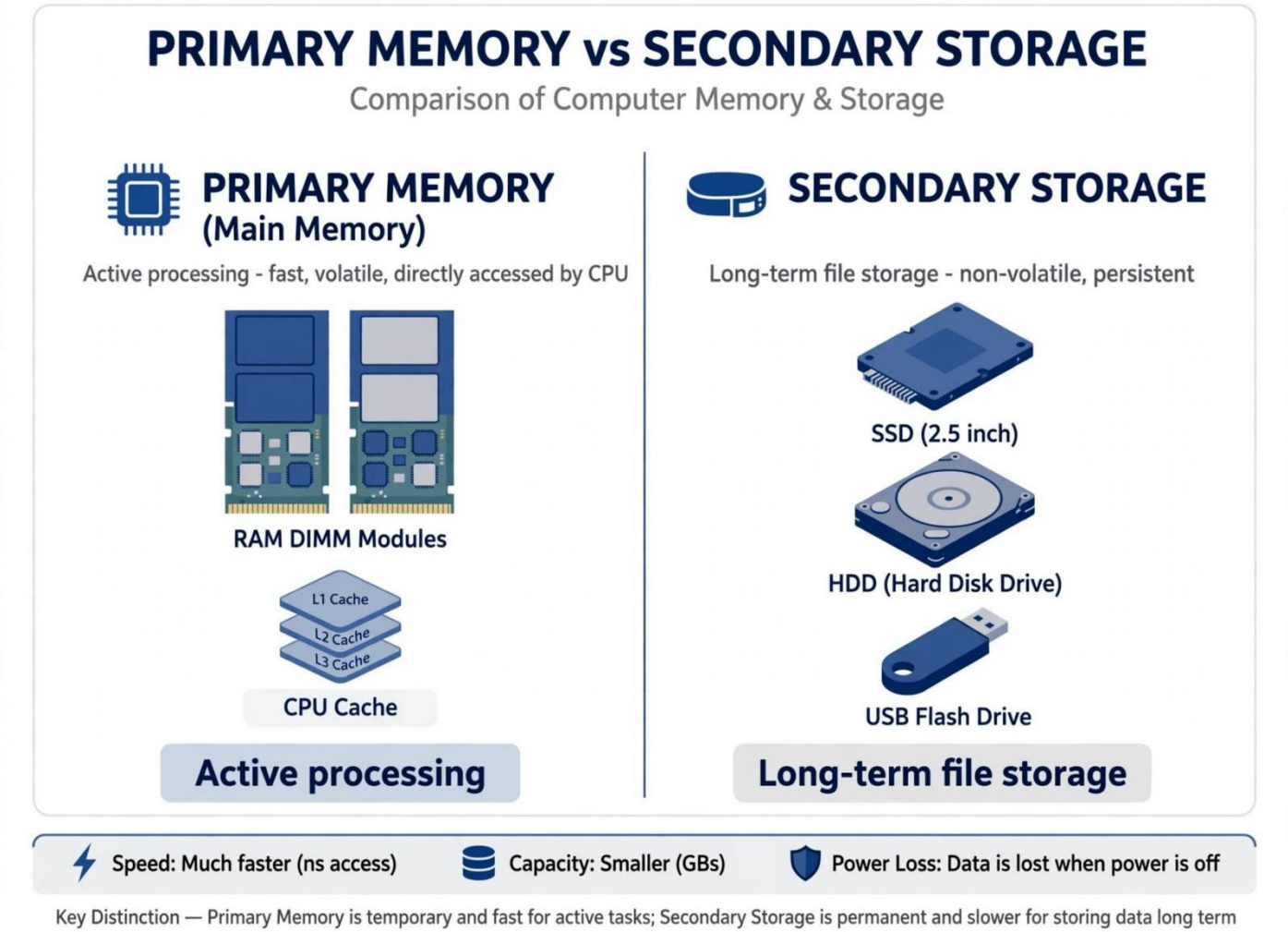


Figure 5.1: Primary Memory and Secondary Storage

Topic 60: Random Access Memory (RAM)
Random Access Memory (RAM) is volatile working memory that temporarily holds the data and instructions a computer is actively using.
RAM allows the processor to access active information quickly without repeatedly retrieving it from long-term storage.
The amount of RAM can affect multitasking and performance when applications require more memory than the system has readily available. However, adding RAM does not automatically make every task faster.
RAM capacity is commonly measured in gigabytes (GB).
Example: If a user runs a browser, a word processor, and a video-editing application together, these programs use RAM to hold their active data.

RAM contents are normally lost when the computer loses power, so important work should be saved to non-volatile storage.

Topic 61: Types of RAM
Random Access Memory (RAM) is available in different types, designed for different purposes and technologies.
The two major categories are:

- **Static RAM (SRAM):** Stores data using electronic circuits called flip-flops. It is fast and does not require periodic refreshing while powered, but it is more expensive and less dense than DRAM. It is commonly used for CPU cache.
- **Dynamic RAM (DRAM):** Stores data in tiny capacitors that must be refreshed regularly. It is commonly used as a computer's main memory because it offers high capacity at a lower cost.

Modern computers commonly use synchronous DRAM (SDRAM) and its Double Data Rate (DDR) generations, such as DDR4 and DDR5.
RAM types must match the requirements of the motherboard and processor platform.

Topic 62: RAM Capacity and Memory Performance
RAM capacity describes how much data the memory can hold at a time. It is commonly measured in gigabytes (GB).
Memory performance also depends on factors such as data-transfer rate, latency, memory-channel configuration, and compatibility.
A computer with insufficient RAM may rely more heavily on storage-based virtual memory, which can slow down demanding tasks.
When selecting RAM, consider:

- The motherboard's supported memory type.
- Maximum supported capacity.
- Supported memory speeds.
- The number of memory slots.
- The requirements of the intended applications.

More RAM can improve multitasking when the existing capacity is insufficient, but additional capacity does not guarantee a proportional increase in speed.

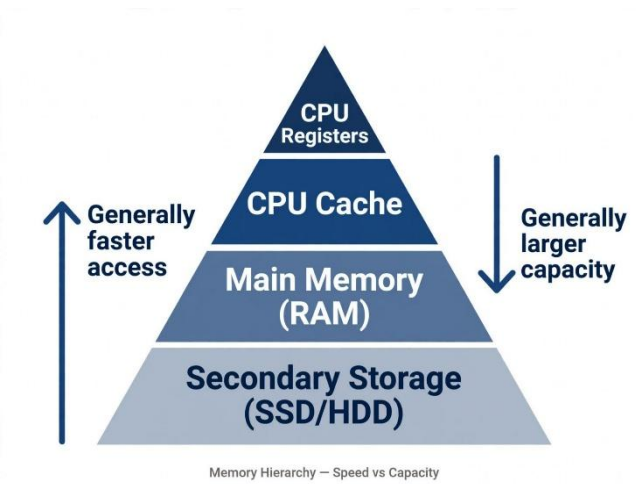
Topic 63: Cache Memory
Cache memory is a small, high-speed memory that stores copies of data and instructions likely to be needed soon by the processor.
It reduces the time the CPU spends waiting for information from slower memory.
Processor cache is commonly organized into levels:

- **L1 cache:** Usually the smallest and closest to the processing core.
- **L2 cache:** Generally larger than L1.
- **L3 cache:** Often larger and shared among multiple processor cores, depending on the design.

Not every processor uses exactly the same cache arrangement.
Example: If a processor repeatedly uses the same instructions or data, cache memory may allow it to access them more quickly.

Topic 64: Registers and CPU Memory Hierarchy
Registers are very small storage locations inside the CPU that hold values needed for immediate processing
They may contain operands, addresses, instructions, or intermediate results.
Computer memory systems are arranged in a hierarchy based on factors such as speed, capacity, and cost.
A simplified hierarchy is:

1. CPU registers
2. CPU cache
3. Main memory (RAM)



4. Secondary storage, such as SSDs and HDDs

Registers and cache generally provide faster access than RAM, while secondary storage usually offers much greater capacity.

Figure 5.2: Simplified Computer Memory Hierarchy

Topic 65: Read-Only Memory and Firmware Storage

Read-Only Memory (ROM) is a non-volatile memory category traditionally used to store instructions that remain available without power.

Modern computer firmware is often stored in flash memory, which can be updated electronically. Firmware provides low-level instructions that help initialize hardware and support startup processes.

Firmware storage is different from RAM because it retains its contents when the system is turned off.

Example: A motherboard stores firmware that helps initialize its hardware before the operating system begins loading.

Firmware updates should be performed carefully, using the manufacturer's instructions and the correct file for the device.

Topic 66: Introduction to Data Storage

Data storage refers to the methods and devices used to preserve digital information for later use.

Storage can hold operating systems, applications, documents, photographs, videos, databases, and backups.

Common storage technologies include:

- **Magnetic storage:** Used by traditional hard disk drives.
- **Flash storage:** Used by SSDs, USB drives, and memory cards.
- **Optical storage:** Used by CDs, DVDs, and Blu-ray discs.

Each technology differs in capacity, speed, durability, portability, and cost.

The appropriate storage method depends on the purpose, required capacity, access frequency, and importance of the data.

Topic 67: Hard Disk Drive (HDD)

A Hard Disk Drive (HDD) is a non-volatile storage device that stores data magnetically on rotating platters. A mechanical actuator moves read/write heads to access information on the platters. Because HDDs contain moving parts, they can be affected by physical shock and mechanical wear.

HDDs are commonly used to store large amounts of data, including documents, videos, software, and archives.

Advantages:

- Often provides a relatively low cost per gigabyte.
- Available in large capacities.
- Useful for storing extensive collections of files.

Limitations:

- Usually slower than modern SSDs.
- Contains moving mechanical components.
- Can be damaged by severe physical impact.

An HDD is a storage device, not a replacement for RAM.

Topic 68: Solid-State Drive (SSD)

A Solid-State Drive (SSD) is a non-volatile storage device that uses flash memory to store data electronically.

Unlike an HDD, an SSD has no rotating platters or moving read/write heads. This design generally allows faster data access and better resistance to ordinary movement-related shock.

Common benefits include:

- Faster system startup and application loading.
- Quiet operation.
- Low power consumption in many designs.
- No mechanical moving parts.

SSDs are available in different form factors and interfaces, including SATA and NVMe-based drives.

Not every SSD uses the same interface or provides the same performance. Compatibility should be checked before purchasing or installing one.

Topic 69: HDD and SSD — A Comparison
HDDs and SSDs both provide long-term storage, but their underlying technologies differ.

Feature	HDD	SSD
Storage technology	Magnetic platters	Flash memory
Moving parts	Yes	No
Typical access speed	Slower	Faster
Noise	May produce mechanical noise	Silent in normal operation
Physical shock resistance	Generally lower	Generally higher
Cost per gigabyte	Often lower	Often higher
Common use	Large-capacity storage	Operating systems, applications, and everyday storage

Actual performance and pricing depend on the specific drive, capacity, interface, and market.
Example: A computer may use an SSD for the operating system and an HDD for storing a large collection of media files.

Topic 70: Optical Storage Devices
Optical storage devices use laser technology to read information from or write information to optical discs.
Common optical disc formats include:

- **CD (Compact Disc):** Commonly used for music and small amounts of digital data.
- **DVD (Digital Versatile Disc):** Offers more capacity than a standard CD.
- **Blu-ray Disc:** Provides greater capacity than standard DVDs and is used for high-definition media and data storage.

Some discs are read-only, while others can be written once or rewritten, depending on the format.
Optical drives may be built into older computers or connected externally. Many modern laptops do not include them.
Optical media can be useful for distribution and archiving, but discs should be stored and handled carefully to avoid scratches and damage.

Topic 71: USB Flash Drives
A USB flash drive is a small, portable storage device that uses flash memory and connects through a USB interface.
It is commonly used to transfer documents, presentations, photographs, videos, and other files between compatible devices.
Advantages:

- Portable and easy to carry.
- Reusable.
- Available in different storage capacities.
- Requires no moving mechanical parts.

Limitations:

- Easy to misplace.
- Can be physically damaged.
- May spread malware if used carelessly.
- Not suitable as the only copy of important files.

Always use trusted devices, scan suspicious files when appropriate, and safely eject removable storage when the operating system recommends it.

Topic 72: Memory Cards

Memory cards are compact removable storage devices commonly used in cameras, mobile devices, and other electronic equipment.
Common formats include SD, microSD, and CompactFlash. Different formats and versions may have different capacities, speeds, and compatibility requirements.
Memory cards are useful for storing photographs, videos, and other digital files.
Before buying one, check:

- Supported card format.
- Maximum supported capacity.
- Required speed class for recording.
- Device compatibility.

A card that physically fits a slot may still be incompatible or unable to meet the device's performance requirements.

Topic 73: Cloud Storage
Cloud storage allows users to store and access files on remote servers through an internet connection.
A cloud service may synchronize files across computers, tablets, and phones, making it easier to access work from different locations.
Common uses include sharing documents, collaborating with others, and keeping an additional copy of important files.
Benefits:

- Access from supported devices.
- Convenient file sharing.
- Synchronization and collaboration.
- Storage capacity that may be expandable through a service plan.

Limitations:

- Some features require internet access.
- Free storage may be limited.
- Account security and privacy require attention.
- Service availability depends on the provider.

Use strong passwords, enable available account-protection features, and review sharing permissions before storing sensitive information.

Topic 74: Network-Attached Storage (NAS)
Network-Attached Storage (NAS) is a dedicated storage system connected to a network. It allows authorized users and devices to access shared files.
A NAS may contain one or more storage drives and provide features such as file sharing, user permissions, automated backups, and remote access.
Common uses include:

- Sharing files within a home or office.
- Centralizing documents and media.
- Creating backup destinations.
- Managing access for different users.

NAS systems vary in performance, capacity, security features, and complexity.
A NAS is not automatically a complete backup solution. If important data exists only on the NAS, drive failure, theft, malware, or other incidents may still cause data loss.

Topic 75: Storage Capacity and Data Units
Digital storage capacity measures how much data a device can hold.
Common units include:

- **Bit (b):** A basic unit of digital information, represented as 0 or 1.
- **Byte (B):** Usually consists of 8 bits.
- **Kilobyte (KB):** Commonly 1,000 bytes in decimal notation.
- **Megabyte (MB):** 1,000 kilobytes in decimal notation.
- **Gigabyte (GB):** 1,000 megabytes in decimal notation.
- **Terabyte (TB):** 1,000 gigabytes in decimal notation.

Computing also uses binary units: 1 KiB equals 1,024 bytes, 1 MiB equals 1,024 KiB, and 1 GiB equals 1,024 MiB.
Storage manufacturers commonly advertise capacities using decimal units, while some operating systems display capacity using binary-based calculations or different labels. This can make the reported available capacity appear smaller than the advertised number.

Topic 76: Data Transfer Speed and Storage Performance
Data transfer speed describes how quickly information moves between devices or components. Storage performance can also depend on access latency, workload, and the way data is read or written.
Transfer rates are commonly expressed in megabytes per second (MB/s) or gigabytes per second (GB/s).
Network and communication speeds are often expressed in megabits per second (Mb/s).
Remember that a byte contains eight bits, so MB/s and Mb/s are not interchangeable.
Storage performance can be affected by:

- Drive technology and interface.
- File size and number of files.
- Read or write workload.
- Available free space and device condition.
- Other tasks running on the computer.

Advertised maximum speeds do not always represent everyday performance.

Topic 77: File Compression and Archives
File compression reduces the amount of space needed to store or transfer data.
Two major forms are:

- **Lossless compression:** Preserves the original data so it can be recovered exactly.
- **Lossy compression:** Removes selected information to reduce file size, often used for images, audio, and video.

An archive file may combine multiple files and folders into one package. Some archive formats also compress their contents.
Common formats include ZIP, 7z, and RAR, although software support varies.
Example: Compressing a folder into a ZIP archive can make it easier to share several documents together.
Compression results depend on the file type. Already-compressed files may not become much smaller.

Topic 78: File Backup and Recovery
A backup is an additional copy of data kept so that files can be restored if the originals are lost, damaged, deleted, or made inaccessible.
Backups can be stored on external drives, network storage, or cloud services.
Common backup approaches include:

- **Full backup:** Copies all selected data.
- **Incremental backup:** Copies changes made since the previous backup.
- **Differential backup:** Copies changes made since the last full backup.

A useful backup strategy considers how frequently data changes, how quickly it must be restored, and how much data loss is acceptable.
A backup should be tested periodically to confirm that files can actually be restored.

Topic 79: The 3-2-1 Backup Rule
The 3-2-1 backup rule is a widely used guideline for protecting important data.
It recommends keeping:

- **3 copies** of important data, including the original.
- **2 different types of storage media** or storage systems.
- **1 copy off-site**, separate from the main location.

For example, a user might keep the original files on a computer, a backup on an external drive, and another copy in a secure cloud service.
This approach reduces dependence on a single device or location. Additional safeguards, such as version history and offline copies, may help protect against ransomware and accidental deletion.
The exact setup should match the importance and sensitivity of the data.

Topic 80: Data Security in Storage Devices
Data security protects stored information from unauthorized access, alteration, loss, and misuse.
Important practices include:

- Use strong, unique passwords for accounts that control access to stored data.
- Enable encryption when appropriate.
- Keep operating systems and storage-related software updated.
- Limit access permissions to authorized users.
- Maintain tested backups.
- Be careful when disposing of or transferring old storage devices.
- Avoid connecting unknown or untrusted removable devices.

Deleting a file normally removes its reference from the file system; it does not always securely erase every copy of its data.
For devices containing sensitive information, use a suitable secure-erasure or disposal method recommended by the manufacturer or a qualified service provider.

Chapter 6: Input and Output Devices

Topic 81: Introduction to Input Devices
An input device is a hardware component used to enter data, instructions, or commands into a computer.
Input devices allow users to interact with software and provide information for processing.
Common input devices include:

- Keyboard
- Mouse
- Touchpad
- Scanner

- Microphone
- Webcam
- Barcode reader
- Touchscreen

Different input devices are designed for different types of information. A keyboard enters text, a microphone captures sound, and a scanner converts physical documents into digital images.
Example: When a student types an assignment using a keyboard, the keyboard sends the entered characters to the computer for processing.

Topic 82: Keyboard
A keyboard is an input device used to enter letters, numbers, symbols, and commands into a computer.
Most keyboards contain several groups of keys:

- **Alphanumeric keys:** Letters, numbers, and punctuation marks.
- **Function keys:** Keys such as F1–F12, whose functions depend on the software and system.
- **Modifier keys:** Shift, Ctrl, and Alt, used with other keys.
- **Navigation keys:** Arrow keys, Home, End, Page Up, and Page Down.
- **Numeric keypad:** A separate group of number and arithmetic keys on many full-size keyboards.
- **Special keys:** Enter, Backspace, Delete, Esc, and Tab.

Keyboards may be wired or wireless. Laptop keyboards are usually integrated into the device.

Topic 83: Keyboard Layouts and Key Functions
A keyboard layout defines how characters and functions are arranged on a keyboard.
Common layouts include QWERTY, AZERTY, and QWERTZ. The layout used depends on language, region, and user preference.
Some important keys are:

- **Enter:** Confirms an action or starts a new line.
- **Backspace:** Deletes the character before the cursor in many text fields.
- **Delete:** Removes selected content or the character after the cursor in many text editors.
- **Shift:** Types uppercase letters or alternate symbols when supported.
- **Caps Lock:** Toggles uppercase letter entry.
- **Tab:** Moves between fields or inserts indentation in some applications.
- **Esc:** Cancels or exits certain operations.
- **Arrow keys:** Move the cursor or selection.

Key functions can vary by application and operating system.

Topic 84: Keyboard Shortcuts
Keyboard shortcuts allow users to perform common operations quickly by pressing one key or a combination of keys.
Common Windows shortcuts include:

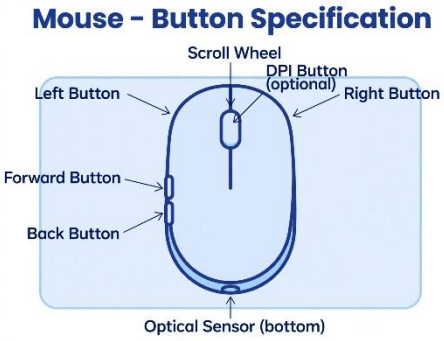
Shortcut	**Common Function**
Ctrl + C	Copy selected content
Ctrl + X	Cut selected content
Ctrl + V	Paste copied or cut content
Ctrl + Z	Undo the previous action
Ctrl + S	Save the current file
Ctrl + A	Select all content in the active area
Ctrl + P	Open the print dialog in many applications
Alt + Tab	Switch between open applications
Windows + E	Open File Explorer
Windows + L	Lock the computer

Shortcuts may behave differently depending on the active application. Some applications use different shortcuts for particular operations.

Topic 85: Mouse

A mouse is a pointing input device used to control the pointer on a computer screen.

A typical mouse includes a left button, a right button, and a scroll wheel. Some models include additional buttons.



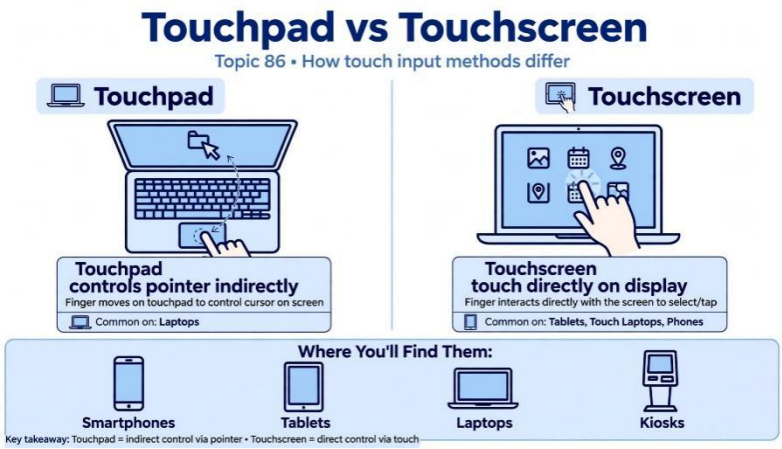
Common mouse actions include:

- **Click:** Press and release a mouse button once.
- **Double-click:** Click twice quickly, often used to open files or folders.
- **Right-click:** Opens a context menu in many applications.
- **Drag and drop:** Hold a button while moving an item, then release it at the desired location.
- **Scroll:** Move through a document or webpage using the wheel

Optical and laser mice use light-based sensing to detect movement. Mice may connect through USB or wireless technologies.

Topic 86: Touchpad and Touchscreen

A touchpad is a touch-sensitive pointing surface commonly found on laptops. It allows users to move the pointer and perform gestures without a separate mouse. Common touchpad actions include tapping to click, dragging, scrolling with two fingers and using supported multi-finger gestures.



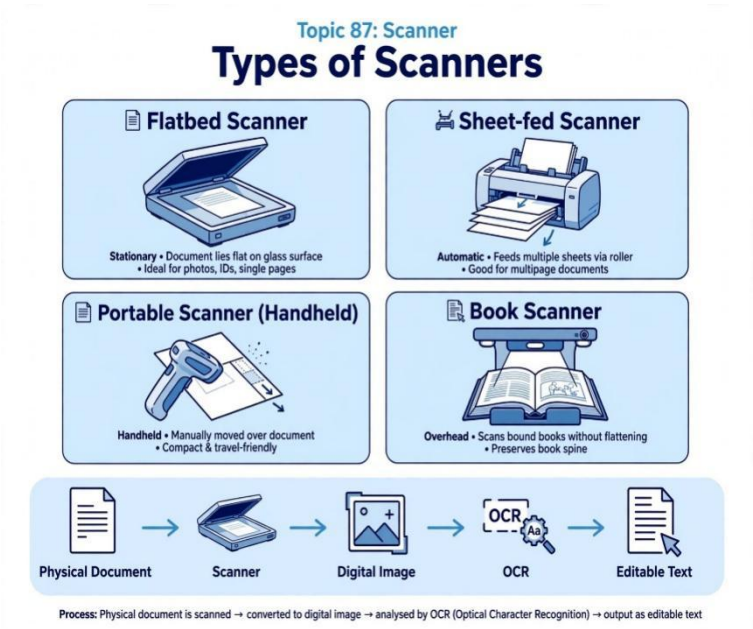
A touchscreen allows users to interact directly with displayed content by touching the screen with a finger or compatible stylus. Touchscreens are used in smartphones, tablets, laptops, kiosks, and some desktop monitors.

The main difference is that a touchpad controls an on-screen pointer, while a touchscreen detects touch directly on the display.

Topic 87: Scanner

A scanner is an input device that captures text, photographs, drawings, or other physical material and converts it into a digital image. Common types include:

- **Flatbed scanner:** Uses a glass surface to scan documents and photographs.
- **Sheet-fed scanner:** Pulls sheets through a scanning mechanism.
- **Portable scanner:** Designed for convenient scanning while travelling.
- **Book scanner:** Designed to capture pages from books with reduced handling



Scanners are used in offices, schools, libraries, and businesses to digitize records.

A scanned page is usually an image. To make printed text searchable or editable, Optical Character Recognition (OCR) software may be required.

Topic 88: Microphone and Audio Input

A microphone is an input device that converts sound into an electrical or digital signal that a computer can process.
Microphones are used for voice calls, online classes, recordings, speech recognition, gaming, and video production.
Common types include:

- **Built-in microphone:** Integrated into a laptop, phone, or webcam.
- **USB microphone:** Connects through a USB port.
- **Analog microphone:** Often connects through a compatible audio input.
- **Wireless microphone:** Transmits audio through a supported wireless system.

Sound quality depends on microphone design, placement, room noise, connection type, and recording settings.
Example: During an online class, a microphone captures the teacher's voice and sends the audio to the computer for transmission.

Topic 89: Webcam and Digital Camera
A webcam is a camera designed to capture images or video for use with a computer or other compatible device.
Webcams are commonly used for video meetings, online teaching, interviews, streaming, and recording.
Some webcams include built-in microphones. Others require a separate audio input device.
Digital cameras capture photographs and videos in digital form. Images can be transferred to a computer through a cable, memory card, or wireless connection, depending on the device.
Image quality depends on factors such as sensor size, lens quality, lighting, resolution, and image-processing capabilities.
Users should review camera permissions and avoid sharing video from private spaces without considering privacy.

Topic 90: Barcode and QR Code Readers
A barcode reader captures information represented by a printed or displayed code and converts it into data a computer system can process.
A traditional barcode usually represents information through a pattern of lines and spaces. A QR code is a two-dimensional code that can store different kinds of information.
These readers are commonly used in retail shops, libraries, warehouses, hospitals, and inventory systems.
Examples:

- A shop scans a product barcode to identify an item.
- A student scans a QR code to open a website or access an event registration page.

A code reader does not guarantee that the encoded information is safe. Users should verify unfamiliar links before opening them.

Topic 91: Biometric Input Devices
Biometric input devices capture physical or behavioural characteristics for identification or verification.
Examples include:

- Fingerprint readers
- Facial recognition cameras or sensors
- Iris recognition systems
- Voice recognition systems

Biometric systems compare captured information with previously enrolled data to determine whether it meets the system's matching criteria.
They are used in supported phones, laptops, security systems, and access-control applications.
Biometric systems are not perfect. Performance can depend on sensor quality, environmental conditions, and the matching method. Biometric information is sensitive and should be protected through appropriate security and privacy measures.

Chapter 7: Output Devices

Topic 92: Introduction to Output Devices
An output device presents the results of computer processing in a form that users can understand or use. Output may be visual, printed, or audio-based.
Common output devices include:

- Monitor

- Printer
- Speakers
- Headphones
- Projector
- Plotter

A monitor displays text and images, speakers produce sound, and a printer creates a physical copy of digital content.
Some devices perform both input and output functions. A touchscreen, for example, displays information and detects user touch.

Topic 93: Monitor
A monitor is an output device that displays text, images, video, and graphical interfaces generated by a computer.
Monitors are used for office work, education, programming, design, entertainment, and many other tasks.
Important monitor specifications include:

- **Screen size:** Usually measured diagonally in inches.
- **Resolution:** The number of pixels used to form the image.
- **Refresh rate:** The number of times the display can refresh per second, measured in hertz (Hz).
- **Response time:** A measure of how quickly a pixel changes state under specified test conditions.
- **Brightness:** The amount of light produced by the display.
- **Panel technology:** Such as IPS, VA, or OLED-based display technology, each with different characteristics.

The best monitor depends on the user's tasks, budget, workspace, and visual preferences.

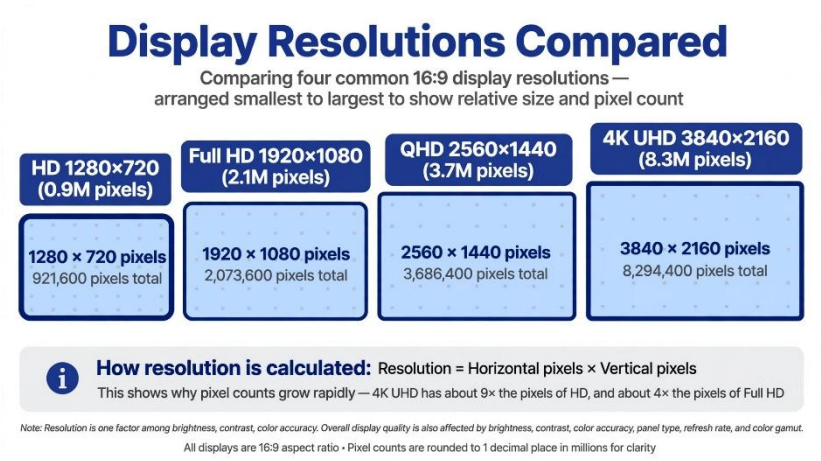
Topic 94: Types of Monitors
Monitors use different display technologies and designs.
Common types include:

- **CRT monitor:** An older display that uses a cathode-ray tube. It is bulky and heavier than most modern monitors.
- **LCD monitor:** Uses liquid crystals with a backlight.
- **LED-backlit LCD monitor:** An LCD monitor that uses LEDs for its backlight. Many modern monitors marketed as LED monitors belong to this category.
- **OLED display:** Uses organic light-emitting diodes, allowing individual pixels to produce light.
- **Touchscreen monitor:** Supports direct touch input as well as visual output.

These categories are not all mutually exclusive. For example, a touchscreen can use LCD or OLED technology.
Display selection should consider image quality, intended use, energy consumption, connectivity, and ergonomic requirements.

Topic 95: Screen Resolution and Display Quality
Screen resolution describes the number of pixels used to display an image, usually expressed as horizontal pixels by vertical pixels.
Common resolutions include:

- **HD:** Often 1280 × 720.
- **Full HD (1080p):** 1920 × 1080.
- **QHD:** Commonly 2560 × 1440.
- **4K UHD:** 3840 × 2160.



Higher resolution can display more detail, but the visible result also depends on screen size, pixel density, viewing distance, scaling, and content quality.
Refresh rate, colour reproduction, contrast, brightness, and viewing angles also influence the overall display experience.

Figure 7.1: Common Display Resolutions

Topic 96: Printer

A printer is an output device that produces a physical copy of digital text, images, or other documents on paper or another suitable medium.

Printers are used in schools, offices, homes, shops, and industrial environments.

Common printer types include:

- **Inkjet printer:** Sprays tiny droplets of ink onto the printing surface.
- **Laser printer:** Uses an electro-photographic process and toner.
- **Thermal printer:** Uses heat to produce output on suitable thermal media.
- **Dot-matrix printer:** Uses pins striking an ink ribbon to form characters and images.

Printer selection depends on print quality, speed, colour requirements, running costs, connectivity, and expected workload.

Topic 97: Types of Printers and Their Uses

Printers can also be grouped according to how they produce output.

Impact printers physically strike an inked ribbon against paper. Dot-matrix printers are a common example.

Non-impact printers produce output without striking the paper with a printing mechanism. Inkjet, laser, and many thermal printers belong to this broad group.

Typical uses include:

- **Inkjet:** Home printing, photographs, and occasional colour documents.
- **Laser:** High-volume text documents and office printing.
- **Thermal:** Receipts, labels, and supported point-of-sale applications.
- **Dot-matrix:** Certain multipart forms and specialized business tasks.

These are general tendencies, not strict rules. The best choice depends on the specific model and printing requirements.

Topic 98: Speakers and Headphones

Speakers and headphones are output devices that convert audio signals into sound.

Speakers produce sound that can be heard by people in the surrounding area. They are commonly used for music, presentations, alerts, and video playback.

Headphones deliver sound directly to the listener. They may be wired or wireless and can include microphones for communication.

Audio quality depends on the device, sound source, connection, settings, and listening environment.

Users should keep volume at a comfortable level and take breaks during extended listening.

Topic 99: Projector

A projector is an output device that displays an enlarged image or video on a suitable surface, such as a projection screen or wall.

Projectors are commonly used in classrooms, conference rooms, training sessions, and presentations.

Important features include:

- **Brightness:** Influences visibility under different lighting conditions.
- **Resolution:** Determines the detail the projector can display.
- **Contrast:** Affects the distinction between dark and bright areas.
- **Connectivity:** Determines which computers and media devices can connect.
- **Throw distance:** Influences how far the projector must be from the screen.

Projectors use different technologies, including LCD and DLP, while some models use other display methods.

For a clear presentation, use suitable lighting, correct focus, and a compatible video connection.

Topic 100: Plotters and Large-Format Output

A plotter is a device designed to produce large-format drawings, graphics, or other specialized output.

Modern large-format printing systems can produce detailed engineering drawings, maps, architectural plans, posters, and banners.

Traditional pen plotters moved pens across a surface to draw lines. Many modern devices described as plotters use inkjet or other printing technologies.

Plotters and large-format printers are used in architecture, engineering, construction, design, advertising, and geographic information work.
Example: An engineering office may use a large-format printer to produce a detailed building plan that can be read at full size.
The suitable device depends on the required output size, media type, accuracy, colour requirements, and production volume.

Chapter 8: Operating Systems Fundamentals

Topic 101: Introduction to Operating Systems

An Operating System (OS) is system software that manages computer hardware and provides an environment for running applications.

It acts as an intermediary between users, applications, and the computer's hardware. Without an operating system, most general-purpose computers would be difficult to use for everyday tasks.
Common operating systems include Microsoft Windows, macOS, Linux distributions, Android, and iOS.
Operating systems differ according to their design, supported devices, features, and intended uses.
Example: Microsoft Windows allows users to open applications, manage files, connect devices, and change system settings through a graphical interface.

Topic 102: Functions of an Operating System
An operating system performs several important functions to keep a computer running efficiently.
Its main functions include:

- **Process management:** Coordinates running programs and tasks.
- **Memory management:** Allocates and manages memory used by programs.
- **File management:** Organizes files and folders on storage devices.
- **Device management:** Coordinates communication with supported hardware.
- **User interface:** Provides ways for users to interact with the computer.
- **Security and access control:** Helps protect accounts, resources, and data.
- **Networking:** Supports communication with networks and other devices.

The exact functions and implementation vary across operating systems.

Topic 103: Types of Operating Systems
Operating systems can be classified according to their purpose, design, and the devices they support.
Common categories include:

- **Desktop operating systems:** Designed for personal computers, such as Windows and macOS.
- **Mobile operating systems:** Designed for smartphones and tablets, such as Android and iOS.
- **Server operating systems:** Designed to support network services, applications, and shared resources.
- **Embedded operating systems:** Used in dedicated devices such as industrial controllers and appliances.
- **Real-time operating systems:** Designed to meet timing requirements for tasks that must respond within specified deadlines.

These categories may overlap. An operating system can support multiple types of workloads or devices.

Topic 104: Graphical User Interface (GUI)
A Graphical User Interface (GUI) allows users to interact with a computer through visual elements instead of relying entirely on typed commands.
Common GUI elements include windows, icons, menus, buttons, pointers, and dialog boxes.
A GUI makes many computer operations easier to discover and perform, especially for users who are learning to use a computer.
Example: Double-clicking a folder icon in Windows opens the folder in File Explorer.
Most desktop operating systems provide a GUI, although command-line interfaces may also be available.

Topic 105: Command-Line Interface (CLI)
A Command-Line Interface (CLI) allows users to interact with a computer by typing commands into a text-based environment.
A CLI can be useful for administration, troubleshooting, automation, programming, and performing repetitive tasks.
Examples include Command Prompt and PowerShell in Windows, as well as terminal shells in Linux and macOS.
Commands must follow the syntax supported by the relevant shell or program. Incorrect commands may produce errors or affect files and settings.
Example: The dir command in Windows Command Prompt displays files and folders in the current directory.
Users should understand a command before running it, especially if it changes or deletes data.

Topic 106: Multitasking and Multiuser Operating Systems

Multitasking allows an operating system to manage multiple tasks so users can work with several applications during the same period.
For example, a user may listen to music, download a file, and edit a document while the operating system coordinates these activities.
A **multiuser operating system** supports multiple user accounts or users accessing system resources, sometimes simultaneously.
User accounts can have different permissions, allowing administrators to control access to files, applications, and settings.
These capabilities depend on the operating system and its configuration.

Topic 107: Device Drivers and System Utilities
A device driver is software that enables an operating system to communicate with and control a particular hardware device.
Examples include drivers for graphics adapters, printers, audio devices, and network adapters.
A system utility is a program designed to maintain, inspect, configure, or protect a computer.
Common utilities include storage-management tools, backup utilities, diagnostic tools, and security software.
Drivers and utilities serve different purposes: drivers help hardware work with the operating system, while utilities help users manage or maintain the system.
Use trusted sources, such as the operating system's update service or the device manufacturer's official support page, when obtaining drivers.

Topic 108: Booting and Shutting Down a Computer
Booting is the process of starting a computer and preparing it to run its operating system.
During startup, the system initializes hardware, runs firmware procedures, and loads the operating system into memory.
A simplified startup sequence is:

1. The computer receives power.
2. Firmware initializes and checks essential hardware.
3. The system identifies a suitable boot device.
4. The operating system loader begins loading the operating system.
5. The sign-in screen or desktop becomes available.

Shutting down closes running applications as appropriate and powers off the computer through the operating system.
Save your work before shutting down. Use Restart when the computer needs to start again, and avoid disconnecting power while the system is performing important operations.

Topic 109: Windows Operating System
Microsoft Windows is a family of operating systems developed by Microsoft for personal computers and other devices.
Windows provides a graphical interface, file management, application support, device management, networking, security features, and system settings.
Different Windows versions offer different features and hardware requirements.
Windows is widely used in homes, schools, offices, and businesses because it supports a broad range of software and devices.
The appearance and location of some features can vary by Windows version, edition, and configuration.
This book primarily describes common Windows desktop concepts, noting where steps may differ.

Topic 110: Windows Versions and Editions
Microsoft has released multiple versions of Windows over time. Each version introduces changes to the interface, features, security, and hardware support.
Examples of major desktop releases include Windows 7, Windows 8.1, Windows 10, and Windows 11.
An operating-system **version** identifies a particular release or generation, while an **edition** identifies a product variant with a particular set of features.
For example, Windows Home and Windows Pro are editions of supported Windows releases. Their available administrative and security features can differ.

When installing software or troubleshooting a computer, check the installed Windows version, edition, and system requirements rather than relying only on the computer's brand or age.

Chapter 9: Windows Desktop and Basic Navigation

Topic 111: Introduction to the Windows Desktop

The Windows desktop is the main workspace displayed after a user signs in, unless the system opens another screen or application first.
It commonly contains a background image or colour, desktop icons, and a taskbar. Users can open applications, access files, and organize shortcuts from this environment.
The desktop helps users reach frequently used programs and locations.
Its appearance can be customized through supported Windows settings, including background, themes, and some icon options.
Example: A student may keep a shortcut to a frequently used learning application on the desktop for quick access.

Topic 112: Main Components of the Windows Desktop
The Windows desktop contains several interface elements that help users navigate the system.
Important components include:

- **Desktop background:** The image or colour behind desktop icons.
- **Desktop icons:** Shortcuts, files, folders, or system items displayed on the desktop.
- **Taskbar:** Provides access to Start, search, pinned applications, running applications, and system indicators.
- **Start button:** Opens the Start menu.



Figure 1.1 — Simplified Windows 11-style desktop interface showing common components. Labels identify key elements for educational reference.

- **Notification area:** Shows selected status indicators and background-app icons.
- **Open windows:** Display the applications and documents currently being used.

The exact arrangement can differ between Windows versions and personal settings.

Figure 9.1: Main Components of the Windows Desktop

Topic 113: Desktop Icons and Shortcuts
A desktop icon is a small graphic representing an application, file, folder, or system item.
A shortcut is a link that provides quick access to an item stored elsewhere. Deleting a shortcut normally does not delete the original file or application.
Common desktop icons may include the Recycle Bin, folders, documents, and application shortcuts.
Users can often create a shortcut by right-clicking an item and selecting an available shortcut or send-to option. Exact steps depend on the item's type and Windows version.
Keeping the desktop organized makes frequently used items easier to find.

Topic 114: Opening and Closing Applications
Applications are programs used to perform tasks such as writing documents, browsing the internet, editing images, and playing media.

In Windows, applications can commonly be opened through the Start menu, taskbar, desktop shortcuts, or search.
To close an application, use its Close button or the appropriate application command. Save important work before closing.
Closing a window may not always end every background process associated with the application.
If an application stops responding, wait briefly and use the operating system's available recovery or task-management options. Unsaved changes may be lost when a nonresponsive application is forced to close.

Topic 115: Working with Windows

A window is a rectangular area that displays an application, document, folder, or dialog.
Common window controls include:

- **Minimize:** Hides the window from the desktop while leaving the application available.
- **Maximize:** Expands the window to occupy most or all of the available desktop area.
- **Restore Down:** Returns a maximized window to a resizable size.
- **Close:** Closes the window or application, depending on its behaviour.

Users can move a window by dragging its title bar and resize it by dragging a suitable edge or corner, when resizing is supported.
Windows can often be arranged side by side to make comparison and multitasking easier.

Topic 116: The Start Menu

The Start menu provides access to applications, search, settings, power options, and other supported Windows features.
Its layout has changed across Windows versions. In modern Windows releases, users may see pinned applications, a list of installed applications, recommended items, and account-related options.
The Start menu can help users find applications without remembering where each program is stored.
Example: To open a calculator, select Start and locate Calculator, or search for its name.
The available sections and controls depend on the Windows version and configuration.

Topic 117: Opening the Start Menu

The Start menu can usually be opened in either of these ways:

1. Select the Start button on the taskbar.
2. Press the Windows logo key on the keyboard.

After the menu opens, select a pinned application, browse the application list where available, or use search to find a program or setting.
To close the menu without launching anything, select the Start button again, press the Windows key, or click an appropriate area outside the menu.
Keyboard and mouse methods offer convenient alternatives depending on the task.

Topic 118: Pinned Applications and the All Apps List

Pinned applications are shortcuts displayed in an easily accessible area of the Start menu or taskbar.
Pinning an application does not normally move the program itself; it creates a convenient way to open it.
The All Apps list, available in supported Windows versions, provides access to installed applications that may not be pinned.
Users can commonly pin an application by opening its context menu and selecting the relevant pin option.
Removing an item from a pinned area usually removes the shortcut from that area rather than uninstalling the application.
Menu labels and steps vary by Windows version.

Topic 119: Power Options in Windows

Windows provides power options that allow users to manage how a computer operates when it is not being actively used.
Common options include:

- **Sleep:** Places the computer in a low-power state so work can usually be resumed quickly.
- **Hibernate:** Saves the current system session to storage and powers down more fully, where supported.
- **Shut down:** Closes the current session and turns off the computer.

- **Restart:** Shuts down and starts the computer again.

Available options depend on the device, firmware, and system settings.
Sleep is useful for short breaks, while shutting down may be suitable when the computer will not be used for an extended period. Restart is often useful after certain updates or troubleshooting steps.
Save important work before using any power option.

Topic 120: Sign Out, Lock, Sleep, and Shut Down

These options serve different purposes and should not be treated as interchangeable.

- **Sign out:** Ends the current user session and returns Windows to a sign-in screen or user-selection screen. Open applications in that session may close.
- **Lock:** Protects the current session behind the sign-in screen while leaving applications running.
- **Sleep:** Places the computer in a low-power state for quick resumption.
- **Shut down:** Closes the current session and powers off the computer.
- **Restart:** Reboots the system, closing and then reopening the operating system.

Example: Lock the computer when briefly leaving your desk but wanting applications to remain open. Sign out when another user needs to access their own session.
Choose the option that matches your needs, and save work before signing out, shutting down, or restarting.

CHAPTER 10: WINDOWS DESKTOP AND PERSONALIZATION

Topic 121: Introduction to Desktop Personalization
Desktop personalization means changing the appearance and arrangement of the Windows desktop according to the user's preferences. It allows users to create a comfortable and organized working environment.
Windows provides several personalization options, including changing the desktop background, selecting themes, choosing colors, adjusting the lock screen, and managing desktop icons.
Personalization mainly affects the appearance of the operating system. Some options may also improve accessibility and make the computer easier to use.

Topic 122: Changing the Desktop Background
The desktop background is the image, solid color, or slideshow displayed behind desktop icons. Users can select a background that suits their preferences.
Steps to change the desktop background in Windows 11:

1. Right-click an empty area of the desktop.
2. Select **Personalize**.
3. Open the **Background** settings.s
4. Choose Picture, Solid Color, or Slideshow.
5. Select an image or color according to your choice.
6. Adjust the picture display option if available.

The selected background appears on the desktop. The available options may vary slightly between Windows versions.

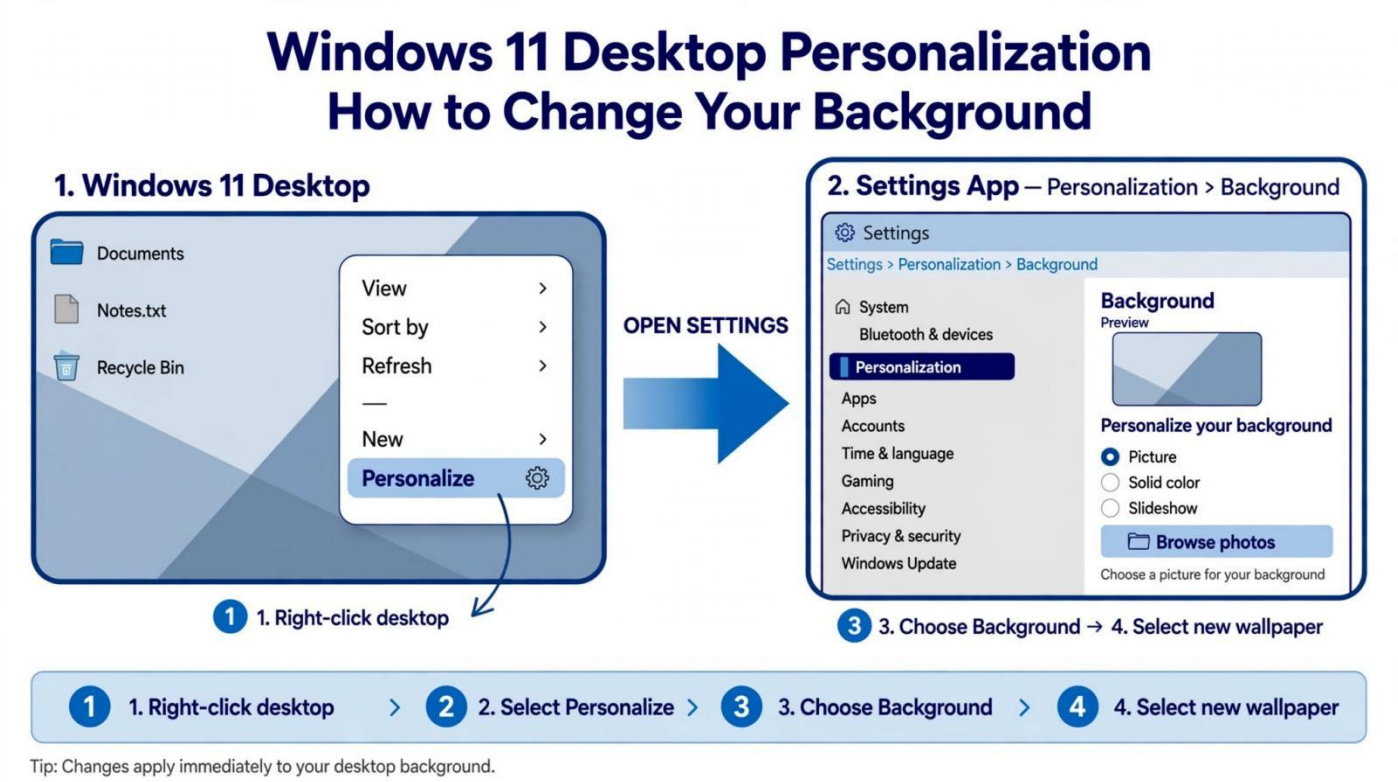


Figure 9.2 — Windows Desktop Personalization Settings

Topic 123: Understanding Windows Themes
A theme is a collection of visual settings that work together to create a consistent appearance in Windows. Depending on the Windows version and theme, it may include a desktop background, colors, sounds, and mouse cursor settings.
Themes help users customize the operating system without changing every appearance setting separately.
To select a theme:

1. Open Settings.
2. Select Personalization.
3. Open Themes.
4. Choose an available theme.

Some themes can also be downloaded from the Microsoft Store or other official Microsoft sources. Availability depends on the Windows version and device configuration.

Topic 124: Changing Windows Colors and Modes

Windows allows users to customize interface colors and choose between light and dark appearance modes. The light mode generally displays light backgrounds with dark text, while the dark mode uses darker backgrounds for many supported interface elements.

To change color settings in Windows 11:

1. Open Settings.
2. Select Personalization.
3. Choose Colors.
4. Select Light, Dark, or an available custom mode.
5. Adjust the accent color if desired.

The accent color highlights selected interface elements, such as certain buttons and window borders. Not every application follows Windows color settings in the same way.

Topic 125: Screen Resolution and Display Scaling

Screen resolution refers to the number of pixels used to display an image on a screen. A resolution of 1920 × 1080 means the display image contains 1920 pixels horizontally and 1080 pixels vertically.

Display scaling changes the size of text, icons, and other interface elements without necessarily changing the screen's resolution.

To adjust display settings:

1. Open Settings.
2. Select System.
3. Choose Display.
4. Locate the Scale and Display resolution options.
5. Select an appropriate setting.

The recommended resolution is generally a good starting point because it is selected for the display's capabilities. Available resolutions and scaling options depend on the monitor, graphics hardware, and Windows configuration.

Topic 126: Managing Desktop Icons

Desktop icons provide quick access to files, folders, applications, and certain system locations. Users can organize these icons to make frequently used items easier to find.

Windows can display standard system icons, such as This PC and Recycle Bin, when the corresponding options are enabled.

To display or hide standard desktop icons in Windows 11:

1. Open Settings.
2. Select Personalization.
3. Choose Themes.
4. Open Desktop icon settings.
5. Select the icons you want to display.
6. Click Apply and then OK.

The available system icons may include This PC, Recycle Bin, Network, and the user's files folder.

Topic 127: Arranging and Aligning Desktop Icons

Arranging desktop icons helps maintain an organized workspace. Windows provides options for aligning icons and controlling how they are positioned.

To arrange desktop icons:

1. Right-click an empty area of the desktop.
2. Point to View.
3. Select an available icon-size option if required.
4. Enable or disable Auto arrange icons according to your preference.
5. Enable or disable Align icons to grid as needed.

Auto arrange icons organizes icons automatically. **Align icons to grid** helps position icons neatly in an invisible grid.

When Auto arrange is disabled, users can usually move desktop icons manually to preferred positions.

Topic 128: Creating and Managing Desktop Shortcuts

A desktop shortcut is a link that provides quick access to an application, file, folder, or another location. A shortcut does not normally create a second copy of the original item.

Shortcuts are useful for frequently used programs and documents.

To create a shortcut using the desktop context menu:

1. Right-click an empty area of the desktop.
2. Select New.
3. Choose Shortcut.
4. Enter or browse to the location of the item.
5. Click Next.
6. Enter a name for the shortcut.
7. Click Finish.

To rename a shortcut, right-click it and select Rename. To remove it, select Delete.

Deleting a shortcut normally does not delete the original file or application. However, users should carefully distinguish shortcuts from original files before deleting items.

Topic 129: Using Task View and Multiple Desktops

Task View helps users see open windows and manage virtual desktops. Virtual desktops allow users to separate different groups of windows into distinct workspaces.

For example, a student may use one desktop for online classes and another for assignments.

To open Task View:

1. Press Windows + Tab.
2. Review the open windows.
3. Select an existing virtual desktop or create a new one using the available controls.
4. Switch between desktops as needed.

Virtual desktops help organize work, but they do not create separate Windows accounts. They generally share the same user account, files, and applications.

The exact appearance and controls may vary between Windows versions.

Topic 130: Arranging Windows with Snap Features

Windows Snap features help users arrange application windows on the screen. They are particularly useful when working with multiple documents or applications simultaneously.

For example, a user can place a web browser on one side of the screen and a word processor on the other.

Common ways to arrange windows include:

- Dragging a window to a screen edge or corner.
- Using available Snap Layouts.
- Using keyboard shortcuts to position windows.

In supported Windows versions, pressing Windows + Z opens Snap Layouts for the active window. Pressing Windows + Left Arrow or Windows + Right Arrow can help position a window to the left or right, depending on the current layout and configuration.

Snap features improve multitasking by reducing the need to resize and reposition windows manually.

Topic 131: Understanding Windows Widgets

Widgets provide small information panels that display selected information, such as weather, news, calendar details, or other supported content.

Their availability and features depend on the Windows version, region, account settings, and installed applications.

In Windows 11, users can generally open the Widgets board from the taskbar when the feature is available.

Widgets may allow users to customize the information they see. However, some content requires an internet connection, and some features may require signing in to an account.

Users should review privacy settings and choose widgets according to their needs.

Topic 132: Notifications and Quick Settings

Windows provides tools for viewing notifications and accessing frequently used system controls.

Notifications display alerts from supported applications and system features. They may include reminders, messages, and application updates.

Quick Settings in Windows 11 provides convenient access to supported controls, such as Wi-Fi, Bluetooth, volume, and brightness on compatible devices.

Users can open Quick Settings by selecting the relevant area of the taskbar, commonly the network, sound, and battery area. Keyboard shortcuts and controls may differ between versions.
Notifications can often be managed through Settings, allowing users to choose which applications may send alerts.

Topic 133: Refreshing the Windows Desktop

Refreshing the desktop updates its displayed contents and can help Windows show changes that have not appeared immediately.
To refresh the desktop:

1. Right-click an empty area of the desktop.
2. Select Refresh.

On supported Windows configurations, pressing F5 while the desktop is active may also refresh it.
Refreshing does not normally increase the computer's processing speed or repair system problems. It simply requests an update of the displayed desktop contents.
If the desktop repeatedly fails to display changes, the underlying issue may require separate troubleshooting.

Topic 134: Understanding the Desktop Context Menu

A context menu displays commands related to the item or area selected by the user. On the desktop, it provides access to several organization and personalization options.

To open the desktop context menu:

1. Right-click an empty area of the desktop.
2. Review the available commands.
3. Select the required option.

Depending on the Windows version, the menu may include View, Sort by, Refresh, New, Display settings, and Personalize.
Right-clicking a file or application shortcut displays a different context menu because the available commands depend on the selected item.
In Windows 11, some commands may appear under Show more options.

Topic 135: Windows Accessibility Features

Accessibility features help people use computers according to their individual needs and preferences. They may assist users with visual, hearing, mobility, or other access requirements.
Common Windows accessibility features include:

- **Magnifier:** Enlarges portions of the screen.
- **Narrator:** Reads supported on-screen information aloud.
- **High Contrast and Contrast Themes:** Improve the visibility of interface elements for some users.
- **Text Size:** Allows supported interface text to be enlarged.
- **On-Screen Keyboard:** Provides a visual keyboard that can be operated using a mouse or another supported input method.
- **Voice Access or related voice-control features:** Allow supported computer operations through spoken commands.

To explore these options in Windows 11, open Settings and select Accessibility.
The best settings depend on the user's needs. Some features may differ by Windows version or language.

CHAPTER 10: WINDOWS START MENU AND SEARCH

Topic 136: Introduction to the Windows Start Menu

The Start menu is a central Windows interface for accessing applications, searching for content, opening certain system locations, and reaching power options.
In Windows 11, the Start menu commonly includes pinned applications, access to the full application list, and recommended or recently accessed content, depending on configuration.
Users can open the Start menu by selecting the Start button on the taskbar or pressing the Windows key on the keyboard.
Its appearance and available features vary between Windows versions.

Topic 137: Understanding the Start Menu Components

The Start menu contains several areas that help users find and open applications.
In Windows 11, common elements include:

- **Search:** Helps locate applications, files, settings, and other supported content.
- **Pinned applications:** Provides quick access to selected applications.
- **All apps:** Displays an alphabetical list of installed applications.

- **Recommended section:** May display recently opened files or suggested content.
- **User account controls:** Provides access to certain account-related options.
- **Power options:** Provides supported options for restarting, shutting down, or managing the computer's power state.

The exact layout depends on the Windows version, updates, account settings, and device configuration.

Topic 138: Opening and Closing the Start Menu

The Start menu can be opened using either the mouse or keyboard.

To open the Start menu:

1. Click the Start button on the taskbar.
2. Alternatively, press the Windows key.

To close the Start menu:

1. Press the Esc key.
2. Click outside the Start menu.
3. Alternatively, select an application or another interface element.

The Windows key is useful for opening the Start menu quickly without moving the mouse.

Topic 139: Using Pinned Applications and All Apps

Pinned applications provide convenient access to frequently used programs. The All apps section provides a broader list of installed applications.

To open a pinned application:

1. Open the Start menu.
2. Locate the application.
3. Select its icon.

To find an application through All apps:

1. Open the Start menu.
2. Select All apps.
3. Browse the application list.
4. Select the required application.

Depending on the Windows version and application, users may also be able to pin or unpin applications through a context menu.

Pinning an application provides easier access but does not install the application or create a separate copy of it.

Topic 140: Searching for Applications and Files in Windows

Windows Search helps users locate applications, files, settings, and supported online information. It can save time when users do not remember where an item is stored.

To search for an application:

1. Open the Start menu or select the Search control.
2. Type the application name.
3. Review the results.
4. Select the required application.

To search for a file:

1. Open Windows Search or File Explorer.
2. Enter a distinctive part of the file name.
3. Review the available results.
4. Open the correct file.

Search results depend on indexing, permissions, search settings, and the location of the content. Some files may take longer to appear or may not be included in the current search scope.

Using specific words usually makes searches more effective. For example, searching for computer assignment may be more useful than searching for file when looking for a particular assignment.

CHAPTER 11: WINDOWS START MENU AND SEARCH

Topic 141: Searching for Windows Settings

Windows Search allows users to find system settings without navigating through multiple menus. It is useful when a user needs to change display options, manage accounts, connect to a network, or adjust other computer settings.

Steps to search for a setting:

1. Open the Start menu or select the Search control.
2. Type the name of the setting, such as Display or Sound.
3. Review the search results.
4. Select the relevant result to open the setting.

Search results may vary depending on the Windows version, installed features, and search configuration. If the required setting does not appear, it can usually be located manually through the Settings application.

Topic 142: Using Search Filters

Search filters help users narrow down results to a particular category, such as applications, documents, settings, or web results. The available filters depend on the Windows version and search interface.

For example, when looking for a document, selecting a file-related category may reduce unrelated results.

How to use search filters:

1. Open Windows Search.
2. Enter a search term.
3. Select an available category or filter.
4. Review the results.
5. Open the item that matches your needs.

Using a specific search term and the appropriate filter can make it easier to locate information.

Topic 143: Finding Recently Opened Files

Windows and supported applications may provide access to recently opened files. This feature helps users return to documents they have worked on without searching for them again.

Recent files may appear in File Explorer's Home area, application menus, or the Start menu's Recommended section, depending on the Windows version and configuration.

To locate a recent file:

1. Open File Explorer.
2. Select Home, if available.
3. Review the recent files section.
4. Select the required file.

Recent-file lists do not necessarily contain every file opened on the computer. Their contents depend on application support, privacy settings, and Windows configuration.

Topic 144: Pinning Applications to the Start Menu

Pinning an application places it in the Start menu's pinned section, providing convenient access.

Steps to pin an application in Windows 11:

1. Open the Start menu.
2. Select All apps.
3. Find the application.
4. Right-click its entry.
5. Choose Pin to Start if the option is available.

To remove a pinned application, right-click it and choose Unpin from Start.

Unpinning an application removes its shortcut from the pinned section. It does not normally uninstall the application.

Topic 145: Organizing Pinned Applications

Organizing pinned applications helps users find frequently used programs quickly. Windows 11 allows users to rearrange supported pinned items within the Start menu.

To organize pinned applications:

1. Open the Start menu.
2. Locate a pinned application.
3. Drag it to a preferred position if supported.
4. Repeat the process for other applications.

Users can place frequently used applications near the beginning of the pinned section for easier access.

The available organization options may vary by Windows version. A well-organized Start menu reduces the time required to locate common applications.

Topic 146: Managing Recommended Items

The Recommended section in Windows 11 may display recently opened files, recently installed applications, or other suggested items. It is designed to help users return to content they have used recently.

The items shown depend on Windows settings and the user's activity.

Users can manage certain Start menu recommendations through **Settings > Personalization > Start**.

Available controls may differ between Windows releases.

Changing recommendation settings can affect which items appear, but it does not necessarily delete the original files.

Topic 147: Searching from the Taskbar

Windows Search can be accessed from the taskbar when its Search control is enabled. Depending on the configuration, the control may appear as a search box, a search icon, or another supported presentation.

To search from the taskbar:

1. Select the Search control.
2. Type the name of an application, file, or setting.
3. Review the results.
4. Select the required item.

If the Search control is not visible, users may be able to change its appearance through the taskbar settings.

Taskbar search is particularly useful when users want to locate an item without opening File Explorer or browsing the Start menu manually.

Topic 148: Searching for Files by Name

Searching by file name is one of the simplest ways to locate a saved document, image, spreadsheet, or other file.

Steps:

1. Open File Explorer.
2. Navigate to the folder you want to search, or select a suitable starting location.
3. Select the search box.
4. Type all or part of the file name.
5. Review the results.

For example, entering computer_notes may help locate a document with a similar name.

The search scope matters. Searching within one folder may produce different results from searching across a larger location. Some files may also be unavailable because of permissions, indexing settings, or storage locations.

Topic 149: Understanding Windows Search Indexing

Windows Search indexing is a process that records information about selected files and their properties so that searches can return results more quickly.

An index may include file names, locations, and other supported information. Depending on the indexing configuration, Windows may also index file contents for certain file types and locations.

Indexing can improve search performance, particularly when working with many files. However, not every location or file is necessarily included.

Users can review indexing options through the Windows search-related settings available on their system.

Changing the indexed locations may affect which results appear and how quickly they are returned.

Topic 150: Search Privacy and Online Results

Windows Search may provide local results and, depending on configuration, online suggestions or web results. Search experiences can vary according to the Windows version, region, account, and privacy settings.

Users should understand the difference between searching their own files and searching for information online.

Privacy practices include:

- Reviewing Windows privacy and search settings.
- Avoiding unnecessary sharing of sensitive information.
- Checking whether an application or service requires account sign-in.
- Using trusted sources when opening online results.
- Keeping Windows and security software updated.

Search features should be configured according to the user's needs and privacy preferences.

CHAPTER 11: WINDOWS TASKBAR AND SYSTEM TRAY

Topic 151: Introduction to the Windows Taskbar

The taskbar is a Windows interface element that provides access to applications, open windows, and selected system controls. It is usually located along the bottom of the screen, although its appearance and configuration can vary.

Common taskbar elements include the Start button, Search control, pinned applications, indicators for open applications, and the system tray area.

The taskbar helps users switch between programs and access frequently used features without returning to the desktop each time.

Topic 152: Understanding Taskbar Components

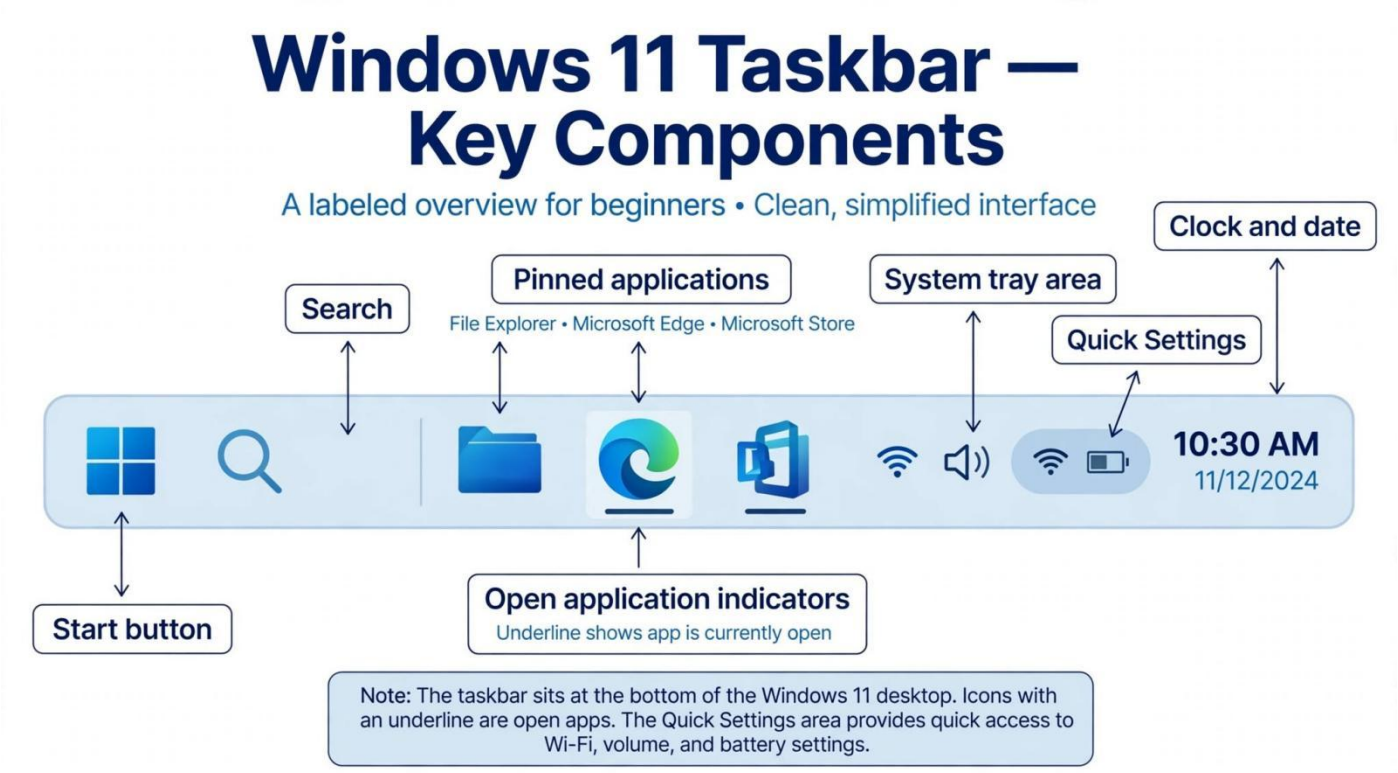
The Windows taskbar contains several components that support everyday computer use.

Common components include:

- **Start button:** Opens the Start menu.
- **Search control:** Helps locate applications, files, and settings.
- **Pinned applications:** Provides quick access to selected programs.
- **Open application indicators:** Helps identify applications that are running or have open windows.
- **System tray area:** Displays selected status icons and background application indicators.
- **Clock and date:** Displays the current system time and date.
- **Notification and system controls:** Provides access to supported notifications and controls, depending on the Windows version.

The exact arrangement and appearance of these components may differ between Windows releases.

Figure 11.1: Main Components of the Windows Taskbar



Topic 153: Pinning Applications to the Taskbar

Pinning an application to the taskbar creates a convenient way to access it. A pinned application can remain on the taskbar even when it is not currently running.

To pin an application in Windows 11:

1. Open the Start menu.
2. Find the application.
3. Right-click the application.
4. Select Pin to taskbar if the option is available.

Some applications may also provide a pinning option through their taskbar icon when they are running.

To remove a pinned application, right-click its taskbar icon and select Unpin from taskbar.
Unpinning an application does not normally uninstall it.

Topic 154: Opening and Switching Between Applications

The taskbar makes it easier to move between open applications.

To open an application:

1. Select its pinned taskbar icon, or open it through the Start menu.
2. Wait for the application window to appear.

To switch between applications:

1. Select the taskbar icon for the application you want to use.
2. If several windows belong to the same application, choose the required window when prompted.

Users can also press **Alt + Tab** to switch between open windows.
Switching applications allows users to work with multiple programs without closing them each time.

Topic 155: Understanding the System Tray

The system tray is the taskbar area that displays selected system status icons and background application indicators. In Windows 11, some icons may appear in a hidden-icons menu rather than remaining visible all the time.
Depending on the device and configuration, the area may display network status, sound controls, battery status, security indicators, and other supported icons.
The system tray helps users check important system information and access certain controls.
Users should not assume that every application running in the background will display an icon in this area.

Topic 156: Using Quick Settings

Quick Settings in Windows 11 provides convenient access to supported system controls.
Depending on the device, these controls may include Wi-Fi, Bluetooth, Airplane mode, accessibility options, and display-related features. The available controls vary according to hardware and Windows configuration.

To open Quick Settings:

1. Select the network, sound, or battery area of the taskbar.
2. Review the available controls.
3. Select the feature you want to manage.
4. Close the panel when finished.

On many Windows 11 devices, **Windows + A** opens Quick Settings.
Quick Settings provides a faster way to access supported controls than navigating through the full Settings application.

Topic 157: Checking Wi-Fi and Network Status

The taskbar's network control helps users check their connection status and access available network options.

To check Wi-Fi status:

1. Open Quick Settings from the taskbar.
2. Locate the Wi-Fi control.
3. Check whether Wi-Fi is enabled.
4. If necessary, open the available network list.
5. Select the appropriate network and connect using the required credentials.

A connected Wi-Fi symbol does not always guarantee internet access. A device may connect to a router while the internet connection itself is unavailable.
If a connection fails, users should check the network name, password, router status, and relevant network settings.

Topic 158: Adjusting Sound Volume from the Taskbar

Windows allows users to adjust audio volume through the taskbar's sound controls.

Steps:

1. Select the sound icon or open Quick Settings.
2. Locate the volume slider.
3. Move the slider to increase or decrease the volume.
4. If necessary, choose the appropriate output device.

Some devices provide separate volume controls for applications or additional audio settings.
If sound is not working, check whether the correct output device is selected, whether the application is muted, and whether the speakers or headphones are connected properly.
The available controls depend on the device and installed audio hardware.

Topic 159: Checking Battery Status on a Laptop

Laptop users can use the taskbar's battery indicator to check the current battery level and, when available, charging status.

Depending on the Windows version and device, selecting the battery or power area may show additional information and power controls.

To check battery status:

1. Locate the battery indicator in the taskbar.
2. Review the displayed charge level or charging information.
3. Connect the charger if the battery needs charging.
4. Review the available power settings if necessary.

Battery information may not appear on desktop computers that do not have a supported battery.

Actual battery life depends on battery condition, screen brightness, running applications, hardware, and power settings.

Topic 160: Managing the Clock, Date, and Calendar

Windows displays the system time and date in the taskbar. Selecting the clock area may open a calendar or related date-and-time interface, depending on the Windows version.

To review date and time settings:

1. Open Settings.
2. Select Time & language.
3. Open Date & time.
4. Review the time zone and automatic date-and-time options.
5. Make changes when necessary and permitted.

Incorrect date or time settings can cause problems with scheduled tasks, file timestamps, and some secure online connections.

Automatic time synchronization can help keep the system clock accurate when the device has a suitable connection and the feature is enabled.

CHAPTER 11: WINDOWS TASKBAR AND SYSTEM TRAY

Topic 161: Customizing Taskbar Settings

Taskbar customization allows users to adjust the appearance and behavior of the Windows taskbar. Depending on the Windows version, users may be able to change alignment, manage taskbar items, control system tray icons, and adjust other available options.

Steps to open taskbar settings in Windows 11:

1. Right-click an empty area of the taskbar.
2. Select Taskbar settings.
3. Review the available options.
4. Change the required settings.
5. Close Settings when finished.

Taskbar settings help users create a workspace that suits their preferences. Available options may differ between Windows versions and editions.

Topic 162: Changing Taskbar Alignment

Taskbar alignment determines where the taskbar icons appear along the taskbar. In Windows 11, the Start button and application icons are centered by default on many installations, but supported configurations may allow left alignment.

To change taskbar alignment in Windows 11:

1. Open Settings.
2. Select Personalization.
3. Choose Taskbar.
4. Expand Taskbar behaviors.
5. Locate Taskbar alignment.
6. Select the available alignment option you prefer.

Changing alignment affects the position of taskbar icons, not the location of files or applications on the computer.

The available choices depend on the Windows version.

Topic 163: Showing or Hiding Taskbar Items

Windows may allow users to show or hide certain taskbar items, such as Search, Task View, Widgets, or other supported controls.

To manage taskbar items in Windows 11:

1. Open Settings.
2. Select Personalization.
3. Choose Taskbar.
4. Locate the Taskbar items section.
5. Turn the available controls on or off as required.

Hiding a taskbar item usually removes its visible shortcut from the taskbar. It does not necessarily uninstall or disable the associated feature.

Users can often access the same feature through another method, such as a keyboard shortcut or the Start menu.

Topic 164: Understanding Taskbar Overflow

Taskbar overflow helps manage pinned applications when there is not enough room to display every icon in the available taskbar space.

Depending on the Windows version and configuration, additional icons may appear in an overflow area or menu.

Overflow is particularly useful on smaller displays or when many applications are pinned to the taskbar. Users can reduce clutter by unpinning applications they rarely use and keeping frequently used applications easily accessible.

The exact behavior of taskbar overflow differs between Windows releases.

Topic 165: Managing Notifications in Windows

Notifications are messages displayed by Windows or supported applications. They may inform users about reminders, messages, updates, security events, and other activities.

Users can control which applications are allowed to send notifications.

To manage notifications in Windows 11:

1. Open Settings.
2. Select System.
3. Choose Notifications.
4. Review the available notification settings.
5. Enable or disable notifications for selected applications.

Some applications have additional notification settings within their own interfaces.
Managing notifications can reduce distractions while ensuring that important alerts remain visible.

Topic 166: Using Do Not Disturb and Focus Sessions

Windows provides features that can help reduce interruptions during work or study. Their names and availability may vary by Windows version.
Do Not Disturb can silence or limit certain notifications according to the user's settings.
Focus sessions in supported Windows versions provide tools for concentrating on a task for a selected period. Depending on the version, the feature may integrate with clock, timer, or productivity functions.
To explore these features, open Settings and look under System for Notifications or Focus-related options.
These tools can help users concentrate, but their behavior depends on the settings configured on the computer.

Topic 167: Managing Background Applications

Background applications perform tasks even when their windows are not currently visible. Examples may include applications that synchronize files, check for updates, or provide notifications.
Background activity can be useful, but unnecessary activity may consume memory, processing power, network bandwidth, or battery energy.
Users should manage background activity carefully.

General management practices:

1. Review the applications installed on the computer.
2. Open an application's own settings to review its background behavior.
3. Check Windows power and privacy settings where relevant.
4. Disable unnecessary background permissions only when the option is available and appropriate.
5. Avoid disabling essential security or system services without understanding their purpose.

Background controls differ between applications and Windows versions.

Topic 168: Using Task Manager to Manage Applications

Task Manager is a Windows utility that displays information about running applications, processes, and system resource usage.
It can help users identify an unresponsive application and review whether the computer is using a large amount of CPU, memory, disk, or network resources.

To open Task Manager:

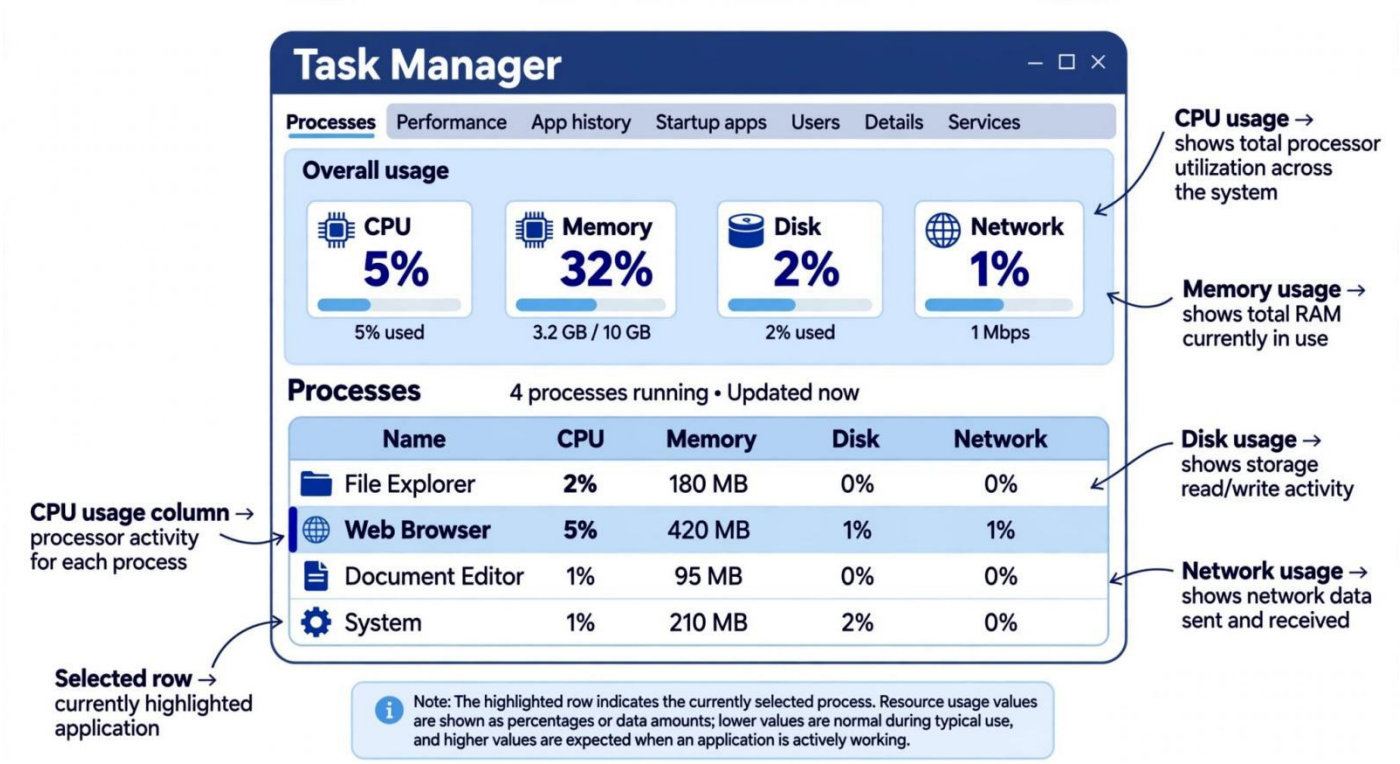
- Press Ctrl + Shift + Esc.
- Alternatively, right-click the Start button and select Task Manager if the option is available.

To close an unresponsive application:

1. Open Task Manager.
2. Locate the application in the processes list.
3. Select the application.
4. Choose End task when necessary.

Ending a task may cause unsaved work to be lost. Users should try to save their work before closing an application whenever possible.

Figure 11.2: Task Manager and System Resource Usage



Topic 169: Understanding Startup Applications

Startup applications are programs configured to launch automatically when a user signs in to Windows. Examples may include communication applications, cloud synchronization tools, and certain utility programs.

Too many unnecessary startup applications can increase the time required for a computer to become ready for use.

To review startup applications:

1. Open Task Manager.
2. Select Startup apps, if available.
3. Review the listed applications.
4. Check their startup impact information where available.
5. Disable an application only if you understand its purpose and do not need it to start automatically.

Disabling a startup entry normally prevents automatic launching at sign-in; it does not uninstall the application.

Avoid disabling essential security software or important device utilities without checking their purpose.

Topic 170: Restarting Windows Explorer

Windows Explorer, also known as File Explorer's underlying shell process in common desktop troubleshooting contexts, supports important parts of the Windows interface, including the taskbar and desktop.

If the taskbar or desktop becomes temporarily unresponsive, restarting Windows Explorer through Task Manager may help restore the interface.

Steps:

1. Press Ctrl + Shift + Esc to open Task Manager.
2. Find Windows Explorer in the processes list.
3. Select it.
4. Choose Restart if that option is available.

The desktop or taskbar may disappear briefly and then return.

Restarting Windows Explorer is not the same as restarting the entire computer. It may help resolve some temporary interface problems, but it will not fix every Windows issue.

CHAPTER 12: WORKING WITH WINDOWS AND APPLICATIONS

Topic 171: Understanding Application Windows

An application window is a visual area in which a program displays its interface and allows users to interact with its features.
For example, a web browser displays websites, while a word processor provides tools for creating documents.
Many Windows applications have common interface elements, such as a title bar, menus or commands, a working area, and window controls. However, the exact design depends on the application.
Understanding application windows helps users navigate programs and manage several tasks efficiently.

Topic 172: Maximizing, Minimizing, and Restoring Windows

Windows provides controls for changing the size and visibility of application windows.

- **Maximize:** Expands a window to fill the available screen area.
- **Minimize:** Hides a window from the desktop while normally keeping the application running.
- **Restore:** Returns a maximized window to its previous windowed size.

To use window controls:

1. Open an application.
2. Locate the controls near the upper-right corner of the window.
3. Select Minimize, Maximize, or Restore as required.

A minimized application can usually be reopened by selecting its taskbar icon.
These controls help users organize their workspace and move between applications.

Topic 173: Moving and Resizing Windows

Moving and resizing windows allows users to arrange applications according to their needs.

To move a window:

1. Place the pointer over an appropriate area of the title bar.
2. Hold the mouse button.
3. Drag the window to the desired position.
4. Release the mouse button.

To resize a window:

1. Make sure the window is not maximized.
2. Move the pointer to an edge or corner.
3. When the resizing pointer appears, drag the edge or corner.
4. Release the mouse button when the window reaches the desired size.

Some applications restrict resizing or use fixed layouts. Window behavior may also depend on display scaling and Snap settings.

Topic 174: Switching Between Open Windows

Users often need to move between multiple open applications. Windows provides several methods for switching between them.

Method 1: Taskbar

Select the taskbar icon of the application you want to use.

Method 2: Keyboard shortcut

Press Alt + Tab to move between open windows. Continue holding Alt while selecting a different window, then release the keys.

Method 3: Task View

Press Windows + Tab to view open windows and available virtual desktops.
Switching between windows allows users to compare information, copy content between applications, and perform related tasks without repeatedly closing and reopening programs.

Topic 175: Using Snap Layouts to Arrange Applications

Snap Layouts help users organize multiple windows on the screen. Supported layouts may place two, three, or four windows in different arrangements, depending on the display size and Windows version.

To use Snap Layouts in Windows 11:

1. Open an application.
2. Move the pointer over the Maximize button, or press Windows + Z.
3. Review the available layouts.
4. Select a layout.
5. Choose positions for the other open applications if prompted.

Snap Layouts are useful for tasks such as comparing documents, following online lessons while taking notes, or working with a browser and spreadsheet together.

The available layouts depend on screen size, window dimensions, and Windows configuration.

Topic 176: Using Multiple Applications Together

Multitasking means performing or managing more than one task within the same period. On a computer, this often involves switching between or viewing multiple applications.

For example, a student may keep an online lecture open while writing notes in a word processor.

Practical multitasking tips:

- Keep related applications open together.
- Use Snap features to organize windows.
- Save work regularly.
- Close applications that are no longer needed.
- Avoid opening unnecessary programs when system resources are limited.

Multitasking can improve productivity, but too many open applications may make work confusing or affect performance on computers with limited resources.

Topic 177: Understanding the Clipboard

The clipboard is a temporary storage area used by Windows and applications to hold content that has been copied or cut.

It can store items such as text, images, and other supported content.

Common clipboard commands include:

- **Copy (Ctrl + C):** Copies selected content without removing the original.
- **Cut (Ctrl + X):** Places selected content on the clipboard and normally removes it from its original location when the action is completed.
- **Paste (Ctrl + V):** Inserts clipboard content into a supported destination.

Clipboard behavior depends on the application and type of content.

Users should be careful when copying sensitive information because clipboard history or other applications may make copied content accessible beyond the immediate task.

Topic 178: Using Clipboard History

Clipboard history allows users to access multiple recently copied items instead of relying only on the most recent clipboard item.

In supported Windows versions, users can usually open clipboard history by pressing Windows + V.

If clipboard history is not enabled, Windows may display an option to turn it on.

To use clipboard history:

1. Copy text or another supported item.
2. Press Windows + V.
3. Review the available items.
4. Select the item you want to paste.

Available features depend on Windows configuration and the types of items copied. Clipboard history should be used carefully on shared computers because it may contain sensitive information.

Topic 179: Using Undo and Redo

Undo and Redo commands allow users to reverse or reapply certain actions in supported applications.

Undo reverses a recent action, such as typing text or changing formatting.

Redo reapplies an action that was undone.

Common keyboard shortcuts include:

- Ctrl + Z for Undo.
- Ctrl + Y for Redo in many Windows applications.

Some applications use different shortcuts or provide additional history controls.

Undo and Redo are useful when editing documents, images, spreadsheets, and other files. However, they do not reverse every operation, and their behavior depends on the application.

Users should save important work regularly rather than relying only on Undo.

Topic 180: Saving and Switching Between Application Tasks

Saving work is an essential part of using computer applications. It helps preserve changes and reduces the risk of losing information if a program closes unexpectedly.

Many applications use Ctrl + S as the Save shortcut.

General saving procedure:

1. Open or create a document or project.
2. Select Save or press Ctrl + S.

3. If prompted, choose a folder and enter a file name.
4. Select the appropriate file format if required.
5. Confirm the save operation.

When switching between applications, users should save important work before moving to another task, especially when working with large documents or unstable software.

The exact saving process depends on the application. Some cloud-based applications save changes automatically, but users should still confirm that their work has been saved successfully.

CHAPTER 12: WORKING WITH WINDOWS AND APPLICATIONS

Topic 181: Understanding Application Menus and Toolbars

Application menus and toolbars provide access to the commands and features of a program. Their appearance varies according to the application.

A menu organizes commands into categories, while a toolbar provides convenient access to frequently used functions through buttons or icons.

For example, a word processor may provide commands for opening files, formatting text, inserting images, and printing documents.

To use an application menu:

1. Open the application.
2. Locate the menu or command area.
3. Select the required menu or button.
4. Choose the appropriate command.
5. Follow any additional instructions displayed by the application.

Understanding menus and toolbars helps users locate features and work more efficiently.

Topic 182: Understanding Dialog Boxes

A dialog box is a small window that allows users to enter information, choose options, or confirm an action.

For example, a Save As dialog box allows a user to choose a folder, enter a file name, and select a file format.

Common dialog box controls include text fields, checkboxes, radio buttons, drop-down lists, and command buttons.

To use a dialog box:

1. Read the information displayed.
2. Enter the required details or select the appropriate options.
3. Review your selections.
4. Select OK, Apply, Save, Cancel, or another relevant button.

The available controls depend on the purpose of the dialog box. Users should review important options before confirming changes.

Topic 183: Understanding File Associations

A file association connects a file type with an application that can open it. Windows often uses file extensions to help identify file types.

For example, .txt commonly identifies plain-text files, while .pdf identifies Portable Document Format files.

When a user opens an associated file, Windows usually launches the selected default application.

Users may be able to change default applications through **Settings > Apps > Default apps** in supported Windows versions.

Changing a file association does not normally change the file's contents. It changes which application Windows uses to open that type of file by default.

Topic 184: Choosing Default Applications

Default applications are programs Windows uses automatically for particular activities or file types.

For example, a user may choose a preferred web browser for opening web links or a preferred application for viewing photographs.

To manage default applications in Windows 11:

1. Open Settings.
2. Select Apps.
3. Choose Default apps.
4. Search for the application or file type.
5. Select the available default option.
6. Confirm the change if prompted.

The available choices depend on the applications installed on the computer.

Choosing suitable default applications can make everyday tasks more convenient and help users work with their preferred tools.

Topic 185: Installing Applications Safely

Installing an application adds software to a computer so that it can be used. Applications may be obtained through official application stores, software developers' websites, or approved organizational sources.

General installation procedure:

1. Identify the application you need.
2. Visit its official source or a trusted distribution platform.
3. Check system requirements and compatibility.
4. Download the installer if required.
5. Open the installer and review the instructions.
6. Accept only the permissions and options you understand.
7. Complete the installation.
8. Open the application to confirm that it works.

Avoid installers from suspicious websites or unexpected messages. Check the publisher and review optional offers carefully.

Installation steps differ between traditional desktop programs and applications distributed through a store.

Topic 186: Uninstalling Unnecessary Applications

Uninstalling removes an application from the computer through its supported removal process. It can help free storage space and reduce clutter.

To uninstall an application in Windows 11:

1. Open Settings.
2. Select Apps.
3. Choose Installed apps.
4. Find the application.
5. Select its menu or available options.
6. Choose Uninstall and follow the instructions.

Some applications provide their own uninstallers or require administrative permission.

Users should avoid removing unfamiliar system components or device drivers without understanding their purpose. Removing essential software may affect system functionality.

Uninstalling an application may not automatically delete every personal file or setting associated with it.

Topic 187: Updating Applications

Software updates may provide security fixes, bug corrections, compatibility improvements, and new features. Applications may update automatically, through an application store, or through the developer's update mechanism.

General update procedure:

1. Open the application or its official update source.
2. Locate the update option.
3. Check the available update information.
4. Install the update using the supported process.
5. Restart the application or computer if required.

Users should obtain updates from trusted sources. Updates to important software should be handled carefully, especially on computers used for work or school.

An update does not guarantee that every problem will be resolved, but keeping supported software current can reduce security risks.

Topic 188: Understanding Application Compatibility

Application compatibility refers to whether software can operate correctly with a particular operating system, hardware configuration, or set of dependencies.

An application may require a specific Windows version, processor architecture, amount of memory, graphics capability, or additional runtime component.

Before installing an application, check:

- Supported operating system and version.
- Minimum and recommended hardware requirements.
- Available storage space.
- Required permissions.
- Any necessary supporting software.

Older applications may sometimes run using compatibility settings, but this does not guarantee correct operation.

Users should consult the software developer's documentation when an application does not work as expected.

Topic 189: Managing Application Permissions

Application permissions determine which resources or features an application can access. Depending on the application and Windows configuration, these may include the camera, microphone, location, contacts, or files.

Permissions help users control how applications interact with the computer and personal information.

To review privacy permissions in Windows 11:

1. Open Settings.
2. Select Privacy & security.
3. Choose the relevant permission category.
4. Review the available controls.
5. Allow access only when appropriate for the application's purpose.

Some desktop applications manage permissions through their own settings rather than through every Windows privacy control.

Users should consider why an application needs a permission before granting it.

Topic 190: Troubleshooting Unresponsive Applications

An application may become unresponsive because of a software error, insufficient resources, a problem with a file, or another system issue.

When this happens, users should allow the application a short time to respond before taking further action.

Basic troubleshooting steps:

1. Wait briefly to see whether the application recovers.
2. Avoid repeatedly clicking buttons while it is unresponsive.
3. If possible, save important work when the application becomes responsive again.
4. If necessary, use Task Manager to close the unresponsive application.
5. Reopen the application and check whether the problem continues.
6. Install appropriate updates or consult the developer's support information if needed.

Force-closing an application may cause unsaved changes to be lost. If the problem occurs repeatedly, further troubleshooting may be necessary.

CHAPTER 13: MOUSE, TOUCHPAD AND KEYBOARD SKILLS

Topic 191: Introduction to Keyboard Skills

The keyboard is an input device used to enter text, numbers, symbols, and commands into a computer. Developing keyboard skills helps users work more accurately and efficiently. These skills are useful for writing documents, entering data, programming, searching the internet, and controlling applications.

A standard computer keyboard contains letter keys, number keys, function keys, modifier keys, navigation keys, and other special-purpose keys.

Keyboard layouts and key arrangements may differ by language, device, and region.

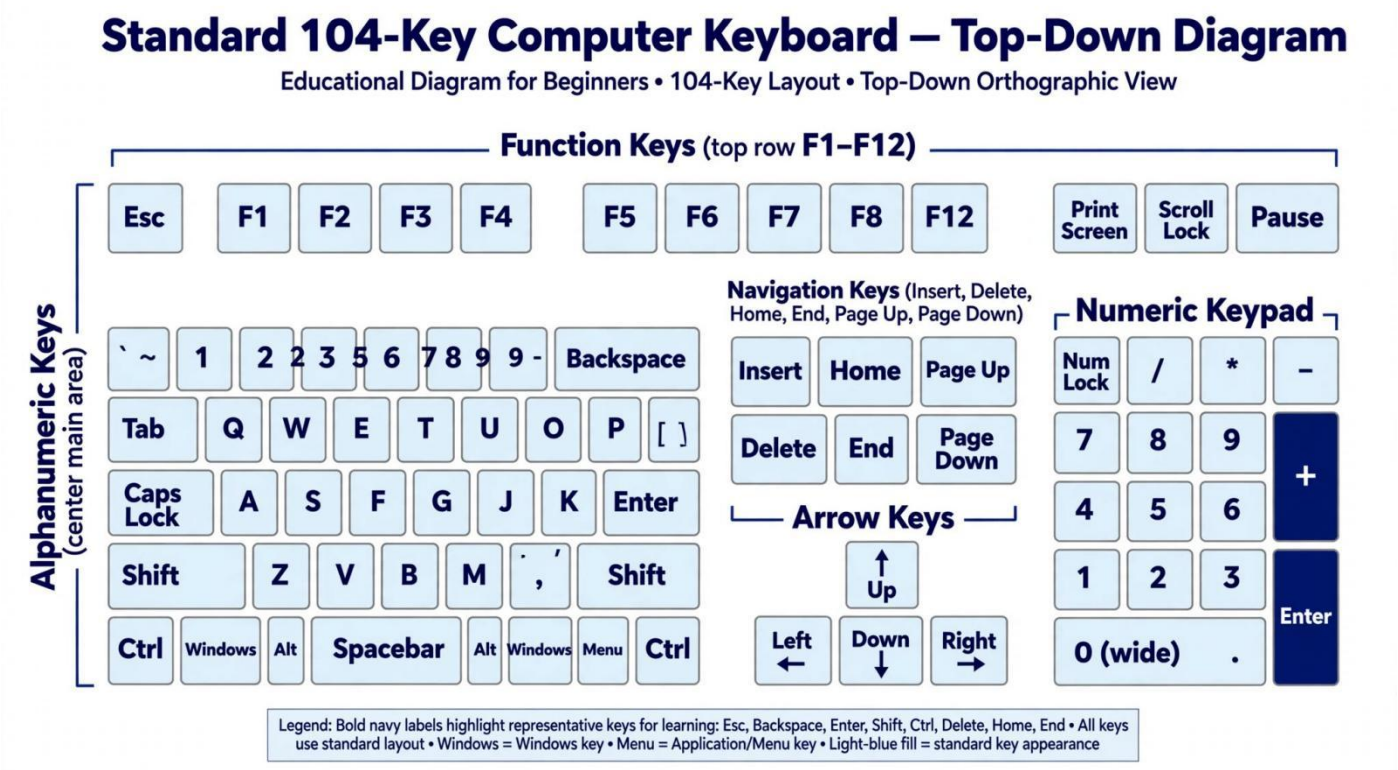
Topic 192: Understanding Keyboard Sections

A typical full-size computer keyboard contains several groups of keys, each designed for particular tasks.

- **Alphanumeric keys:** Enter letters, numbers, and common symbols.
- **Function keys:** Provide application- or system-related functions.
- **Modifier keys:** Such as Ctrl, Alt, and Shift, modify the action of another key.
- **Navigation keys:** Include keys such as Home, End, Page Up, and Page Down.
- **Arrow keys:** Move the cursor or selection in supported interfaces.
- **Numeric keypad:** Provides a convenient way to enter numbers on keyboards that include one.
- **Special-purpose keys:** Include Enter, Esc, Backspace, and Delete.

Not every keyboard includes all these sections. Laptop keyboards often combine functions or provide a smaller layout.

Figure 13.1: Main Sections of a Computer Keyboard



Topic 193: Correct Finger Placement on a Keyboard

Correct finger placement can help users develop consistent typing habits. In touch typing, users learn to type without looking at the keyboard continuously.

On a conventional QWERTY keyboard, the home row for the left hand is commonly A, S, D, and F, while the right-hand home row is J, K, L, and the semicolon key. The thumbs are generally used for the Spacebar. The raised markers on F and J help users locate the home row by touch on many keyboards.

Beginners should start slowly, focus on accuracy, and gradually build typing speed through regular practice. Hand comfort and a relaxed posture are more important than forcing the fingers into an uncomfortable position.

Topic 194: Using Common Keyboard Keys

Several keyboard keys perform essential functions across many applications.

- **Enter:** Confirms an action or creates a new line in many text editors.
- **Spacebar:** Inserts a space between words.
- **Backspace:** Removes the character before the insertion point in many text fields.
- **Delete:** Removes selected content or the character after the insertion point, depending on the application.
- **Esc:** Cancels or closes certain menus and operations.
- **Tab:** Moves focus between controls or inserts a tab character in supported applications.
- **Shift:** Produces uppercase letters or alternate symbols when used with other keys.

The effect of a key depends on the active application and the current selection.
Learning these keys helps users enter information and navigate programs more effectively.

Topic 195: Understanding Modifier Keys

Modifier keys change how other keys behave when pressed together. Common modifier keys include Ctrl, Alt, Shift, and the Windows key.
For example, Ctrl + C commonly copies selected content, while Ctrl + V pastes clipboard content.
Modifier keys are useful for shortcuts, navigation, text editing, and system commands.
Some keyboards include additional modifier keys, and shortcuts may differ between applications or operating systems.
Users should learn shortcuts gradually and practice them in appropriate applications.

Topic 196: Essential Keyboard Shortcuts

Keyboard shortcuts allow users to perform common tasks without selecting every command with a mouse.
Frequently used Windows shortcuts include:

- **Ctrl + C:** Copy selected content.
- **Ctrl + X:** Cut selected content.
- **Ctrl + V:** Paste clipboard content.
- **Ctrl + Z:** Undo an action in many applications.
- **Ctrl + S:** Save in many applications.
- **Alt + Tab:** Switch between open windows.
- **Windows + E:** Open File Explorer.
- **Windows + L:** Lock the computer.
- **Windows + D:** Show or hide the desktop by minimizing or restoring windows as supported.
- **Windows + V:** Open clipboard history when the feature is supported and enabled.

Shortcuts may behave differently depending on the active application, keyboard layout, and Windows version.

Topic 197: Using Function Keys

Function keys are commonly located along the top row of a keyboard and are labeled F1 through F12. Their functions depend on the application and system configuration.
For example, F1 often opens Help in supported applications, while F5 commonly refreshes content in certain interfaces.
On many laptops, function keys may also control brightness, volume, wireless features, or other hardware functions. The Fn key may be required to access one set of functions instead of another.
Users should consult the device documentation to understand their keyboard's function-key behavior.

Topic 198: Using the Numeric Keypad

The numeric keypad is a group of keys found on many full-size keyboards, usually on the right side. It provides convenient number entry and mathematical operators.
The Num Lock key controls numeric keypad behavior on many keyboards. When numeric mode is enabled, the keypad typically enters numbers. When it is disabled, some keys may perform navigation functions.
Laptop keyboards may not include a separate numeric keypad.
The numeric keypad is particularly useful for spreadsheets, accounting tasks, data entry, and other work involving frequent numerical input.

Topic 199: Switching Keyboard Languages and Layouts

A computer may support multiple keyboard languages and layouts. These settings determine how physical key presses are interpreted when entering text.
For example, a user may switch between an English keyboard layout and another language layout when writing in different languages.

In supported Windows versions, users can add keyboard layouts through **Settings > Time & language > Language & region**.
After adding a layout, users may be able to switch between installed layouts using the language indicator or a supported keyboard shortcut.
The selected layout should match the user's typing needs. A keyboard layout changes text input behavior; it does not necessarily change the physical labels printed on the keyboard.

Topic 200: Improving Typing Accuracy and Speed

Typing accuracy means entering the intended characters correctly, while typing speed measures how quickly text is entered. Both skills are useful for schoolwork, office tasks, programming, and digital communication.
Beginners should prioritize accuracy before attempting to increase speed.

Practical tips for improving typing:

1. Sit comfortably and maintain a relaxed posture.
2. Keep the keyboard at a comfortable height.
3. Practice using the home-row keys.
4. Look at the screen to notice mistakes.
5. Use short, regular practice sessions.
6. Learn common keyboard shortcuts gradually.
7. Take breaks when needed.

Typing speed improves with practice, but progress differs between individuals. Consistent, comfortable practice is more beneficial than forcing speed or continuing when the hands feel strained.

CHAPTER 13: MOUSE, TOUCHPAD AND KEYBOARD SKILLS

Topic 201: Understanding the Computer Mouse

A computer mouse is a pointing input device used to control the pointer on the screen. It allows users to select items, open applications, interact with menus, and perform other tasks.
Most traditional mice have a left button, a right button, and a scroll wheel. Some models include additional buttons for navigation or customizable functions.
A mouse may connect to a computer through a USB cable or wirelessly using Bluetooth or a USB receiver. The mouse is widely used with desktop computers and can also be connected to many laptops.

Topic 202: Mouse Clicking Techniques

Clicking is one of the most common ways to interact with a graphical user interface.
Common mouse actions include:

- **Single-click:** Selects an item or activates a control, depending on the interface.
- **Double-click:** Commonly opens files, folders, and desktop shortcuts in Windows.
- **Right-click:** Opens a context menu for the selected item or area.
- **Click and hold:** Keeps a mouse button pressed while performing another action, such as dragging.

Double-click speed and behavior may be adjusted in supported Windows settings.
Users should click carefully and avoid pressing buttons unnecessarily. The effect of a click depends on the selected item and the active application.

Topic 203: Using Drag and Drop

Drag and drop is a technique for moving an item by selecting it with a pointing device, holding the button, moving the pointer, and releasing the button at the destination.
It can be used to move files, reposition desktop icons, rearrange supported items, and interact with many applications.

Basic procedure:

1. Place the pointer over the item.
2. Press and hold the left mouse button.
3. Move the pointer to the destination.
4. Release the button.

The result depends on the application and destination. For example, dragging a file between folders on the same drive may move it, while dragging it to another drive may copy it by default.
Users should check the destination and resulting action, particularly when working with important files.

Topic 204: Scrolling with a Mouse

Scrolling allows users to move through content that extends beyond the visible area of a window.

Most traditional mice include a scroll wheel between the main buttons. Turning the wheel moves the page or document vertically in many applications.
Some mice also support horizontal scrolling or additional scrolling gestures.

To scroll through a document:

1. Place the pointer over the document or page.
2. Turn the scroll wheel in the desired direction.
3. Continue until the required content is visible.

Scrolling behavior may depend on the application, mouse settings, and device hardware.

Topic 205: Understanding Mouse Pointer Shapes

The mouse pointer changes appearance to indicate different actions or conditions. Recognizing these shapes helps users understand how the interface responds.
Common pointer shapes include:

- **Arrow:** Indicates the normal pointer.
- **Hand:** Often indicates a clickable link or interactive element.
- **I-beam:** Indicates a text insertion area.
- **Resize pointer:** Indicates that an object or window can be resized.
- **Busy indicator:** Indicates that Windows or an application is processing something.

The exact appearance depends on the selected pointer scheme, Windows version, and application.
Pointer changes provide visual feedback, but users should still check the context before selecting or moving an item.

Topic 206: Adjusting Mouse Settings

Windows provides settings for changing mouse behavior according to the user's preferences.
Depending on the device and Windows version, available options may include pointer speed, primary button selection, scrolling behavior, and double-click speed.

To open mouse settings in Windows 11:

1. Open Settings.
2. Select Bluetooth & devices.
3. Choose Mouse.
4. Review the available settings.
5. Adjust the required options.

Users can experiment with pointer speed to find a comfortable setting. A pointer that moves too quickly or too slowly may make precise selection more difficult.
Additional options may be available through related mouse settings or manufacturer software.

Topic 207: Understanding the Laptop Touchpad

A touchpad is a pointing device commonly built into laptops. It allows users to control the pointer by moving a finger across its surface.
Many touchpads support tapping, clicking, scrolling, and multi-finger gestures.
A touchpad is useful when an external mouse is unavailable, especially while travelling or working in a limited space.
Its available features depend on the laptop hardware, touchpad driver, and operating system settings.

Topic 208: Using Touchpad Gestures

Touchpad gestures allow users to perform actions through finger movements.
Common gestures on supported Windows laptops include:

- **Single-finger movement:** Moves the pointer.
- **Single-finger tap:** Often performs a left-click.
- **Two-finger tap:** Often performs a right-click.
- **Two-finger movement:** Commonly scrolls through content.
- **Pinch gestures:** May zoom in or out in supported applications.
- **Three- or four-finger gestures:** May switch between applications or desktops, depending on configuration.

Gestures can differ between touchpads and Windows versions.

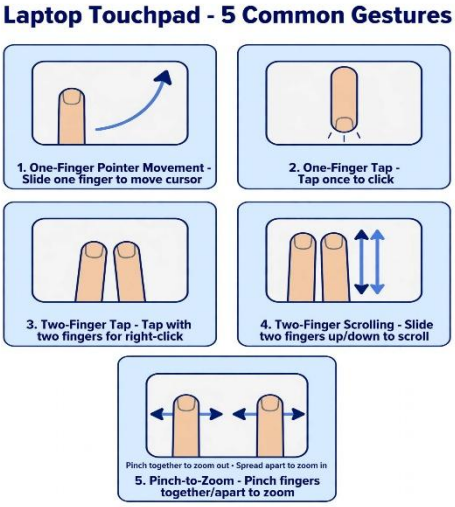


Figure 13.2: Common Laptop Touchpad Gestures

To review touchpad settings in Windows 11:

1. Open Settings.
2. Select Bluetooth & devices.
3. Choose Touchpad.
4. Review the available gestures and sensitivity options.

Users should practice gestures slowly and adjust settings if accidental touches occur.

Topic 209: Troubleshooting Mouse and Touchpad Problems

Mouse and touchpad problems may occur because of low battery power, connection issues, incorrect settings, dirt, driver problems, or hardware faults.

Basic troubleshooting steps:

1. Check that the mouse is connected properly.
2. Replace or recharge batteries in a wireless mouse if necessary.
3. Clean the mouse sensor and touchpad surface gently.
4. Review pointer and touchpad settings.
5. Restart the computer if the problem continues.
6. Check for appropriate driver updates when necessary.

If an external mouse works but the laptop touchpad does not, the touchpad may be disabled or experiencing a separate problem.

Avoid using liquids directly on the touchpad or mouse. Follow the manufacturer's cleaning instructions.

Topic 210: Choosing Between a Mouse and a Touchpad

A mouse and a touchpad both control the pointer, but they differ in design and use.

A mouse is often preferred for desktop computers, detailed selection, and tasks requiring precise pointer movement. A touchpad is integrated into many laptops and is convenient when portability matters.

The best choice depends on the task, user's comfort, available space, and device compatibility.

Some users prefer a mouse for extended office work, while others find touchpad gestures convenient for everyday navigation.

Neither device is universally better for every situation. Users should choose a comfortable and suitable pointing method.

CHAPTER 14: FILE AND FOLDER MANAGEMENT

Topic 211: Introduction to Files and Folders
A file is a named collection of information stored on a computer or another storage device. It may contain text, images, audio, video, software, or other types of data.
A folder is a container used to organize files and, in many file systems, other folders.
For example, a student may create a folder named Computer Studies and store assignment documents, presentation files, and images inside it.
Files and folders help users organize information, locate documents, and manage stored data efficiently.
Good file organization is important for schoolwork, office tasks, software development, and personal data management.

Topic 212: Understanding File Names and Extensions
A file name identifies a file and helps users distinguish it from other files. A file extension is the suffix that commonly follows the final period in a file name and indicates its format.
Examples include:

- assignment.docx — commonly a Microsoft Word document.
- marks.xlsx — commonly an Excel workbook.
- photo.jpg — a JPEG image.
- presentation.pptx — a PowerPoint presentation.
- notes.txt — a plain-text file.
- report.pdf — a PDF document.

Extensions help Windows and applications identify file types, although the extension alone does not guarantee that a file's contents are valid or safe.
Users should choose meaningful file names and avoid changing extensions unless they understand the consequences.

Topic 213: Creating a New Folder
Creating folders is a simple way to organize related files in one location.
To create a folder in Windows 11:

1. Open File Explorer.
2. Navigate to the location where the folder is needed.
3. Right-click an empty area.
4. Select New and then Folder, where available.
5. Type a meaningful folder name.
6. Press Enter.

A new folder can be created in locations such as Documents, Desktop, or another folder where the user has permission to save items.
Descriptive folder names make it easier to find information later. For example, Computer_Notes is more informative than New Folder.

Topic 214: Opening and Closing Files and Folders
Opening a folder displays its contents. Opening a file launches an application that can handle its format or displays the file within an existing application.
To open a folder or file:

1. Navigate to its location in File Explorer.
2. Double-click the folder or file.
3. Wait for Windows or the associated application to open it.

A file may not open if the required application is unavailable, the file is damaged, or the user lacks permission.
To close an opened application, use its Close button or the appropriate application command. Closing a window does not necessarily delete the file.

Topic 215: Selecting Files and Folders
Selecting an item identifies it so that a user can perform an action, such as copying, moving, renaming, or deleting it.
Common selection methods in File Explorer:
- Click an item to select it.
- Hold Ctrl while selecting additional items to select multiple separate items.
- Select one item, hold Shift, and select another item to select a range in a suitable list view.
- Press Ctrl + A to select all items in the current view.

Selection behavior may differ in some applications or interface layouts.
Users should check which items are selected before performing operations that change or remove files.

Topic 216: Renaming Files and Folders
Renaming changes the name of a file or folder without normally changing its contents.
To rename an item in Windows 11:
1. Open File Explorer.
2. Select the file or folder.
3. Right-click it and choose Rename, or use a supported keyboard shortcut.
4. Enter the new name.
5. Press Enter.

In many File Explorer configurations, F2 is a shortcut for renaming the selected item.
When renaming a file, take care not to change its extension accidentally. Changing an extension does not convert the file into another format and may make it harder to open.

Topic 217: Copying Files and Folders
Copying creates another instance of a file or folder at a different location while leaving the original in place.
To copy an item:
1. Select the file or folder.
2. Press Ctrl + C.
3. Open the destination folder.
4. Press Ctrl + V.
5. Wait for the copy operation to finish.

Users can also use the Copy and Paste commands from the relevant context menu.
Copying is useful for creating working copies, sharing documents, and organizing files. However, a copied file is not automatically a complete backup of all related data.

Topic 218: Moving Files and Folders
Moving transfers a file or folder from one location to another. Unlike copying, moving normally leaves the item at the destination rather than keeping the original in its previous location.
To move an item:
1. Select the file or folder.
2. Press Ctrl + X.
3. Open the destination folder.
4. Press Ctrl + V.
5. Confirm that the item appears at the destination.

Moving can also be performed through supported drag-and-drop actions or context menu commands.
The result of dragging an item may depend on whether the source and destination are on the same drive or different drives. Users should verify the destination before removing the original copy.

Topic 219: Deleting Files and Folders
Deleting removes a file or folder from its current location. In many ordinary Windows file operations, deleted items are moved to the Recycle Bin, where they may be restored.

However, items deleted from certain removable drives, network locations, or through permanent deletion commands may bypass the Recycle Bin.

To delete an item normally:

1. Select the file or folder.
2. Press Delete or choose Delete from the context menu.
3. Confirm the action if prompted.

To remove an item permanently, Windows may provide additional commands, but users should use them cautiously.

Before deleting important information, check that the correct item is selected and that another copy is available if needed.

Topic 220: Understanding the Recycle Bin

The Recycle Bin is a Windows feature that temporarily stores many items deleted through ordinary file operations. It allows users to restore eligible items or remove them permanently.

To restore an item:

1. Open the Recycle Bin.
2. Locate the required file or folder.
3. Right-click the item.
4. Select Restore.

The item is generally returned to its original location when that location is available.

To empty the Recycle Bin:

1. Open the Recycle Bin.
2. Review its contents.
3. Choose Empty Recycle Bin if you are sure the items are no longer needed.

Emptying the Recycle Bin removes its stored items from that recovery location. Restoring permanently deleted data may require separate recovery methods, and success is not guaranteed.

The Recycle Bin is useful for recovering accidental deletions, but it should not be treated as a backup system.

CHAPTER 14: FILE AND FOLDER MANAGEMENT

Topic 221: Understanding File and Folder Paths
A file path describes the location of a file or folder within a storage system. It helps users identify where an item is stored.
For example, C:\Users\Student\Documents\Assignment.docx represents a file named Assignment.docx inside the Documents folder of the Student user account on the C: drive.
A typical Windows path may contain a drive letter, folder names, and a file name. Network locations may use a different path format.
Understanding paths is useful when saving files, sharing locations, troubleshooting problems, and working with command-line tools.

Topic 222: Understanding the File Explorer Address Bar
The File Explorer address bar displays the current location and allows users to navigate to other folders or locations.
In Windows 11, the address bar commonly appears near the top of the File Explorer window.
To use the address bar:
1. Open File Explorer.
2. Select the address bar.s
3. Type a folder path or supported location.
4. Press Enter.
5. Review the destination.

Users can also select the displayed folder names in the address bar to navigate to parent locations.
The address bar is useful when the exact location of a folder is known and browsing through multiple folders would take longer.

Topic 223: Searching for Files by Type
Searching by file type can help users find documents, images, audio files, and other items without knowing their complete names.
For example, a user looking for PDF documents may search for files ending in .pdf.
Basic procedure:
1. Open File Explorer.
2. Navigate to the folder or drive you want to search.
3. Select the search box.
4. Enter a file extension or suitable search term.
5. Review the results.

In supported Windows search interfaces, users can also use filters or queries to narrow results by file type.
The results depend on the selected search location, indexing settings, and file permissions.

Topic 224: Sorting Files and Folders
Sorting changes the order in which files and folders appear in a list. It helps users locate items more easily.
Common sorting options include:
- **Name:** Arranges items alphabetically or in reverse order.
- **Date modified:** Arranges items according to their last modification time.
- **Type:** Groups items according to their file types.
- **Size:** Arranges files according to their storage size.

To sort items in File Explorer:
1. Open the required folder.
2. Select the available Sort option.
3. Choose a sorting criterion.
4. Select ascending or descending order if available.

Sorting changes how items are displayed; it does not normally rename, move, or modify the files themselves.

Topic 225: Grouping Files and Folders

Grouping organizes displayed items into categories based on a selected property. For example, files may be grouped by type, date modified, or another supported characteristic.

Grouping differs from sorting. Sorting changes the order of items, while grouping separates them into labeled sections.

To group items:

1. Open File Explorer.
2. Navigate to a folder.
3. Open the View or Sort controls, depending on the Windows version.
4. Select Group by.
5. Choose a suitable property.

Grouping can be helpful when a folder contains many different files. The available grouping options depend on the current folder view and Windows version.

Topic 226: Viewing File Properties

File properties provide information about a file or folder. Depending on the item, this information may include its name, type, location, size, creation date, and modification date.

To view file properties:

1. Open File Explorer.
2. Right-click the required file or folder.
3. Select Properties.
4. Review the available tabs and information.
5. Select OK or close the dialog box when finished.

Some files provide additional properties, such as document details, image dimensions, or security information.

File properties can help users identify items, check their size, and investigate basic storage or access issues.

Topic 227: Understanding File Attributes

File attributes are properties that describe or affect how a file is handled by the operating system. In Windows, common file attributes include Read-only, Hidden, and system-related attributes.

A Read-only setting can help prevent ordinary editing or overwriting in some applications, but it is not a complete security mechanism.

The Hidden attribute can make an item less visible in normal File Explorer views. It does not encrypt the item or prevent all access.

Some attributes are managed by Windows or applications and should not be changed without understanding their purpose.

File attributes are different from file permissions, which determine what actions a user or process is allowed to perform.

Topic 228: Showing Hidden Files and Folders

Some files and folders are hidden from ordinary File Explorer views to reduce clutter or protect important system-related items from accidental changes.

Hidden items are not necessarily private or secure. Users should avoid modifying unfamiliar system files.

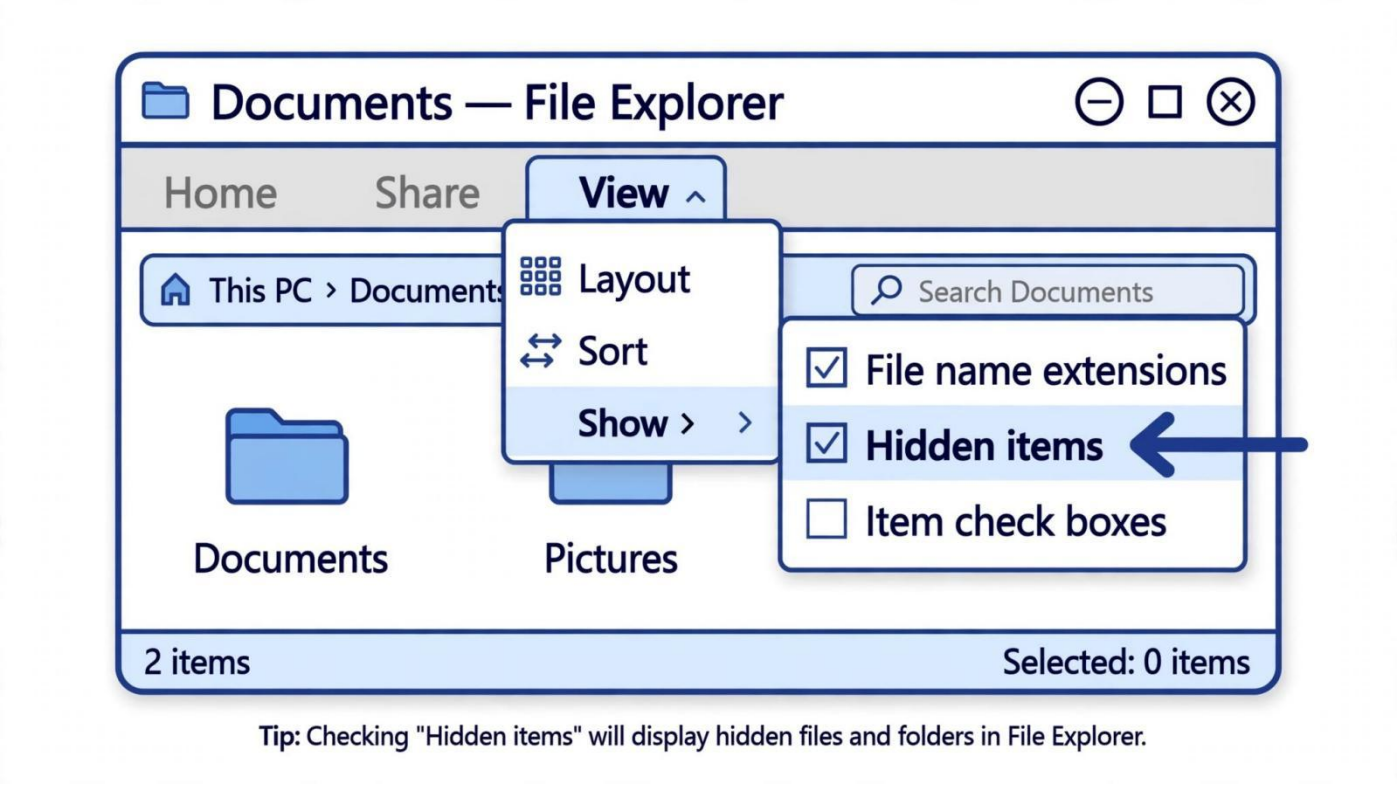
To show hidden items in Windows 11:

1. Open File Explorer.
2. Select View.
3. Open Show.
4. Enable Hidden items if the option is available.

To hide these items again, repeat the process and disable Hidden items.

The precise menu arrangement may differ between Windows versions.

Figure 14.1: Viewing Hidden Items in File Explorer



Topic 229: Understanding Compressed Files

A compressed file uses a storage format or compression method to reduce the size of data or combine multiple files into a single package.

Compression can make files easier to store and transfer. However, the amount of space saved depends on the type of data and the compression method.

Some formats support lossless compression, which allows the original data to be reconstructed exactly. Other formats, particularly certain media formats, may use lossy compression that discards some information.

Compressed archives are commonly used to distribute software packages, documents, and collections of files.

Topic 230: Creating and Extracting ZIP Archives

ZIP is a widely used archive format that can store multiple files and folders in one package. ZIP archives may also reduce storage requirements through compression.

To create a ZIP archive in supported Windows versions:

1. Select the files or folders.
2. Right-click the selection.
3. Choose the available Compress to ZIP file option.
4. Enter a name for the archive if prompted.
5. Press Enter.

The exact command may differ between Windows versions.

To extract a ZIP archive:

1. Locate the ZIP file.
2. Right-click it.
3. Select Extract All or the available extraction command.
4. Choose a destination if prompted.
5. Complete the extraction process.

Before extracting archives from unknown sources, check that they are trustworthy. Compressed files can contain harmful software just like other downloaded files.

CHAPTER 15: FILE EXPLORER

Topic 231: Introduction to File Explorer

File Explorer is the Windows application used to browse and manage files, folders, and supported storage locations.

It allows users to open folders, copy and move items, search for files, view properties, and manage connected storage devices.

File Explorer is essential for everyday computer use because it provides a graphical way to organize stored information.

Users can open it through the taskbar, Start menu, or a supported keyboard shortcut.

Topic 232: Opening File Explorer

Windows provides several methods for opening File Explorer.

Method 1: Keyboard shortcut

Press Windows + E.

Method 2: Taskbar

Select the File Explorer icon if it is pinned to the taskbar.

Method 3: Start menu

1. Open the Start menu.
2. Search for File Explorer or locate it in the application list.
3. Select the application.

File Explorer usually opens to a default location such as Home, depending on the Windows version and settings.

Knowing multiple ways to open File Explorer helps users access files more quickly.

Topic 233: Understanding the File Explorer Interface

The File Explorer interface contains several areas that help users navigate and manage stored information.

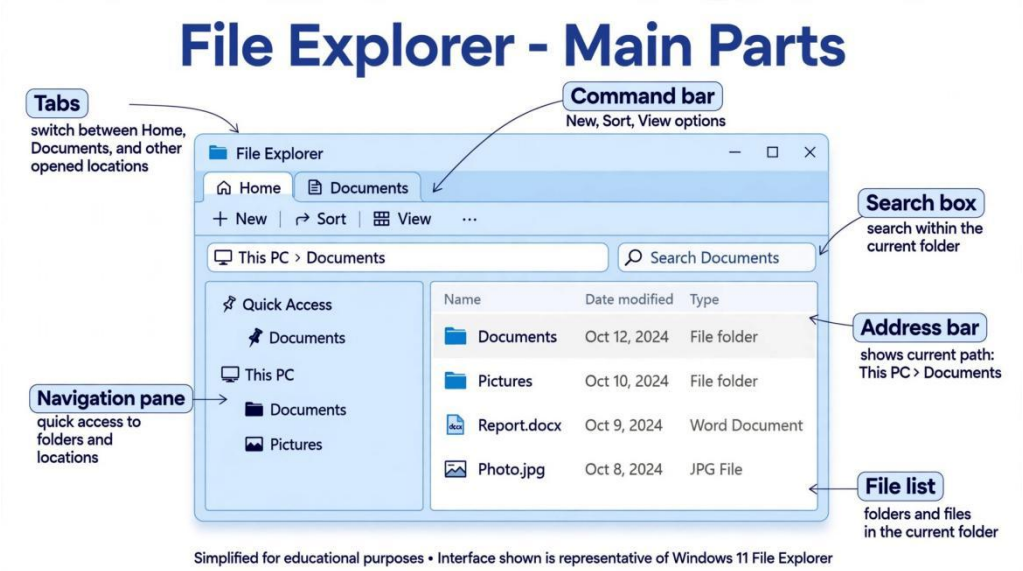
Common elements in Windows 11 include:

- **Tabs:** Allow supported locations to be opened in separate tabs.
- **Address bar:** Shows the current location and supports navigation.
- **Search box:** Helps locate files within the applicable search scope.
- **Navigation pane:** Provides access to locations such as Home, OneDrive when configured, This PC, and other folders.
- **File list or content area:** Displays the items in the current location.
- **Command bar:** Provides common file-management actions.
- **View controls:** Allow users to change how items are displayed.

The precise interface may vary by Windows version, configuration, and window size.

Understanding these components makes it easier to navigate File Explorer efficiently.

Figure 15.1: Main Components of Windows File Explorer



Topic 234: Understanding Home and Quick Access

File Explorer provides convenient entry points for frequently used locations and recent content. Their names and exact behavior vary between Windows versions.

In Windows 11, the Home area may display pinned locations, favorite folders, recent files, and other available content.

Earlier Windows versions commonly used the name Quick access for a similar navigation feature.

These features help users return to frequently used folders without navigating through the entire folder structure each time.

The content displayed depends on Windows settings, account configuration, and the locations available to the user.

Topic 235: Using This PC

This PC is a File Explorer location that provides access to the computer's available storage drives and certain common user folders.

Depending on the system configuration, it may display the local disk, additional partitions, removable drives, and connected storage devices.

To open This PC:

1. Open File Explorer.
2. Locate This PC in the navigation pane, if available.
3. Select it.
4. Review the listed drives and locations.

A drive may not appear if it is disconnected, disabled, uninitialized, or unavailable to the current user.

Users should identify a drive carefully before formatting it or changing its partitions, because such operations may destroy data.

Topic 236: Understanding Drives and Storage Locations

A drive is a storage device or a logical storage area that Windows can access. Drives may be internal, removable, optical, or network-based.

Windows commonly assigns letters such as C: and D: to certain drives or partitions. However, drive letters are labels and do not indicate the physical size or importance of a device.

For example, C: often contains the Windows installation, while another drive letter may refer to a data partition or removable device.

Network storage may be accessed through a shared location or mapped drive.

Users should understand where files are stored before moving, deleting, or formatting data.

Topic 237: Understanding Libraries and Common Folders

Windows provides common user folders for organizing personal information. These may include Documents, Downloads, Pictures, Music, and Videos.

Their actual locations can vary depending on Windows configuration, user settings, and cloud synchronization.
Some Windows versions also support libraries, which provide a combined view of content from selected folders.
Organizing files in appropriate folders helps users locate information and maintain a consistent structure.
For example, school assignments may be stored in Documents, downloaded installers in Downloads, and photographs in Pictures.
Users should check the actual folder location before deleting or moving content.

Topic 238: Using the Downloads Folder
The Downloads folder is commonly used to store files downloaded through web browsers and other applications.
Downloaded content may include documents, images, software installers, compressed archives, and other files.
To access Downloads:
1. Open File Explorer.
2. Select Downloads from the navigation pane if available.
3. Review the files in the folder.
4. Open, organize, move, or delete items as appropriate.

The Downloads folder can become cluttered over time. Regularly organizing its contents makes important files easier to find.
Before opening downloaded software or documents from unfamiliar sources, verify that they are trustworthy.

Topic 239: Changing File Explorer Views
File Explorer provides different ways to display files and folders. The best view depends on the type of content and the user's task.
Common views may include:
- **Details:** Displays information such as name, date modified, type, and size.
- **List:** Displays items in a compact list.
- **Tiles:** Displays items with selected identifying information.
- **Large or small icons:** Changes the size of displayed item icons.
- **Content:** Displays additional information for supported items.

To change the view:
1. Open File Explorer.
2. Navigate to the required folder.
3. Open the View controls.
4. Select an available layout.

Available view names and controls may vary between Windows versions and folder types.
The Details view is particularly useful when comparing file sizes, modification dates, or file types.

Topic 240: Showing File Name Extensions
File name extensions help identify file formats and can make file management clearer. Windows may hide known file extensions in its default settings.
For example, a file may appear as assignment rather than assignment.docx when extensions are hidden.
To show file name extensions in Windows 11:
1. Open File Explorer.
2. Select View.
3. Open Show.
4. Enable File name extensions if the option is available.

The menu arrangement may differ between Windows versions.
Displaying extensions helps users distinguish file types and notice unusual names. However, a file extension alone does not prove that a file is safe or that its contents match the extension.
Users should avoid changing extensions simply to make a file appear to have a different format.

Topic 241: Using the Navigation Pane

The Navigation Pane is the section on the left side of Windows File Explorer that helps users move between folders, drives, and other storage locations. It provides a convenient way to access frequently used areas without returning to the main This PC view each time.
Depending on the Windows version and settings, the Navigation Pane may display Home, Gallery, OneDrive, This PC, network locations, and connected drives.

How to Use the Navigation Pane

1. Open File Explorer by pressing **Windows + E**.
2. Look at the left side of the window.
3. Select a folder or storage location to open it.
4. Use the arrow beside a folder, when available, to expand or collapse its contents.
5. Select another location to move to a different folder.

The Navigation Pane makes file management easier by providing quick access to different locations. Its appearance may vary depending on the Windows version and configuration.

Topic 242: Using Tabs in File Explorer

Tabs in File Explorer allow users to open multiple folders within a single window. This feature is available in supported versions of Windows 11 and makes it easier to work with several locations without opening many separate windows.
For example, a user can open the Documents folder in one tab and the Downloads folder in another. The user can switch between these tabs whenever necessary.

How to Open a New Tab

1. Open File Explorer.
2. Press **Ctrl + T** to open a new tab.
3. Navigate to the folder you want to access.
4. Select another tab to switch between open locations.
5. Use the plus (+) button, when available, to create another tab.

Tabs help organize work and reduce desktop clutter. However, tabs are not the same as separate windows: each tab is a different browsing location within the same File Explorer window.

Topic 243: Opening Folders in New Tabs

Opening folders in separate tabs helps users compare or manage files stored in different locations. Instead of opening a new File Explorer window for every folder, users can keep several locations available in one window.
The available method for opening a folder in a new tab may depend on the Windows version and interface. Users can open a new tab first and then navigate to the required folder.

Example

Suppose a student wants to copy an assignment from Downloads to Documents. The student can open Downloads in one tab and Documents in another. This makes it easier to view both locations while working.

Basic Procedure

1. Open File Explorer.
2. Create a new tab using **Ctrl + T**.
3. Navigate to the first folder.
4. Create another tab and navigate to the second folder.
5. Switch between the tabs as needed.

Remember that opening a folder in a tab does not automatically copy or move its files. File operations must be performed separately.

Topic 244: Using Cloud Storage Locations in File Explorer

Cloud storage allows users to store files on remote servers and access them through the internet. Some cloud storage services can integrate with File Explorer, allowing cloud folders to appear alongside ordinary folders. For example, Microsoft OneDrive can appear in File Explorer when it is installed and configured. Other cloud storage services may provide similar integration through their applications.
Cloud files may be available in different ways. Some are downloaded and stored locally, while others may appear in File Explorer but require an internet connection before their contents can be opened.

Basic Steps for Accessing Cloud Files

1. Install or open the relevant cloud storage application.
2. Sign in using the appropriate account.
3. Complete the setup process if required.
4. Open File Explorer.
5. Select the cloud storage location if it appears in the Navigation Pane.
6. Open the required folder or file.

Cloud synchronization helps users access files across devices. However, synchronization is not always the same as an independent backup. Accidental deletions or unwanted changes may also synchronize across devices.

Users should protect their accounts with strong authentication and avoid storing sensitive information in accounts they do not control.

Topic 245: Accessing Removable Drives

A removable drive is a storage device that can be connected to and disconnected from a computer. Common examples include USB flash drives, external hard drives, and some memory card readers.

Removable drives are useful for transferring files, creating backups, and moving documents between computers.

How to Access a Removable Drive

1. Connect the drive to a compatible port or card reader.
2. Wait for Windows to detect the device.
3. Open File Explorer.
4. Select This PC.
5. Locate the drive under the available devices and drives.
6. Open it to view its folders and files.

A drive may not appear immediately if Windows cannot recognize it, the port is faulty, or the device has a technical problem. In such cases, check the connection and try another compatible port if available.

Never assume that every connected drive is safe. Files from unknown removable media may contain malicious software. Use updated security software and scan suspicious files before opening them.

Topic 246: Safely Ejecting USB Drives

Safely removing a USB drive helps reduce the risk of data loss or file-system problems. A computer may still be writing information to a drive after a file appears to have finished copying.

How to Eject a USB Drive

1. Save and close files stored on the drive.
2. Close applications that are using files from it.
3. Open File Explorer.
4. Locate the USB drive.
5. If an Eject option is available, select it.
6. Wait for Windows to confirm that the device can be removed, when such a confirmation appears.
7. Disconnect the drive.

The exact method may vary by device and Windows configuration. Some removable devices use a notification-area option, while others provide an Eject command in File Explorer.

If Windows reports that the device is in use, close any applications or windows accessing it and try again. Do not disconnect a drive while files are being transferred.

Topic 247: Accessing Network Locations and Shared Folders

A network location is a storage resource or folder that can be accessed through a computer network. Organizations often use shared folders to allow authorized employees to access common documents.

For example, a school may maintain a shared folder containing lesson plans, notices, and administrative documents.

A shared folder can be accessed only when the computer has suitable network connectivity and the user has the required permissions.

Basic Procedure

1. Connect to the appropriate network.

2. Open File Explorer.
3. Locate the shared folder through the provided network path or approved organizational instructions.
4. Enter the required credentials if requested.
5. Open the folder and access files permitted by the owner.

Some network folders may be mapped to drive letters so that users can access them more conveniently.
Network access and mapping procedures depend on the organization's configuration.
Users should not attempt to bypass access restrictions. Shared folders may contain confidential information, and permissions help protect that information from unauthorized access.

Topic 248: Understanding File Explorer Folder Options

Folder Options provides settings that control certain aspects of File Explorer. Depending on the Windows version, these settings may be accessed through File Explorer's options or folder options interface.
Users can configure aspects of browsing behavior, such as how folders open, whether certain navigation options are displayed, and how files and folders are presented.

How to Access Folder Options

1. Open File Explorer.
2. Open the available menu or toolbar options.
3. Locate **Options** or **Folder Options**, depending on the Windows version.
4. Review the available settings.
5. Change only the options you understand.
6. Select Apply or OK, when available, to save changes.

Folder Options can help users customize file browsing according to their preferences. However, changing certain settings can make the interface unfamiliar or expose files that were previously hidden.
For this reason, beginners should change one setting at a time and observe its effect before making further changes.

Topic 249: Understanding Folder View Settings

Folder view settings control how files and folders appear in File Explorer. Different views help users choose the most suitable way to browse their data.
Common views include:

- **Extra Large Icons:** Displays large file and folder icons.
- **Large Icons:** Displays relatively large icons.
- **Medium Icons:** Displays medium-sized icons.
- **Small Icons:** Displays compact icons.
- **List:** Arranges items in a list.
- **Details:** Displays file information in columns.
- **Tiles:** Displays items with additional visual information.
- **Content:** Shows selected details alongside each item.

The exact view names and available options may differ by Windows version and folder type.

How to Change a Folder View

1. Open the folder in File Explorer.
2. Locate the View menu or corresponding toolbar option.
3. Select the preferred layout.
4. Choose Details when you need to compare file names, sizes, types, or modification dates.

The Details view is particularly useful for organizing documents because it displays multiple properties in a structured format. Icon views are often more convenient when browsing pictures or visually identifiable files.

Topic 250: Copying File Paths

A file path identifies the location of a file or folder within a storage system. It helps users and applications locate a specific item.
For example:
C:\Users\Student\Documents\Assignment.docx
In this example, C: identifies the drive, Users and Student identify folders, Documents is the folder containing the file, and Assignment.docx is the file name.

How to Copy a File Path

1. Open File Explorer.
2. Navigate to the required file or folder.

3. Select the item.
4. Use the available context-menu command to copy its path, when provided.
5. Alternatively, use the address bar to copy a folder's location.
6. Paste the copied path into a suitable application.

Windows versions may offer a **Copy as path** command in the context menu. On supported systems, holding Shift while opening the context menu may reveal additional options.

A copied path is useful for technical support, documentation, and locating files. However, a path that includes a user's account name or private folder structure may reveal personal information. Avoid sharing such paths publicly when they contain sensitive details.

Topic 251: Sharing Files Through Windows

Windows provides several ways to share files, depending on the device, network, installed applications, and account configuration. A file can be shared through a cloud storage service, an approved network folder, or a supported nearby-sharing feature.

Before sharing a file, users should consider who needs access and whether the information is confidential.

General File-Sharing Procedure

1. Locate the file in File Explorer.
2. Select the file.
3. Open the Share option from the context menu or toolbar, when available.
4. Choose a supported sharing method.
5. Review the recipient and any access settings.
6. Complete the sharing process.

Cloud services may allow users to create links with different permission levels, such as viewing or editing. A link that allows anyone who receives it to access a file may expose the information more widely than intended.

Users should grant only the access necessary for the task and remove access when it is no longer required.

Topic 252: Using Nearby File-Sharing Features

Nearby file sharing allows compatible devices to exchange files without requiring the user to send them through a traditional email attachment. Availability and requirements depend on the operating system version, device capabilities, and installed sharing applications.

Windows may provide Nearby Sharing or other supported sharing options. The exact name, setup, and supported devices can vary. Some workflows may also require Bluetooth, Wi-Fi, or a compatible application.

General Procedure

1. Confirm that both devices support the same sharing method.
2. Enable the required settings on both devices.
3. Keep the devices within the supported range.
4. Select the file to share.
5. Choose the available nearby-sharing option.
6. Select the intended recipient.
7. Confirm receipt on the destination device, if required.

Before accepting a file, verify that it comes from a trusted source. Receiving a file does not automatically mean that the file is safe to open.

If the sharing option is unavailable, consult the instructions for the particular Windows version or application rather than assuming that all computers support the same feature.

Topic 253: Understanding File and Folder Permissions

File and folder permissions determine which users can perform particular actions on stored information. Permissions help protect data from accidental changes and unauthorized access.

Common permission concepts include:

- **Read:** Allows a user to view a file or folder's contents.
- **Write:** Allows a user to create or change information, subject to the applicable permissions.
- **Modify:** Commonly allows reading, creating, changing, and deleting items.
- **Full Control:** Can include changing permissions and other administrative actions, depending on the permission system.

The available permission options depend on the file system, Windows edition, storage location, and account type. Network shares and cloud services may apply additional access rules.
For example, a teacher may be allowed to view a shared school document but may not have permission to modify it.
Users should not change permissions without understanding the consequences. Removing access from the wrong account can prevent legitimate users or applications from working correctly.

Topic 254: Troubleshooting File Explorer
File Explorer may sometimes become slow, stop responding, fail to display a drive, or show an unexpected error. Basic troubleshooting can help identify and resolve common problems.
Common Problems and Solutions

- **File Explorer does not open:** Try opening it with **Windows + E**. If the problem continues, restart the computer.
- **A folder opens slowly:** Check whether it contains many files or items stored on an unavailable network or cloud location.
- **A USB drive is missing:** Check the connection, try another compatible port, and confirm whether Windows detects the device.
- **Search results are incomplete:** Confirm the search location and check the search terms.
- **File Explorer stops responding:** Close other applications if possible and restart File Explorer through Task Manager when appropriate.
- **A file cannot be opened:** Check its extension, the application associated with it, and whether the file is damaged or access is restricted.

Restarting File Explorer Through Task Manager

1. Press **Ctrl + Shift + Esc** to open Task Manager.
2. Locate Windows Explorer in the process list, if it is shown.
3. Select it.
4. Choose Restart, when available.

Restarting Windows Explorer may temporarily refresh the desktop and taskbar. Save your work first, and do not end unrelated processes without understanding their purpose.
If the problem persists, check for Windows updates and consult reliable technical support.

Topic 255: Creating an Effective Folder Structure
A folder structure is an organized arrangement of folders and subfolders used to store files logically. A good structure makes information easier to find, maintain, back up, and share.
For example, a student might organize academic work as follows:

- Documents
  - College
    - Semester 1
    - Semester 2
  - Assignments
  - Projects
  - Certificates

A web developer might use a different structure:

- Development
  - Client Projects
  - Personal Projects
  - Practice Files
  - Documentation
  - Backups

Guidelines for Organizing Files

1. Create a main folder for each major area of work.
2. Use subfolders for related projects or subjects.
3. Choose short, meaningful folder names.
4. Use a consistent naming style.
5. Separate active projects from archived work.

6. Avoid creating unnecessary levels of subfolders.
7. Back up important files regularly.

For dated documents, a consistent format such as 2026-10-12_Report.docx can help files sort chronologically by name.

An effective folder structure should match the user's needs. The goal is not to create the largest number of folders but to make information easy to locate and manage.

Chapter 16: Windows Settings and Control Panel

Topic 256: Introduction to Windows Settings

Windows Settings is the main interface for changing many system preferences and managing computer features. It allows users to configure display options, connected devices, networks, accounts, accessibility, privacy, applications, updates, and other system functions.

The Settings interface has evolved across Windows versions. Windows 11 uses a category-based interface, while older Windows versions may present options differently.

How to Open Settings

A common method is to press **Windows + I**. Users can also open the Start menu and select Settings.

Once Settings opens, users can select a category or use the search field to find a particular option.

Settings makes system configuration easier because many common controls are presented through a graphical interface. However, some advanced or legacy configuration tools may still be available through other Windows interfaces.

Before changing important settings, users should understand the purpose of the option and consider whether it may affect other users or applications on the computer.

Topic 257: Navigating Settings Categories

Windows Settings organizes system options into categories. The exact categories and their arrangement vary by Windows version.

Common Windows 11 Settings categories include:

- **System:** Display, sound, notifications, power, storage, and related controls.
- **Bluetooth & devices:** Connected devices, Bluetooth, printers, and related options.
- **Network & internet:** Wi-Fi, Ethernet, VPN, and network configuration.
- **Personalization:** Background, themes, colors, Start menu, and taskbar options.
- **Apps:** Installed applications, default apps, and optional features.
- **Accounts:** Sign-in options, account information, and related settings.
- **Time & language:** Date, time, language, and regional options.
- **Gaming:** Supported gaming features and related controls.
- **Accessibility:** Options designed to make the computer easier to use.
- **Privacy & security:** Privacy controls, permissions, and security-related settings.
- **Windows Update:** Update status and available update controls.

Some categories may appear differently or be unavailable depending on the Windows edition, device, organizational policies, or account permissions.

To navigate Settings, open the relevant category and select the required option. Read the descriptions provided on the screen before changing a configuration.

Topic 258: Searching Within Windows Settings

The Settings search feature helps users find configuration options without manually exploring every category. It is especially useful when a user knows the purpose of a setting but does not know where it is located.
For example, searching for display resolution can help locate display-related controls, while searching for default apps can help locate application settings.

How to Search Settings

1. Open Windows Settings using **Windows + I**.
2. Locate the search field.
3. Enter a keyword or phrase describing the required option.
4. Review the available results.
5. Open the most relevant result.
6. Read the displayed information before changing the setting.

Search results may vary according to the Windows version and system configuration. Some options may also be restricted by administrator policies.
Using clear search terms usually produces better results. Instead of entering a broad word such as computer, search for a specific task such as change screen brightness.

Topic 259: Introduction to Control Panel

Control Panel is a traditional Windows interface used to manage many computer settings. It has been part of Windows for many years and continues to provide access to certain configuration tools.
Depending on the Windows version, Control Panel may contain options related to programs, user accounts, hardware, devices, network settings, and system configuration.
Some functions have moved to the modern Settings application, while others remain available through Control Panel or related legacy interfaces.

How to Open Control Panel

1. Open the Start menu.
2. Type Control Panel in the search field.
3. Select Control Panel from the results.

Control Panel may display options by category or through larger or smaller icons. Category view groups related tasks, while icon views list individual configuration tools.
Because Windows continues to evolve, the available options and recommended configuration methods may change. Users should follow current Windows guidance when performing important system tasks.

Topic 260: Windows Settings vs. Control Panel

Windows Settings and Control Panel both provide ways to configure a Windows computer, but they differ in design, organization, and the types of controls they expose.
Windows Settings is the modern configuration interface. It provides a more consistent design and includes many controls for personalization, connected devices, accounts, privacy, updates, and system preferences.
Control Panel is the traditional interface. It remains useful for certain legacy tools and configuration tasks that may not be presented in the same way through Settings.

Key Differences

Feature	Windows Settings	Control Panel
Interface	Modern, category-based design	Traditional Windows interface
Navigation	Search and grouped categories	Categories or icon-based lists
Common uses	Display, personalization, accounts, updates, privacy	Legacy configuration tools and certain system options
Availability	Main configuration interface in modern Windows	Retained for selected functions
Layout	Designed for modern Windows experiences	Reflects the traditional Windows configuration model

The two interfaces are complementary rather than completely separate systems. Some tasks may open related tools outside the interface where the user started.

For everyday tasks, begin with Windows Settings. If a required option is not available there, search for the specific Windows feature or consult the official documentation for the installed Windows version.

Chapter 16: Windows Settings and Control Panel

Topic 261: Managing Display Settings

Display settings control how information appears on a computer screen. Users can adjust screen resolution, scaling, orientation, brightness, and other available options to improve readability and usability.

Common display options include:

- **Resolution:** Determines the number of pixels used to display the image.
- **Scaling:** Changes the size of text, applications, and interface elements without necessarily changing the resolution.
- **Orientation:** Controls whether the display is shown in landscape or portrait mode.
- **Brightness:** Adjusts screen brightness on supported displays.
- **Multiple Displays:** Controls how Windows uses connected monitors.

How to Open Display Settings

1. Open Settings by pressing **Windows + I**.
2. Select **System**.
3. Choose **Display**.
4. Review the available options.
5. Change the required setting and confirm it when prompted.

The available options depend on the display hardware, graphics drivers, and Windows version. A resolution recommended by Windows is usually a suitable starting point.

Topic 262: Managing Sound Settings

Sound settings allow users to control audio input and output devices. Output devices include speakers and headphones, while input devices include microphones.

Windows may provide options for selecting devices, adjusting volume, controlling application audio, and troubleshooting sound problems.

How to Configure Sound

1. Open Settings.
2. Select **System**.
3. Choose **Sound**.
4. Select the required output device.
5. Adjust the volume.
6. Choose an input device if microphone access is needed.
7. Test the device when a test option is available.

If sound is not working, check the volume level, selected output device, physical connections, and application settings. Wireless devices may also require a working Bluetooth connection.

Some sound features depend on the hardware and installed drivers. Not every computer provides the same controls.

Topic 263: Managing Power and Battery Settings

Power settings control how a computer uses energy and how it behaves when inactive. These settings are especially useful for laptops because they can influence battery life.

Common options include screen timeout, sleep behavior, and power modes. Some desktop computers and laptops may also provide additional manufacturer-specific controls.

How to Review Power Settings

1. Open Settings.
2. Select **System**.
3. Open **Power & battery** or the equivalent power section available on the device.
4. Review screen and sleep options.
5. Adjust the settings according to your needs.

A shorter screen timeout can reduce unnecessary display use. Sleep mode helps a computer resume work more quickly than a full shutdown, although its behavior varies by device.

Power-saving settings may reduce performance in some circumstances. Choose settings that balance energy use with the computer's workload.

Topic 264: Managing Storage Settings

Storage settings help users understand how disk space is being used. Windows can display the space occupied by applications, temporary files, documents, and other categories.

When a drive becomes nearly full, the computer may have less room for updates, downloads, temporary files, and application operations.

How to Check Storage Usage

1. Open Settings.
2. Select **System**.
3. Choose **Storage**.
4. Review the storage categories.
5. Identify files or applications that can be removed safely.
6. Use the available cleanup tools where appropriate.

Windows may provide temporary-file cleanup options and Storage Sense. Before removing files, check what they contain and whether they are still needed.

Do not delete unfamiliar system folders manually. Removing essential files may cause applications or Windows features to stop working correctly.

Topic 265: Understanding Storage Sense

Storage Sense is a Windows feature that can automatically free up disk space by managing selected temporary and unnecessary files. Its availability and options depend on the Windows version.

Users can configure it to run automatically or use it manually, depending on the available settings.

How to Review Storage Sense

1. Open Settings.
2. Select **System**.
3. Open **Storage**.
4. Locate Storage Sense.
5. Review its configuration.
6. Enable or adjust it only after understanding which files it may remove.

Storage Sense may manage temporary files and, depending on the configuration, items in the Recycle Bin or Downloads folder.

Users should carefully review any option involving Downloads or other personal folders. Files that appear old may still be important.

Storage Sense is a cleanup tool, not a substitute for a backup. Important documents should be stored safely before performing major cleanup operations.

Topic 266: Managing Installed Applications

Installed applications are programs available for use on a computer. Examples include web browsers, office software, media players, and development tools.
Windows Settings can provide a list of installed applications and, depending on the application and Windows version, options to uninstall, modify, or manage them.

How to Review Installed Applications

1. Open Settings.
2. Select **Apps**.
3. Open **Installed apps** or the equivalent section.
4. Search for the application.
5. Review the available options.

Uninstalling an application removes it from the computer, but the process may not remove every personal file associated with it. Some applications store settings or documents separately.
Before uninstalling software, confirm that it is not required for Windows, another application, or an important work process. If the application contains personal projects or settings, back them up first.

Topic 267: Managing Default Applications

Default applications are the programs Windows uses automatically to open particular file types or handle certain links and activities.
For example, a default web browser opens supported web links, while a default PDF application opens PDF documents.
The default application can influence how a file opens, but it does not change the file's contents.

How to Change a Default Application

1. Open Settings.
2. Select **Apps**.
3. Choose **Default apps**.
4. Search for an application or file extension.
5. Select the relevant option.
6. Choose the preferred application from the available list.

Windows may organize default settings by application or by file type. The procedure varies by version.
Choose an application that supports the file format. Assigning an unsuitable application may prevent the file from opening correctly until the default is changed again.

Topic 268: Managing Optional Windows Features

Optional Windows features are additional components that can be enabled or disabled according to a computer's needs. Some are designed for specialized compatibility, development, administration, or system functionality.
The available features depend on the Windows edition and configuration.

How to Find Optional Features

1. Open Settings.
2. Search for **Optional features**.
3. Open the relevant result.
4. Review the available components.
5. Add or remove a feature only when you understand its purpose and requirements.

Some features may require administrator permission, an internet connection, or a restart.
Do not enable unfamiliar components simply because they are available. Certain features are intended for specific technical tasks and may introduce additional security or maintenance considerations.

Topic 269: Managing Language and Keyboard Settings

Language and keyboard settings determine which languages Windows displays and which keyboard layouts users can use to type text.
A display language affects parts of the Windows interface, while a keyboard layout controls how physical keys produce characters.
For example, a user may install an additional keyboard layout to type in another language without changing the entire Windows display language.

How to Review Language Settings

1. Open Settings.
2. Select **Time & language**.
3. Open **Language & region** or the equivalent section.
4. Review the installed languages.
5. Add or configure a language if necessary.
6. Review keyboard options for the selected language.

Windows may provide a keyboard language indicator in the taskbar. On supported configurations, **Windows + Space** can switch between installed keyboard layouts.

Installing a language pack or changing the display language may require additional downloads or signing out before all changes take effect.

Topic 270: Managing Date, Time, and Regional Settings

Date, time, and regional settings affect how Windows displays dates, times, numbers, and certain location-related information.

Incorrect date and time settings may cause problems with scheduled tasks, some websites, file timestamps, and security checks.

How to Review Date and Time

1. Open Settings.
2. Select **Time & language**.
3. Choose **Date & time**.
4. Review the time zone.
5. Check whether automatic time settings are enabled.
6. Correct any inaccurate settings when permitted.

Regional settings may also control date formats, number formats, and other locale-specific preferences.

For example, two systems may represent the same date differently depending on their regional formats. A format such as 2026-10-12 can reduce ambiguity in technical documents because the year, month, and day appear in a consistent order.

Use the correct time zone for the computer's location, particularly when working with online meetings, scheduled tasks, and timestamped records.

Topic 271: Managing Privacy and Application Permissions

Privacy settings help users control how applications access certain device features and information. Depending on the Windows version, permissions may cover the camera, microphone, location, contacts, notifications, and other resources.

An application generally needs appropriate permission to access a protected feature through the relevant Windows permission system. However, permission controls may differ for traditional desktop applications and applications installed through different methods.

How to Review Permissions

1. Open Settings.
2. Select **Privacy & security** or the equivalent section.
3. Choose the relevant permission category.
4. Review the available access controls.
5. Disable access that is unnecessary, where appropriate.

For example, a video-conferencing application may need microphone and camera access to support calls. A basic calculator usually does not need either feature for its main function.

Grant permissions according to an application's purpose. If a feature stops working after a permission is disabled, review the setting before reinstalling the application.

Topic 272: Managing Windows Security Settings

Windows Security provides access to several built-in security protections. Available features depend on the Windows version, device configuration, and installed security software.

Common protection areas include virus and threat protection, firewall and network protection, application and browser control, and device security.

How to Open Windows Security

1. Open the Start menu.

2. Search for **Windows Security**.
3. Open the application.
4. Review the status of the available protection areas.
5. Follow the recommended actions when appropriate.

Security warnings should be investigated rather than ignored. At the same time, users should verify unexpected warnings carefully and avoid installing software from untrusted sources.
Windows Security does not remove the need for safe browsing, software updates, strong passwords, and regular backups. Security works best when multiple protective practices are used together.

Topic 273: Understanding Windows Update Settings
Windows Update helps keep Windows current with supported security fixes, reliability improvements, and other updates.
Updates can address vulnerabilities and compatibility problems. Depending on the device and update type, installation may require a restart.
How to Check for Updates

1. Open Settings.
2. Select **Windows Update**.
3. Choose **Check for updates**, when available.
4. Review the results.
5. Follow the instructions for installing applicable updates.

The exact update schedule and available options vary by Windows edition, device, and organizational policy.
Before a major update, save important work and ensure that important files are backed up. Laptops should have sufficient battery power or be connected to a suitable power source.
Users should obtain updates through Windows Update or trusted official channels rather than unfamiliar download websites.

Topic 274: Understanding Recovery Settings
Recovery settings provide tools for addressing certain Windows problems. Available options vary by Windows version and system configuration.
Depending on the device, recovery features may include startup repair, advanced startup options, recovery media options, and resetting the computer.
A recovery option is not always a simple repair. Some procedures can remove applications, change settings, or delete personal files.
Before Using Recovery Tools

1. Identify the problem you are trying to solve.
2. Back up important documents and project files.
3. Check whether the device uses encryption and whether the necessary recovery information is available.
4. Review the description of the proposed recovery action.
5. Seek technical assistance if the consequences are unclear.

Resetting a computer may provide an option to retain certain personal files, but users should never assume that this guarantees the preservation of every file, application, or setting.
Recovery planning is most effective when backups are prepared before a serious problem occurs.

Topic 275: Understanding Activation and Windows Edition
Windows activation is the process through which Windows verifies that a copy of the operating system is properly licensed under the applicable licensing terms.
A Windows edition is a particular version of the operating system, such as Home or Pro. Different editions may provide different features.
Activation status and Windows edition are separate concepts: the edition identifies the installed product variant, while activation status indicates whether Windows has been activated.
How to Check Activation Status

1. Open Settings.
2. Select **System**.
3. Open **Activation**.

4. Review the displayed status and edition information.

The precise layout may differ across Windows versions.
If activation fails after a major hardware change or another system event, use the official troubleshooting options and confirm that the correct license is associated with the device.
Avoid unofficial activation utilities and suspicious product-key offers. They may violate licensing terms or expose the computer to security risks.

Topic 276: Managing Accessibility Settings

Accessibility settings help people use computers according to their individual needs. They provide options for improving visibility, hearing, interaction, and navigation.
Common accessibility features include:

- **Magnifier:** Enlarges parts of the screen.
- **Narrator:** Reads supported on-screen information aloud.
- **Text size:** Changes the size of text in supported areas.
- **High contrast or contrast themes:** Improves the distinction between interface elements.
- **Captions:** Displays text for supported audio or video content.
- **Keyboard accessibility:** Provides features that support users who cannot comfortably use a mouse.
- **Voice access or speech features:** Allows supported voice-based interaction.

How to Open Accessibility Settings

1. Open Settings.
2. Select **Accessibility**.
3. Choose the relevant category.
4. Review the available features.
5. Enable and configure the options that meet your needs.

The available tools depend on the Windows version, language support, and hardware.
Accessibility is useful not only for people with permanent disabilities but also for users experiencing temporary difficulties, such as eye strain or an injury that makes ordinary input methods harder to use.

Topic 277: Creating a System Restore Point

System Restore is a Windows recovery feature that can help return certain system settings and files to an earlier state. Restore points may be created automatically or manually when System Protection is configured.
A restore point is not a complete backup of personal documents. It should not be relied upon to recover deleted photos, assignments, or other individual files.

General Procedure

1. Search Windows for **Create a restore point**.
2. Open the relevant system protection interface.
3. Review the available drive protection settings.
4. Enable System Protection for a drive if appropriate and permitted.
5. Select the option to create a restore point.
6. Enter a meaningful description.
7. Follow the displayed instructions.

The available controls may vary by Windows version, edition, and system configuration.
Restore points can help with certain system changes, but they may not resolve hardware failures, every software problem, or malware infections. Keep separate backups of important personal data.

Topic 278: Using Troubleshooters in Windows

Windows troubleshooters help diagnose and sometimes resolve common system problems. Depending on the Windows version, troubleshooters may be built into Settings or available through other support interfaces.
Troubleshooting tools may cover areas such as audio, networking, printers, Bluetooth, and Windows Update.

How to Find a Troubleshooter

1. Open Settings.
2. Search for the name of the feature causing the problem.
3. Look for an available troubleshooting option.
4. Run the tool if it is appropriate.

5. Follow the instructions and review the result.

Not every Windows installation offers the same troubleshooters. Some older troubleshooters have been replaced, moved, or removed in newer releases.

A troubleshooter may identify a likely cause without fixing the problem completely. If the issue continues, record any error message and investigate the hardware, drivers, application settings, or network connection involved.

Topic 279: Understanding Administrator Permissions

Administrator permissions allow an authorized account to perform certain system-level tasks that ordinary users may not be allowed to perform.

Examples include installing some applications, changing protected system settings, and managing certain user accounts.

Windows may request confirmation through User Account Control when an application attempts to make significant changes. The exact prompt depends on the task, account type, and system configuration.

Important Security Practices

- Use a standard account for everyday activities when practical.
- Approve administrator requests only when you understand the action.
- Do not provide administrator approval to unfamiliar applications.
- Keep account credentials private.
- Ask an authorized administrator before changing settings on a school or workplace computer.

Administrator access should be limited to trusted users and necessary tasks. Using elevated permissions unnecessarily can increase the potential impact of mistakes or malicious software.

Topic 280: Basic Windows Settings Troubleshooting

Windows configuration problems may occur because of incorrect settings, missing permissions, outdated drivers, software conflicts, or system errors.

A structured troubleshooting approach helps users investigate problems without making unnecessary changes.

Recommended Troubleshooting Process

1. **Describe the problem:** Identify the setting or feature that is not working.
2. **Check simple causes:** Confirm connections, account access, and relevant options.
3. **Read error messages:** Record the exact wording when possible.
4. **Search Settings:** Locate the relevant configuration area.
5. **Restart when appropriate:** Save work before restarting the computer.
6. **Check updates:** Review applicable Windows and driver updates through trusted sources.
7. **Undo recent changes:** If the problem began immediately after a configuration change, consider reversing that change.
8. **Seek help when needed:** Consult a qualified technician for persistent or high-risk problems.

Avoid changing multiple settings at once because doing so makes it difficult to identify which change affected the result.

For school, business, and shared computers, follow the organization's IT policies before attempting advanced troubleshooting.

A careful and methodical approach reduces the risk of data loss and helps users solve problems more efficiently.

Chapter 17: Windows Accounts and Sign-In

Topic 281: Introduction to Windows User Accounts

A Windows user account represents a person or user identity that can access a computer. It helps Windows manage personal settings, files, permissions, and access to applications and system resources.

A computer may contain multiple user accounts. For example, a family computer may have separate accounts for different family members, while an office computer may provide accounts for authorized employees.

Each account can have its own desktop settings, documents, and sign-in preferences. However, the level of access depends on the account type and the permissions assigned to it.

Main Purposes of User Accounts

- Separating users' personal files and settings.
- Controlling access to applications and system resources.
- Managing administrator and standard-user permissions.
- Supporting personalized desktop environments.
- Improving accountability on shared computers.

User accounts help organize access to a computer, but they do not automatically encrypt every personal file or protect information against every form of unauthorized access. Appropriate security settings and responsible computer use are also necessary.

Topic 282: Types of Windows User Accounts

Windows supports different account arrangements, including Microsoft accounts and local accounts. The features available to each account depend on the Windows version and configuration.

1. Microsoft Account

A Microsoft account is an online account that can be used with supported Microsoft services and Windows features. Depending on the configuration, it may support synchronization of certain settings and access to cloud services.

2. Local Account

A local account is created for use on a particular Windows installation. It does not require the account itself to be a Microsoft online account.

3. Administrator Account

An administrator account has permissions to perform certain system-level tasks, such as changing protected settings and managing software.

4. Standard User Account

A standard account is intended for everyday activities and generally has fewer system-level permissions than an administrator account.

Microsoft and local accounts describe how an account is associated with an identity, while administrator and standard describe permission levels. These are different concepts and should not be treated as mutually exclusive categories.

Topic 283: Understanding Microsoft Accounts

A Microsoft account is an online identity used to access supported Microsoft services. Depending on the services used and the user's settings, it may provide access to services such as Outlook, OneDrive, Microsoft Store, and selected Windows features.

When a Microsoft account is connected to Windows, certain settings or information may be synchronized if the relevant features are enabled.

Potential Benefits

- Access to supported Microsoft services.
- Convenient sign-in across compatible services.
- Synchronization of selected settings.
- Integration with supported cloud storage and applications.
- Account recovery options, depending on the account configuration.

Important Considerations

A Microsoft account relies on account credentials and appropriate security protections. Users should protect their passwords, enable multifactor authentication where available, and keep recovery information current.

Synchronization does not mean that every file, application, or setting is automatically transferred to every device. The available features depend on the Windows version, account configuration, and service.

Topic 284: Understanding Local Accounts

A local account is an account created for use on a particular computer. Its identity and sign-in credentials are managed locally rather than requiring the account itself to be a Microsoft online account.

Local accounts can be useful on shared computers, in certain offline environments, or when a user prefers a computer-specific account.

Characteristics of Local Accounts

- They can be used to sign in to the configured Windows installation.
- They can have their own desktop settings and files.
- They can be assigned standard-user or administrator permissions.
- They do not automatically provide access to all Microsoft online services.
- Password recovery options may differ from those of an online account.

Local accounts can still use internet-connected applications if those applications support separate sign-in methods. A local Windows account does not necessarily mean that the computer must remain offline.

Users should establish appropriate sign-in protection and keep important personal files backed up.

Topic 285: Creating a Windows User Account

Creating separate user accounts can help people share a computer while maintaining different settings and access permissions.

The procedure varies by Windows version, account type, and whether the computer is managed by a school, business, or another organization.

General Procedure

1. Open Windows Settings.
2. Select **Accounts**.
3. Locate the section for other users or family and other users.
4. Choose the available option to add another user.
5. Follow the instructions for a Microsoft account or the supported local-account setup.
6. Assign the appropriate account permissions.

An administrator may need to approve account creation or changes.

When creating accounts for children, students, or other shared-device users, choose appropriate permissions. Standard-user access is usually more suitable for ordinary daily activities than unrestricted administrator access.

Never create an account for another person without appropriate authorization, and avoid sharing sign-in credentials between users.

Topic 286: Managing User Account Information

User account information identifies the person or identity associated with a Windows session. Depending on the account type, Windows may display a name, profile picture, email address, or other account details. Some account details are managed locally, while others may be associated with an online account.

How to Review Account Information

1. Open Settings.
2. Select **Accounts**.
3. Open the relevant account information section.
4. Review the available details.
5. Make permitted changes using the provided options.

A change to the displayed name does not necessarily change the name of the user's profile folder. Windows may retain the original folder name even after the account's display name changes.

For managed school or workplace devices, some information may be controlled by an administrator. Users should follow the organization's policies when changing account details.

Topic 287: Understanding Windows User Profiles

A user profile stores personal settings and other information associated with a Windows user. It helps Windows provide a consistent working environment when that user signs in.

A profile may contain desktop preferences, application settings, personal folders, and other user-specific data.

A typical profile folder is located under:

C:\Users\

For example, a profile might use a path such as:

C:\Users\Student

The actual folder name depends on how the profile was created and may not match the current display name.

Why User Profiles Matter

- They help separate users' settings.
- They provide locations for personal documents and other files.
- They support application preferences.
- They allow Windows to load a personalized environment at sign-in.

A user profile is not the same as a backup. Profile corruption, accidental deletion, or disk failure can cause problems, so important documents should also be backed up separately.

Topic 288: Signing In to Windows

Signing in establishes an authenticated user session on a Windows computer. After successful sign-in, Windows loads the user's profile, settings, and permitted resources.

Depending on the device and configuration, sign-in may use a password, PIN, biometric authentication, security key, or another supported method.

General Sign-In Procedure

1. Start the computer or wake it from sleep.
2. Wait for the sign-in screen.
3. Select the appropriate account if multiple accounts are displayed.
4. Enter the required credentials or use the configured sign-in method.
5. Wait for Windows to load the desktop.

A successful sign-in does not mean the user has administrator permissions. Account authentication establishes identity, while permissions determine which actions the account may perform.

On shared computers, users should sign out when finished if the session must not remain accessible to the next person.

Topic 289: Understanding Passwords and Passphrases

A password or passphrase is a secret used to authenticate a user. A password is often a relatively short sequence of characters, while a passphrase is generally a longer sequence of words or characters.

A strong password should be difficult for others to guess and should not be reused across important accounts.

Password Security Guidelines

- Use long, unique passwords.
- Avoid names, birthdays, and predictable patterns.
- Do not reuse the same password across multiple services.
- Never share passwords through public messages.
- Use a reputable password manager when appropriate.
- Enable multifactor authentication when available.

Long, unique passphrases can be easier to remember than complicated but short passwords. However, a passphrase made from a common quotation or predictable phrase may still be weak.
Password requirements differ among systems. Follow the applicable requirements while prioritizing length, uniqueness, and secure storage.

Topic 290: Understanding Windows PINs
A Windows Hello PIN is a sign-in method used with supported Windows configurations. It is generally associated with a particular device rather than serving as a universal password for online services.
A PIN may be easier to enter than a long account password, but it should still be difficult for other people to guess.
Important Characteristics
- A PIN is configured through supported Windows sign-in settings.
- It is intended for use with the configured device.
- It is different from the password used for an online account.
- Available PIN options depend on the device and Windows configuration.

How to Review PIN Settings
1. Open Settings.
2. Select **Accounts**.
3. Choose **Sign-in options**.
4. Locate the available PIN settings.
5. Follow the displayed instructions to add, change, or reset the PIN, when permitted.

Do not share a PIN with others. If a PIN is forgotten, use the official recovery options displayed by Windows rather than relying on unfamiliar third-party tools.

Topic 291: Understanding Windows Hello
Windows Hello is a Windows sign-in technology that supports authentication methods such as facial recognition, fingerprint recognition, and PIN-based sign-in on compatible devices.
Its purpose is to provide convenient sign-in while using authentication mechanisms supported by the device.
Windows Hello Methods
- **Facial Recognition:** Uses compatible camera hardware and supported recognition technology.
- **Fingerprint Recognition:** Uses a compatible fingerprint reader.
- **PIN:** Uses a configured device-associated sign-in code.

Not every computer supports every Windows Hello method. Facial recognition generally requires compatible hardware, and fingerprint sign-in requires a supported reader.
Biometric methods can improve convenience, but users should still understand the alternative recovery or sign-in options available on their devices.
Never assume that a standard webcam or ordinary camera automatically supports Windows Hello facial recognition.

Topic 292: Understanding Multifactor Authentication
Multifactor authentication (MFA) requires users to verify their identity using more than one type of authentication factor. It can provide additional protection when a password is compromised.
Authentication factors are commonly grouped into three categories:
- **Something you know:** A password or PIN.
- **Something you have:** A registered security key or authentication device.
- **Something you are:** A supported biometric characteristic, such as a fingerprint.

Using two passwords does not normally count as two different authentication factors because both belong to the same category.

Depending on the service, MFA may involve an authentication application, security key, biometric verification, or another supported method.
Users should never approve unexpected sign-in requests. An unfamiliar authentication request may indicate that someone is attempting to access the account.
MFA reduces certain risks but does not eliminate every form of account compromise. Users should continue protecting their credentials and recovery information.

Topic 293: Locking, Signing Out, and Switching Users
Windows provides several ways to leave a computer session. These actions serve different purposes and should not be confused.
Lock: Secures the current session behind the sign-in screen while keeping the user's applications and work running.
Sign Out: Ends the current user session. Unsaved work may be lost if applications close without saving.
Switch User: Allows another user to sign in while the existing user's session may remain active.
Shut Down: Turns off the computer through the operating system's shutdown process.
How to Lock Windows
Press **Windows + L** to lock the computer.
Locking is useful when leaving a desk temporarily. Signing out is more suitable when the current session should end or when a shared computer is being handed over.
Before signing out, save open documents and complete important file operations. The exact behavior of sessions and power options can vary with system policies and Windows configuration.

Topic 294: Managing Sign-In Options
Windows sign-in options allow users to configure supported authentication methods and certain sign-in behaviors.
Depending on the device and account configuration, available options may include a password, PIN, fingerprint, facial recognition, security key, or other supported methods.
How to Review Sign-In Options
1. Open Settings.
2. Select **Accounts**.
3. Choose **Sign-in options**.
4. Review the available authentication methods.
5. Configure a method using the instructions provided by Windows.

Some methods require additional hardware, and certain options may be unavailable because of organizational policies or account restrictions.
Choose a sign-in method that balances convenience and security. Users should also understand how to recover access if their primary method becomes unavailable.
Do not disable important security protections simply to make sign-in faster, especially on shared or portable computers.

Topic 295: Basic Windows Account Security
Windows account security involves protecting user identities, sign-in credentials, personal files, and access permissions.
A secure account setup reduces the risk of unauthorized access and accidental changes.
Recommended Practices
1. Use a strong, unique password where applicable.
2. Configure an appropriate sign-in method.
3. Enable MFA for supported online accounts.
4. Keep account recovery information current.
5. Use standard-user permissions for ordinary tasks when practical.
6. Lock the computer when leaving it unattended.
7. Install updates through trusted channels.
8. Avoid signing in to sensitive accounts on untrusted devices.
9. Review unexpected sign-in notifications.
10. Back up important documents separately.

On school and workplace computers, follow the organization's account and device policies.
If unauthorized access is suspected, use the account provider's official security and recovery procedures.
Changing a password alone may not resolve every issue if other sessions, recovery methods, or devices remain compromised.

Chapter 18: Software, Apps and Programs

Topic 296: Introduction to Software
Software is a collection of programs, instructions, and related data that tells a computer how to perform tasks. Unlike hardware, software consists of digital instructions rather than physical components.
Software enables users to write documents, browse the internet, communicate, manage information, develop applications, and perform many other activities.
Common examples include operating systems, web browsers, word processors, media players, and accounting applications.
Main Categories of Software

- **System Software:** Supports the operation and management of computer hardware. Operating systems are a major example.
- **Application Software:** Helps users perform specific tasks, such as writing documents or editing images.
- **Programming Software:** Provides tools for writing, testing, and maintaining computer programs.

These categories can overlap in complex systems, but they provide a useful starting point for understanding software.
A computer needs appropriate software to make practical use of its hardware. The software must also be compatible with the operating system and device.

Topic 297: Understanding Applications and Programs
An application or program is software designed to perform one or more tasks. In everyday computer use, the terms *application*, *app*, and *program* are often used interchangeably, although their technical meanings may differ in some contexts.
Examples include:

- A word processor for creating documents.
- A spreadsheet application for calculations and data analysis.
- A web browser for accessing websites.
- A media player for playing supported audio and video files.
- A code editor for writing and editing source code.

Some applications run directly on a computer, while others operate mainly through a web browser. Some combine local and online functions.
An application may consist of multiple files and components. Deleting its visible shortcut does not necessarily uninstall the application itself.
Users should obtain applications from trusted sources and confirm that the software meets their needs before installing it.

Topic 298: Understanding Desktop Applications

A desktop application is software designed to run within a desktop operating system, such as Windows. It is commonly installed on the computer, although some desktop applications also depend on online services.

Examples include desktop word processors, image editors, media players, and software development tools.

Common Characteristics

- They can often be launched from the Start menu or a desktop shortcut.
- They may store files locally or use cloud storage.
- They may support interaction with operating-system features.
- Their performance and capabilities depend on the software and computer hardware.
- They may require installation, updates, or additional components.

Some desktop applications can work without an internet connection, while others require connectivity for important functions.

Before installing a desktop application, check its system requirements, publisher, supported Windows version, and licensing terms.

Topic 299: Understanding Web Applications

A web application is software that users access through a web browser or another compatible client. Many web applications provide interactive features such as account management, document editing, online shopping, communication, and data entry.

Examples include web-based email services, online office tools, and browser-accessible management systems.

A web application commonly relies on a server to process requests and store or retrieve information. Some features may also run locally in the browser.

Common Characteristics

- Accessed through a browser or compatible client.
- Often requires an internet connection.
- May store information on remote servers.
- Can be used across compatible devices.
- May receive updates on the server without requiring users to install a traditional desktop application.

However, not every web application requires a continuous internet connection. Some support offline features, depending on their design.

Users should check the website's authenticity before entering passwords, payment information, or other sensitive data. A professional-looking interface alone does not prove that a website is trustworthy.

Topic 300: Desktop Applications vs. Web Applications

Desktop applications and web applications both help users perform tasks, but they differ in how they are accessed, installed, and maintained.

A desktop application generally runs as software on the local operating system. A web application is typically accessed through a browser and often depends on server-side services.

Key Differences

Feature	Desktop Application	Web Application
Access	Usually launched through the operating system	Usually accessed through a browser or compatible client
Installation	Often requires local installation	Often requires no traditional local installation
Internet requirement	Depends on the application	Often required, although offline features may exist
Updates	May require local updates	Many updates are deployed on the server
Data storage	May use local storage, cloud storage, or both	Often uses server-side storage, but local storage may also be used
Device	Depends on the supported operating	Depends on browser, operating system, and

Feature	Desktop Application	Web Application
compatibility	system	application design

Neither type is automatically better in every situation. A desktop application may be appropriate for specialized local tasks, while a web application may be more convenient for accessing shared information across devices.
The best choice depends on the required features, security, performance, accessibility, cost, and working environment.

Chapter 18: Software, Apps and Programs

Topic 301: Installing Software on Windows
Software installation is the process of adding an application or program to a computer so that it can be used. Depending on the software, installation may involve copying files, configuring settings, adding components, and creating shortcuts.
Windows applications can be installed through Microsoft Store, official websites, or other trusted distribution channels.
General Installation Procedure

1. Check the application's system requirements.
2. Obtain the software from a trusted source.
3. Open the installer or select Install in the relevant store.
4. Review the installation options and license terms.
5. Follow the instructions displayed on the screen.
6. Approve administrator requests only when appropriate.
7. Complete the installation and open the application.

Some applications install without requiring administrator permission, while others need elevated permissions to make system-wide changes.
Never run an unfamiliar installer simply because it promises a useful feature. Verify the publisher and source, and avoid installers that bundle unwanted software.

Topic 302: Installing Applications Through Microsoft Store
Microsoft Store is a digital distribution platform for supported Windows applications, games, and other content. It provides a convenient way to find, install, update, and manage supported applications.
The available catalog and installation options may vary by Windows version, region, account configuration, and device policy.
How to Install an Application

1. Open Microsoft Store from the Start menu.
2. Search for the required application.
3. Select the application from the results.
4. Review its publisher and available information.
5. Select Get, Install, or the equivalent available option.
6. Follow any additional instructions.

Some applications may require an account, payment, or additional permissions. Others may be available without charge.
Downloading through an official store can reduce certain risks, but users should still check the publisher, application details, and requested permissions before installation.

Topic 303: Installing Software from Official Websites
Some applications are distributed through their developers' official websites rather than through an app store. Users may need to download an installer and run it manually.
Safe Installation Procedure
1. Identify the software's official website.
2. Confirm that the website address is correct.
3. Locate the download page for the required application.
4. Select the version compatible with the computer.
5. Download the installer.
6. Review the file and publisher information where available.
7. Run the installer and follow the instructions.
8. Complete the installation and verify that the application opens correctly.

Avoid advertisements and unfamiliar download portals that imitate official download buttons.
A digital signature or publisher name can help verify a file's origin, although users should not rely on a single indicator as proof that software is safe.
For important business or school computers, follow the organization's software installation policies.

Topic 304: Understanding Software System Requirements
System requirements describe the hardware and software needed to run an application. Checking them before installation helps users avoid compatibility problems and disappointing performance.
Requirements may include:
- Supported operating system and version.
- Processor architecture and performance.
- RAM capacity.
- Available storage space.
- Graphics hardware or graphics capabilities.
- Required frameworks, drivers, or other software components.
- Internet connectivity, when needed.

Example
A video-editing application may require more RAM and storage than a basic text editor because it processes large media files.
Minimum requirements describe the basic conditions under which software may run. Recommended requirements generally indicate a more suitable configuration for the intended workload.
Meeting the minimum requirements does not guarantee that every feature will perform smoothly. Actual performance also depends on the size of the workload, background applications, hardware condition, and software optimization.

Topic 305: Understanding Software Installation Options
Some installers provide options that control how an application is installed. These options may include the installation folder, available components, shortcuts, language, or whether the software is installed for one user or multiple users.
Two common installation types are:
Typical or Recommended Installation: Uses the developer's standard configuration and is generally suitable for most users.
Custom Installation: Allows users to select specific components or change supported settings.
The terminology and available options differ among installers.
Important Guidelines
1. Read each installation screen before continuing.
2. Check whether optional software is being offered.
3. Select only the components you need.

4. Review the destination folder if you change it.
5. Avoid modifying advanced options unless you understand them.

A custom installation is not automatically safer or better than a typical installation. Choose the method that meets the task's requirements without adding unnecessary components.

Topic 306: Uninstalling Software from Windows

Uninstalling software removes an application through its supported removal process. This can free storage space and remove programs that are no longer needed.

How to Uninstall an Application

1. Open Settings.
2. Select **Apps**.
3. Open **Installed apps** or the equivalent section.
4. Locate the application.
5. Open its available menu.
6. Select Uninstall, if the option is provided.
7. Follow the removal instructions.

Some traditional desktop applications may use a separate uninstaller or an older Windows management interface.

Before uninstalling software, save your work and back up important files stored within the application. Removing an application may also remove associated components or settings.

Do not manually delete program folders as a substitute for proper uninstallation. Doing so may leave configuration entries, services, or other components behind.

Topic 307: Updating Software Applications

Software updates introduce changes to an application. Updates may fix security vulnerabilities, correct errors, improve compatibility, add features, or enhance performance.

Applications may update automatically, through an app store, through a built-in updater, or by installing a newer version from the developer's official website.

General Update Procedure

1. Identify the application and its installed version.
2. Open its update settings or the relevant store.
3. Check for available updates.
4. Review important release information where available.
5. Install the update through the trusted channel.
6. Restart the application or computer if required.

Not every update changes the visible interface. Some updates primarily improve security or reliability.

Users should avoid update notifications from unfamiliar websites or unexpected pop-ups. Verify the application and update source before downloading software.

Topic 308: Understanding Automatic Software Updates

Automatic updates allow software to download or install updates with limited manual intervention. This helps users maintain applications without remembering to check every program individually.

Automatic updating may be managed by the application, Microsoft Store, Windows, or an organizational administrator.

Benefits

- Reduces the chance of missing important security fixes.
- Helps maintain compatibility.
- Saves time during routine maintenance.
- Can provide bug fixes and improvements promptly.

Considerations

- Updates may use internet data.
- Some updates require a restart.
- A new version may change the interface or behavior.
- Updates can occasionally introduce compatibility problems.

For critical work, maintain backups and review major version changes when appropriate. Avoid permanently disabling security updates simply because an update is inconvenient.

The safest update approach depends on the software, device, and working environment.

Topic 309: Understanding Software Licenses
A software license defines the conditions under which software may be installed, used, copied, modified, or distributed. The license terms depend on the software and the rights granted by its owner.
Common licensing models include:

- **Proprietary Software:** Use is governed by the owner's license terms.
- **Open-Source Software:** Source code is available under a license that grants specified rights to use, study, modify, or distribute it.
- **Freeware:** Software distributed without a purchase price, subject to its applicable license terms.
- **Trial Software:** Software offered for evaluation under defined conditions.
- **Subscription Software:** Software or services provided under a recurring payment arrangement.

These categories are not always mutually exclusive. For example, open-source software can be offered without charge, and proprietary software may have a free edition.
Users should read the applicable license terms rather than assuming that free software can be copied or redistributed without restrictions.

Topic 310: Understanding Freeware, Shareware, and Open-Source Software
Freeware, shareware, and open-source software describe different distribution or licensing approaches.
Freeware is software offered without a purchase price. Its source code may remain unavailable, and its license may restrict copying, modification, or commercial use.
Shareware traditionally refers to software distributed for evaluation, often with limited features or a trial period. Continued use or access to additional features may require payment. The precise terms depend on the developer.
Open-Source Software is distributed under a license that grants rights to access and use source code, subject to the license conditions. Some open-source licenses permit modification and redistribution, while imposing requirements that users must follow.
A program's price does not determine its security or quality. Users should consider the developer's reputation, update practices, documentation, license terms, and the software's suitability for the intended task.

Topic 311: Understanding Software Compatibility
Software compatibility describes whether an application can operate correctly with a particular operating system, device, or software environment.
Compatibility may depend on the Windows version, processor architecture, required libraries, hardware capabilities, drivers, and application dependencies.
An application designed for one operating system may not run directly on another without a compatible version or additional software.
Common Compatibility Problems

- The application requires a newer operating system.
- A required component is missing.
- The program does not support the processor architecture.
- A hardware driver is outdated.
- The application depends on a discontinued technology.

Basic Troubleshooting

1. Check the application's official system requirements.
2. Confirm the installed Windows version.
3. Install supported updates or required components.
4. Check for a compatible version from the developer.
5. Use Windows compatibility options only when appropriate.

Compatibility settings may help with some older applications, but they cannot guarantee that unsupported software will work correctly or securely.

Topic 312: Understanding Software Dependencies
A software dependency is a component, library, service, or other resource that an application needs to function correctly.

For example, an application may require a particular runtime environment, database service, browser component, or programming library.
Dependencies are common in software development because applications often reuse existing components instead of implementing every function from scratch.

Examples

- A program written in Java may require a compatible Java runtime.
- A database application may require a database server.
- A web application may rely on server-side packages or libraries.
- A Windows application may require a supported runtime component.

If a dependency is missing or incompatible, the application may fail to start or may display an error message.
Users should obtain required components from their official developers or trusted distribution channels.
Installing random DLL files or unknown runtime packages from unfamiliar websites can introduce security and stability problems.

Topic 313: Understanding Portable Applications

A portable application is designed to run without a traditional installation process or with minimal system changes. Some portable applications can be launched directly from a folder or compatible removable drive.
Portability depends on the application's design. A portable program may still create configuration files, save user data, or require certain operating-system components.

Potential Benefits

- Convenient use across compatible computers.
- Less need for a traditional installation.
- Easier organization of certain utilities.
- Reduced changes to the host system in some cases.

Limitations

- Some applications require administrative permissions or system components.
- Settings may be stored locally or alongside the program.
- A removable drive can be lost or damaged.
- Not every application is genuinely portable.

Users should obtain portable applications from trusted sources and verify their compatibility. Running software from a USB drive does not automatically make it safe.

Topic 314: Understanding Background Applications and Processes

A background application or process performs work without necessarily displaying a visible window. Background activity may support notifications, synchronization, security, updates, printing, or other functions.
Some background processes belong to Windows, while others are associated with installed applications. Task Manager can help users review running applications and processes, although its information may require technical knowledge to interpret correctly.

Basic Review Procedure

1. Press **Ctrl + Shift + Esc** to open Task Manager.
2. Review the displayed applications and processes.
3. Check resource usage when diagnosing a performance problem.
4. Identify the relevant application before taking action.
5. Close only applications or processes you understand and do not need.

Ending an essential process may cause an application or system feature to stop working. Users should avoid terminating unfamiliar processes solely because their names are unfamiliar.
If a computer frequently runs slowly, investigate resource usage, available storage, startup applications, and possible software problems rather than randomly closing processes.

Topic 315: Troubleshooting Software Installation Problems

Software installation may fail because of insufficient storage, incompatible system requirements, missing dependencies, restricted permissions, damaged installation files, or network problems.
A careful troubleshooting process can help identify the cause.

Common Problems and Solutions

- **Insufficient storage:** Check available disk space and remove unnecessary files safely.
- **Incompatible software:** Confirm the supported operating system and hardware requirements.
- **Permission error:** Use an authorized account and follow the developer's instructions.
- **Incomplete download:** Download the installer again from the official source.
- **Missing dependency:** Install the required component through a trusted source.
- **Installation blocked by security software:** Review the warning and verify the software before deciding what to do.
- **Network failure:** Check the internet connection and retry when the connection is stable.

Do not disable security protections simply to force an installation to continue. If an installer is blocked or produces an unfamiliar warning, verify its origin and consult reliable technical support.

Chapter 19: Internet, Browsers and Online Work

Topic 316: Introduction to the Internet

The internet is a global network of interconnected computer networks that communicate using standardized protocols. It supports services such as websites, email, file transfers, online communication, streaming, and cloud computing.

Devices connect to the internet through network equipment and internet service providers. Data travels between devices and servers using the relevant networking technologies.

The internet is not a single computer or a single website. It is a large system that allows independently managed networks and devices to exchange information.

Common Uses of the Internet

- Accessing websites and online information.
- Sending and receiving email.
- Communicating through messaging and video calls.
- Using cloud applications and storage.
- Learning through online courses.
- Conducting business and electronic transactions.

The internet provides many benefits, but users should consider privacy, security, accuracy, and reliability when interacting with online services.

Topic 317: Understanding the World Wide Web

The World Wide Web, commonly called the Web, is a system of interlinked resources that users can access over the internet through web browsers.

Web pages may contain text, images, audio, video, hyperlinks, and interactive features. Related pages are often organized into websites.

The Web uses technologies such as HTTP and HTTPS to support communication between browsers and web servers.

The internet and the Web are related but not identical. The internet is the underlying network infrastructure, while the Web is one of the services that operates over it. Email and other internet services can function independently of the Web.

Example

When a user opens a website through a browser, the browser requests the required web resources from a server and displays the returned content.

Understanding this distinction helps users learn how websites, browsers, servers, and internet connections work together.

Topic 318: Understanding Web Browsers

A web browser is software used to access and interact with websites and other web resources. It requests content from web servers and displays supported information in a readable, interactive form.

Examples of browsers include Microsoft Edge, Google Chrome, Mozilla Firefox, and Apple Safari.

Common Browser Features

- Address bar for entering website addresses and searches.
- Tabs for opening multiple pages.
- Back and Forward buttons for navigating browsing history.
- Bookmarks or favorites for saving useful pages.
- Downloads interface for managing downloaded files.
- Privacy and security controls.
- Extensions that add supported functions.

Browsers differ in design, compatibility, privacy features, and resource usage. Users should keep their chosen browser updated and install extensions only from trusted sources.

A browser is not the same as a search engine. A browser is the application used to access the Web, while a search engine is an online service that helps users find information.

Topic 319: Understanding Website Addresses and URLs

A Uniform Resource Locator (URL) is an address used to identify and access a resource on a network, commonly on the Web.

Consider this example:

https://www.example.com/products/item.html

The URL contains several parts:

- **Scheme:** https specifies the protocol used to access the resource.
- **Host:** www.example.com identifies the host name.
- **Path:** /products/item.html identifies a resource path on the server.

Some URLs also contain a port, query parameters, or a fragment identifier.

For example, a query string may provide information to a website, while a fragment may point to a specific section of a page.

Users should check website addresses carefully before entering sensitive information. A URL that looks similar to a familiar address may belong to a different domain.

HTTPS protects data in transit between a browser and the server when properly configured, but it does not guarantee that a website's content or business practices are trustworthy.

Topic 320: Understanding Search Engines

A search engine is an online service that helps users find information from websites and other indexed resources. Examples include Google Search, Bing, and other services that support web searching.

Search engines commonly use three major processes:

1. Crawling: Automated systems discover pages by following links or using other permitted discovery methods.

2. Indexing: The search engine analyzes discovered content and stores information about eligible pages in an index.

3. Ranking and Retrieval: When a user submits a query, the search engine identifies relevant indexed results and orders them according to its ranking systems.

Not every page is indexed, and search results can differ depending on the query, language, location, freshness, and other factors.

Tips for Better Searching

- Use specific keywords.
- Add important details to narrow the results.
- Put an exact phrase in quotation marks when appropriate.

- Check the publication date of time-sensitive information.
- Compare information from multiple reliable sources.
- Review the website's author, evidence, and purpose.

Search results are not automatically accurate or neutral. Users should evaluate the quality of the information before relying on it for academic, professional, or personal decisions.

Topic 321: Understanding Browser Tabs and Windows

Browser tabs allow users to open multiple web pages within a single browser window. Each tab can display a different website or page, making it easier to switch between tasks.

A browser window is a separate browser interface that can contain one or more tabs.

Common Tab Operations

- Open a new tab using the browser's new-tab button.
- On many browsers, press **Ctrl + T** to open a new tab.
- Press **Ctrl + W** to close the current tab.
- Use **Ctrl + Tab** to move to the next tab in many browsers.
- Open a new browser window when separate windows are more convenient.

The exact shortcuts may vary by browser and operating system.

Tabs are useful for comparing websites, researching information, and working with several online services. However, keeping too many tabs open can consume memory and make it difficult to find the required page. Users should close unnecessary tabs and avoid entering sensitive information into websites they do not recognize.

Topic 322: Using Bookmarks and Favorites

Bookmarks, also called favorites in some browsers, allow users to save website addresses for quick access later.

They are useful for storing educational resources, frequently visited websites, online tools, and reference pages.

How to Bookmark a Website

1. Open the required website.
2. Locate the browser's bookmark or favorites control.
3. Select the option to add the page.
4. Enter a suitable name if requested.
5. Choose a folder when available.
6. Save the bookmark.

Many browsers provide a shortcut such as **Ctrl + D** to add the current page to bookmarks.

Users can organize bookmarks into folders, rename them, and remove outdated entries.

A bookmark saves a website address, not necessarily a permanent copy of the page. If the website is removed or its address changes, the bookmark may no longer work.

Topic 323: Understanding Browser History

Browser history is a record of websites and pages visited through a browser. It helps users return to previously viewed information without searching for it again.

History features differ among browsers and may depend on privacy settings, synchronization, and browsing mode.

How to View Browser History

1. Open the browser.
2. Open its History menu or page.
3. Search for a previously visited website when the option is available.
4. Select an entry to revisit the page.

In many desktop browsers, **Ctrl + H** opens the history interface.

Users may be able to remove individual entries or clear selected periods of history. However, deleting local browser history does not necessarily remove records held by websites, internet services, or other systems.

On a shared computer, users should understand what browser history reveals about their activities and follow any applicable organizational policies.

Topic 324: Downloading Files from the Internet

Downloading is the process of transferring data from a remote system to a local device. Users commonly download documents, images, software installers, audio files, and other resources.

Downloaded files are often saved in the Downloads folder unless the browser or user selects another location.

How to Download a File

1. Open a trusted website.
2. Locate the intended download.
3. Select the download link or button.
4. Choose a destination if the browser asks.
5. Wait for the transfer to complete.
6. Open the Downloads interface or destination folder to locate the file.

Before opening a downloaded file, check its name, file type, and source. Scan suspicious files with appropriate security software and avoid running unexpected installers.

A successful download does not guarantee that a file is safe, authentic, or complete. For important software, obtain the installer from the developer's official website or a trusted distribution platform.

Topic 325: Managing Downloaded Files

Downloaded files may accumulate over time and consume storage space. Organizing them makes important documents easier to locate and reduces clutter.

Good Download Management Practices

- Review the Downloads folder regularly.
- Move important documents into suitable folders.
- Rename files using clear, descriptive names.
- Delete files that are no longer needed.
- Keep important documents backed up.
- Verify unfamiliar files before opening them.

For example, a student may create separate folders for assignments, study materials, images, and software installers.

File extensions can help identify formats, such as .pdf, .docx, and .zip, but extensions alone do not prove that a file is safe or genuine.

Users should not change a file's extension merely to make it appear to be a different file type. Renaming an extension does not convert the file's actual format.

Topic 326: Understanding Browser Cache

A browser cache stores copies of certain web resources, such as images, stylesheets, and other content, so that pages may load more efficiently when revisited.

When a user opens a website, the browser may reuse cached resources rather than downloading every item again.

Caching can improve performance and reduce repeated data transfers. However, an outdated cached resource can sometimes contribute to display problems after a website changes.

When Clearing the Cache May Help

- A website displays an outdated design.
- A page does not load correctly after an update.
- Troubleshooting instructions specifically recommend clearing cached files.
- The user needs to remove locally stored cached content.

The procedure varies by browser. Users should review the available clearing options before confirming.

Clearing the cache may temporarily make some websites load more slowly because resources must be downloaded again. It is not necessary to clear the cache constantly as a routine security measure.

Topic 327: Understanding Browser Cookies

Cookies are small pieces of data that websites store through a browser. They help websites remember information between requests or visits.

Common uses include maintaining sign-in sessions, remembering preferences, and supporting certain website functions.

Common Cookie Categories

- **Session Cookies:** Usually intended to remain available during a browsing session and may expire when the session ends.
- **Persistent Cookies:** Remain until their expiration time or until they are deleted.
- **First-Party Cookies:** Associated with the website the user is visiting.
- **Third-Party Cookies:** Associated with a different domain in contexts where the browser permits them.

Cookie behavior depends on the website, browser, and privacy settings.
Cookies are not the same as computer viruses, but some can be used for tracking or identifying browser activity. Users can review cookie settings and remove cookies when appropriate.
Deleting cookies may sign users out of websites or remove saved preferences. It does not necessarily delete information already stored on the website's servers.

Topic 328: Managing Browser Privacy Settings
Browser privacy settings help users control aspects of how websites access information, store data, and track activity.
Available controls may include cookie restrictions, tracking protection, site permissions, browsing-data deletion, and privacy-related defaults.
How to Review Privacy Settings

1. Open the browser.
2. Open its Settings menu.
3. Locate the Privacy and Security section or equivalent.
4. Review the available controls.
5. Adjust settings according to your needs.
6. Check whether the changes affect websites you use.

Strict privacy settings may improve protection but can also affect sign-ins, embedded content, or other website functions.
Private browsing does not make a user completely anonymous. Websites, network administrators, and service providers may still observe certain activity, depending on the circumstances.
Users should avoid entering sensitive information on unfamiliar websites and should understand the limits of browser privacy features.

Topic 329: Understanding Private Browsing Mode
Private browsing is a browser mode designed to limit certain information saved locally after a private session ends. Depending on the browser, it may be called Incognito, InPrivate, or Private Browsing.
Private browsing generally limits the retention of local browsing history and certain session data. Its exact behavior varies by browser.
How to Open Private Browsing

1. Open the browser.
2. Open its main menu.
3. Select the private or incognito browsing option.
4. Review the information displayed about the mode.
5. Browse only as appropriate for the device and network being used.
6. Close the private windows when finished.

Private browsing does not necessarily hide activity from websites, employers, schools, network administrators, or internet service providers.
Downloaded files and bookmarks may remain on the device after the private session ends. Users should not treat private browsing as a substitute for secure accounts, encrypted connections, or responsible online behavior.

Topic 330: Understanding Browser Extensions
A browser extension is a small software component that adds features to a web browser. Extensions may support password management, accessibility, translation, productivity, content organization, and other tasks.
Extensions can be useful, but some request access to sensitive information or browsing activity.
Installing an Extension Safely

1. Open the browser's official extension store or trusted distribution channel.
2. Search for the required extension.
3. Check the developer and available information.
4. Review the permissions requested.
5. Install it only if the permissions are appropriate.
6. Review the extension's settings after installation.

Avoid installing several extensions that perform the same task without a clear reason.
An extension that can read or modify website content may have access to information displayed in the browser. Remove extensions that are unnecessary, untrusted, or no longer maintained.
Extensions should be updated through trusted channels and reviewed periodically.

Topic 331: Understanding HTTPS and Secure Websites
HTTPS stands for Hypertext Transfer Protocol Secure. It uses Transport Layer Security (TLS) to help protect communication between a browser and a website.
When properly configured, HTTPS helps protect data in transit against certain forms of interception and unauthorized modification.
Browsers commonly indicate an HTTPS connection through the address bar, although the interface varies.
Important Security Points
- Check the website address before entering sensitive information.
- Avoid clicking unexpected links in messages.
- Be careful with misspelled or misleading domain names.
- Do not assume that HTTPS guarantees honest website content.
- Follow the browser's security warnings.

A fraudulent website can use HTTPS too. The encrypted connection protects communication with the website, but it does not prove that the website belongs to the organization it claims to represent.
For sensitive activities, use the correct official website and verify the domain carefully.

Topic 332: Identifying Suspicious Websites and Links
Suspicious websites and links may attempt to steal passwords, distribute malicious software, collect personal information, or mislead users into making unwanted payments.
Such websites may imitate banks, online stores, educational institutions, or popular services.
Warning Signs
- A domain name that resembles a legitimate website but contains unusual spelling.
- Unexpected requests for passwords or verification codes.
- Urgent threats demanding immediate action.
- Unfamiliar download prompts.
- Offers that appear unrealistic.
- Poorly explained requests for sensitive personal information.

These signs do not prove that a website is fraudulent, but they justify caution.
Safer Practices
1. Verify the domain before signing in.
2. Navigate to important services through a known official address.
3. Avoid unexpected attachments and downloads.
4. Never share authentication codes with someone who contacts you unexpectedly.
5. Report suspicious messages through the appropriate service.

If a website seems suspicious, leave it rather than attempting to test whether it is safe.

Topic 333: Understanding Search Operators
Search operators are special words or symbols that can help refine search engine queries. They allow users to narrow results and find information more efficiently.
Common examples supported by many search engines include:
- **Quotation marks:** Search for an exact phrase, such as "computer fundamentals".
- **Minus sign:** Exclude a term, such as jaguar -car.
- **site:** Restrict results to a particular website or domain, such as site:example.com computer.
- **filetype:** Search for a particular file format where supported, such as filetype:pdf computer basics.

Search engine behavior and supported operators can change over time.
Search operators help narrow results, but they do not guarantee that the results are accurate, complete, or safe.
When researching technical topics, users should compare reliable sources and confirm that the information matches the relevant software version, date, and context.

Topic 334: Evaluating Online Information
Online information varies in accuracy, quality, completeness, and purpose. A website may publish educational content, news, personal opinions, advertisements, or misleading claims.
Users should evaluate information before using it in assignments, professional work, or important decisions.
A Practical Evaluation Method
- **Author:** Is the author identified and suitably qualified?
- **Evidence:** Are claims supported by reliable information?
- **Date:** Is the content current enough for the topic?
- **Purpose:** Is the content intended to inform, persuade, entertain, or sell?
- **Source:** Does the publisher have a credible record?
- **Verification:** Can important claims be confirmed through independent sources?

For example, instructions for an older Windows release may not apply to the current interface. Checking the version and publication date can prevent confusion.
Search ranking, professional design, and a large number of views do not automatically establish credibility.
Careful evaluation is an essential digital literacy skill.

Topic 335: Understanding Cloud-Based Web Applications
A cloud-based web application provides functions through online infrastructure. Users commonly access it through a browser, while some services also provide dedicated applications.
Cloud services may store data on remote servers and process requests using computing resources managed by the service provider.
Examples include online document editors, web-based email, and cloud project-management systems.
Potential Benefits
- Access from compatible devices.
- Easier collaboration between authorized users.
- Centralized service updates.
- Reduced need to install traditional desktop software.
- Flexible access to online resources.

Important Considerations
- Internet connectivity may be required.
- Access depends on the service and account configuration.
- Storage and usage limits may apply.
- Privacy depends partly on the provider's policies and technical controls.
- Service outages can interrupt access.

Users should review account security, sharing permissions, and backup options before storing important information in a cloud application.

Topic 336: Uploading Files to Websites
Uploading is the process of transferring data from a local device to a remote system. Users upload files when submitting assignments, attaching documents to forms, adding images to websites, or sharing files through online services.
General Upload Procedure
1. Open the intended website or application.
2. Sign in if required.
3. Locate the Upload, Attach, or Choose File option.
4. Select the required file from the computer.
5. Confirm the selection.
6. Wait for the transfer to complete.
7. Verify that the website shows the correct file.

Websites may impose restrictions on file type, file size, and the number of files allowed.
Before uploading, check that the file is correct and does not contain unnecessary personal information.
Upload sensitive documents only to trusted services that have a legitimate need for them.
Do not close the browser immediately if the transfer is still in progress. Confirm completion before leaving the page.

Topic 337: Understanding Online Forms
An online form is a digital interface used to collect information through a website or application. Forms may be used for registrations, applications, surveys, bookings, feedback, and other tasks.
Common form elements include:

- **Text Fields:** Accept short text entries.
- **Text Areas:** Accept longer responses.
- **Radio Buttons:** Allow one choice from a group.
- **Checkboxes:** Allow one or more selections.
- **Drop-Down Lists:** Provide a list of options.
- **File Upload Controls:** Allow supported files to be attached.
- **Submit Buttons:** Send the completed information.

Completing a Form Carefully

1. Read the instructions.
2. Enter accurate information.
3. Review required fields.
4. Check spelling and dates.
5. Confirm the selected options.
6. Review attachments.
7. Submit the form and save any confirmation details provided.

Before submitting sensitive information, verify the website and understand why the information is being requested.
Avoid entering unnecessary personal details into forms whose purpose or ownership is unclear.

Topic 338: Understanding Online Accounts and Registration
Online registration is the process of creating an account with a website or digital service. The account may provide access to personalized features, saved information, subscriptions, or other services.
Registration commonly involves providing an email address or another identifier and creating sign-in credentials. Some services may require additional verification.
General Registration Process

1. Open the service's official website or application.
2. Locate the Sign Up or Create Account option.
3. Enter the requested information.
4. Create a strong, unique password if required.
5. Review the service's terms and privacy information.
6. Complete any required verification.
7. Store recovery information securely.

Users should provide only the information necessary for a legitimate purpose and avoid registering through suspicious links.
An account should not be considered secure merely because registration is complete. Users should protect their credentials, review security settings, and use additional authentication when available.

Topic 339: Understanding Online Communication Tools
Online communication tools allow people to exchange information through internet-connected devices. These tools support personal communication, education, collaboration, and business activities.
Common types include:

- **Email:** Supports written messages and file attachments.
- **Instant Messaging:** Supports real-time text communication and media sharing.
- **Video Conferencing:** Supports live audio and video meetings.
- **Discussion Forums:** Organize conversations around topics.

- **Collaboration Platforms:** Support shared documents, tasks, and team communication.

Each tool has different capabilities, privacy settings, and limitations.
Users should choose the appropriate tool for the purpose and audience. Sensitive information should be shared only through suitable channels with authorized recipients.
Good online communication includes clear language, respectful behavior, careful handling of attachments, and awareness that digital messages may be stored or forwarded.

Topic 340: Practising Safe and Responsible Internet Use

Safe and responsible internet use combines security awareness, privacy protection, critical thinking, and respectful online behavior.
Users should understand that information shared online may be copied, stored, forwarded, or viewed by people beyond the intended audience.

Essential Practices

1. Use strong, unique passwords.
2. Enable multifactor authentication where available.
3. Keep browsers and applications updated.
4. Verify websites before entering sensitive information.
5. Avoid suspicious downloads and attachments.
6. Review application permissions.
7. Protect personal and financial information.
8. Respect copyright and licensing conditions.
9. Communicate respectfully with others.
10. Report suspicious activity through appropriate channels.

Users should also evaluate online claims carefully rather than accepting every result or message as accurate.
Responsible internet use helps protect devices, accounts, personal information, and other people. Digital safety is an ongoing practice rather than a single setting or application.

Chapter 20: Computer Networking and Connectivity

Topic 341: Introduction to Computer Networks

A computer network is a group of interconnected computers and other devices that communicate and share resources. Networks allow users to exchange information, access shared files, use network printers, and connect to online services.

A network may contain computers, smartphones, printers, servers, switches, routers, and wireless access points.

Main Purposes of Networking

- Sharing files and information.
- Sharing hardware resources, such as printers.
- Communicating through messaging and video calls.
- Accessing internet services.
- Centralizing data and user management.
- Supporting collaboration between individuals and organizations.

For example, a school may connect office computers, teachers' computers, printers, and a server through a local network.

Networks can differ in size, design, transmission technology, and security requirements. The most suitable network depends on the number of users, required services, available resources, and the environment in which it operates.

Topic 342: Types of Computer Networks

Computer networks are commonly classified according to their geographical coverage and intended use.

1. Personal Area Network (PAN)

A PAN connects devices within a small personal area. For example, a smartphone may connect to wireless headphones through Bluetooth.

2. Local Area Network (LAN)

A LAN connects devices within a limited area, such as a home, office, computer laboratory, or school building.

3. Wireless Local Area Network (WLAN)

A WLAN is a local network that uses wireless communication, commonly Wi-Fi, to connect devices.

4. Metropolitan Area Network (MAN)

A MAN connects networks across a city or a large urban area.

5. Wide Area Network (WAN)
A WAN connects networks across large geographical distances. It may use communication infrastructure operated by different service providers.
The internet is a global system of interconnected networks. It is not simply one large local network; it consists of many networks communicating through shared protocols.

Topic 343: Network Topologies
Network topology describes how devices and connections are arranged within a network. The arrangement may be physical, logical, or both.
Common Network Topologies

- **Star Topology:** Devices connect to a central switch or access point. A failure in one individual connection usually affects that device, while failure of the central device can disrupt much of the network.
- **Bus Topology:** Devices share a common communication line. This arrangement is associated with certain older network designs.
- **Ring Topology:** Devices form a circular communication arrangement.
- **Mesh Topology:** Devices have multiple interconnections, providing alternative communication paths.
- **Tree Topology:** Multiple network segments are organized in a hierarchical structure.

Star topology is common in modern wired local networks because it is relatively easy to manage and troubleshoot.
The best topology depends on cost, scalability, reliability, physical layout, and the network's communication requirements.

Topic 344: Wired and Wireless Networks
Networks can connect devices through physical cables or wireless communication.
Wired Networks
Wired networks commonly use Ethernet cables to connect computers, switches, routers, and other compatible equipment.
Advantages include predictable performance, stable connections, and reduced exposure to wireless interference.
Limitations include cable installation costs, physical restrictions, and the need to manage cables.
Wireless Networks
Wireless networks use radio communication technologies such as Wi-Fi and Bluetooth.
Advantages include mobility, convenient access, and reduced dependence on physical cables.
Limitations may include interference, signal range restrictions, congestion, and security risks if wireless settings are poorly configured.
Many homes and offices use both approaches. Desktop computers and servers may use Ethernet, while phones and laptops connect through Wi-Fi.
The appropriate connection depends on device compatibility, performance needs, security requirements, and the physical environment.

Topic 345: Understanding Network Devices
Network devices help connect systems, direct traffic, and enable communication between devices.
Common Network Devices

- **Network Interface Card (NIC):** Provides a device with a network connection.
- **Switch:** Connects devices within a local network and forwards Ethernet frames using learned hardware addresses.
- **Router:** Forwards packets between different IP networks.
- **Wireless Access Point:** Allows compatible wireless devices to connect to a network.
- **Modem or Network Terminal:** Provides an interface to a communication service, depending on the technology used.
- **Firewall:** Applies rules to permit or block network traffic.
- **Repeater or Extender:** Helps extend a signal or connection in supported network designs.

A home internet device may combine routing, Wi-Fi access, firewall functions, and other features in one unit.
Although these devices may be combined physically, their functions are different. Understanding each role helps users diagnose connectivity problems and plan networks more effectively.

Topic 346: Understanding Ethernet
Ethernet is a family of networking technologies widely used in wired local area networks. It defines communication methods and frame formats that allow connected devices to exchange data.
Ethernet connections commonly use twisted-pair cables with compatible network ports. Fiber-optic Ethernet is also used where distance, speed, or environmental conditions require it.
Important Ethernet Concepts
- **Ethernet Cable:** Carries network signals between compatible devices.
- **Network Port:** Provides a physical connection for supported Ethernet equipment.
- **Link Status:** Indicates whether a physical network link has been established.
- **Network Speed:** Depends on the capabilities of connected devices and the quality and type of connection.
- **Switching:** Allows devices within a local network to communicate through switches.

A cable that fits a port does not guarantee that the connection will operate at the expected speed. Cable quality, port capabilities, configuration, and equipment condition all matter.
Ethernet is often preferred for desktop computers, network printers, and other devices that benefit from stable wired connectivity.

Topic 347: Understanding Wi-Fi
Wi-Fi is a family of wireless networking technologies based on IEEE 802.11 standards. It allows compatible devices to communicate with a wireless access point or router without requiring a separate Ethernet cable for each device.
Wi-Fi is commonly used by laptops, smartphones, tablets, smart televisions, and other connected equipment.
Important Wi-Fi Concepts
- **SSID:** The name used to identify a wireless network.
- **Password or Security Credential:** Helps control access to a protected network.
- **Signal Strength:** Indicates the strength of the received wireless signal.
- **Frequency Band:** Common Wi-Fi bands include 2.4 GHz, 5 GHz, and, on supported equipment, 6 GHz.
- **Interference:** Other devices, walls, distance, and competing networks may affect performance.

A strong signal does not necessarily guarantee a fast internet connection. Performance also depends on congestion, router capabilities, internet service quality, and the device being used.
Users should protect their home Wi-Fi with modern supported security settings, a strong password, and updated router firmware.

Topic 348: Connecting a Computer to a Network
A computer may connect to a network through Ethernet, Wi-Fi, or another supported connection method. The exact procedure depends on the operating system, network equipment, and security configuration.
Connecting Through Wi-Fi
1. Open the computer's network settings.
2. Enable Wi-Fi if it is disabled.
3. View the available wireless networks.
4. Select the correct network name.
5. Enter the required password when prompted.
6. Wait for the connection status to update.
7. Open a trusted website to test internet access.

Connecting Through Ethernet
1. Connect a compatible Ethernet cable to the computer's network port.
2. Connect the other end to an active router or switch port.
3. Wait for the operating system to detect the connection.
4. Check the network status.

5. Test access to the required network resources.

Some networks require additional configuration, such as a proxy, organizational sign-in, or administrator-provided settings.
Users should connect only to networks they are authorized to use.

Topic 349: Understanding IP Addresses

An Internet Protocol (IP) address identifies a network interface for communication using the Internet Protocol. IP addresses help devices send and receive packets across networks.
There are two major IP versions.

IPv4

IPv4 addresses use 32 bits and are commonly written as four decimal numbers separated by periods.
Example: 192.168.1.10

IPv6

IPv6 addresses use 128 bits and are written in hexadecimal groups separated by colons.
Example: 2001:db8::10
The IPv6 example uses a documentation prefix intended for examples rather than ordinary public addressing.
IP addresses may be assigned manually or automatically through a service such as DHCP. Private addresses are commonly used within local networks, while public addresses are used for communication across the public internet.
An IP address should not be confused with a MAC address. An IP address supports communication at the network layer, while a MAC address is used for link-layer communication on relevant network technologies.

Topic 350: Understanding DHCP and DNS

DHCP and DNS are two important services used in computer networks.

Dynamic Host Configuration Protocol (DHCP)

DHCP can automatically provide network configuration information to devices. Depending on the configuration, this may include an IP address, subnet mask or prefix information, default gateway, and DNS server addresses.
Without appropriate configuration, a device may be unable to communicate correctly with the intended network.

Domain Name System (DNS)

DNS translates domain names into information used by internet services, including IP addresses.
For example, a user can enter a website's domain name into a browser instead of memorizing the server's IP address.

How They Differ

- DHCP helps devices obtain network configuration.
- DNS helps devices find information associated with domain names.
- Both services may be provided by a router, an organizational server, or other network infrastructure.

If DNS is unavailable or misconfigured, a computer may have a working network connection but still fail to open websites by their domain names.

Topic 351: Understanding Internet Bandwidth and Latency

Internet performance depends on several factors, including bandwidth, latency, packet loss, network congestion, and the quality of the connection.

Bandwidth

Bandwidth describes the data-carrying capacity of a connection. Internet service plans commonly express download and upload speeds in megabits per second (Mbps).

Latency

Latency is the time required for data to travel between endpoints. It is often measured in milliseconds (ms), commonly through a round-trip measurement such as a ping test.

Packet Loss

Packet loss occurs when packets fail to reach their intended destination. Significant packet loss may interrupt calls, gaming, streaming, and other online activities.

Throughput

Throughput is the actual rate at which useful data is transferred over a connection.

A connection with high bandwidth can still feel slow if latency or packet loss is high. Similarly, a slow website may be caused by the website's server or browser rather than the user's internet connection. Understanding these measurements helps users evaluate network performance more accurately.

Topic 352: Sharing Files and Printers on a Network

Network sharing allows authorized users to access resources located on other devices. Common examples include shared folders, documents, and printers.

In a school office, several computers may use a shared network printer rather than maintaining separate printers at every desk.

Basic Steps for Network Sharing

1. Confirm that the devices are connected to the appropriate network.
2. Identify the folder or printer that needs to be shared.
3. Enable the relevant sharing feature where required.
4. Set suitable user permissions.
5. Locate the shared resource from an authorized device.
6. Test access and verify that only intended users can use it.

The available steps differ between operating systems and network configurations.

Sharing should be limited to the resources that other users genuinely need. Avoid granting broad access to an entire drive or sensitive folder.

For business and educational environments, administrators should apply access controls and follow the organization's data protection policies.

Topic 353: Understanding Network Security

Network security involves protecting network devices, communications, and connected systems against unauthorized access, misuse, disruption, and data theft.

Both technical controls and responsible user behavior are important.

Common Network Security Measures

- Use strong, unique administrative passwords.
- Enable modern wireless security supported by the equipment.
- Install firmware and security updates.
- Use firewalls and suitable access controls.
- Disable unnecessary services and unused remote-access features.
- Separate guest devices from sensitive systems where appropriate.
- Monitor suspicious activity.
- Maintain backups of important data.

Organizations may also use network segmentation, centralized authentication, intrusion detection, and other advanced controls.

A network can remain vulnerable even when it has a password. Outdated devices, weak administrative credentials, insecure services, and incorrectly configured permissions can create additional risks.

Security settings should be reviewed regularly, particularly when devices are added, removed, or reassigned.

Topic 354: Understanding Firewalls

A firewall is a security control that monitors and filters network traffic according to defined rules. Firewalls may be implemented in software, hardware, or a combination of both.

They help reduce unauthorized communication between networks and devices.

Common Firewall Types

- **Host-Based Firewall:** Runs on an individual computer or device.
- **Network Firewall:** Protects or separates network segments.
- **Application-Aware Firewall:** Can apply rules based on application-related information, depending on its capabilities.

A firewall rule may permit or block traffic based on factors such as source, destination, protocol, port, or connection state.

For example, a firewall may block unsolicited incoming connections while allowing a computer to access permitted websites.

A firewall does not eliminate every security risk. It may not prevent users from voluntarily submitting information to a fraudulent website, and it cannot replace software updates, secure authentication, or reliable backups.
Users should not disable firewall protection merely because an application displays an unfamiliar network message.

Topic 355: Troubleshooting Network Problems
Network troubleshooting is the process of identifying and resolving problems that prevent devices from communicating or accessing online services.
Common symptoms include missing Wi-Fi connections, limited connectivity, slow downloads, interrupted video calls, and websites that fail to load.
A Systematic Troubleshooting Process
1. Identify the exact problem and when it began.
2. Check whether the issue affects one device or several.
3. Confirm that Wi-Fi or Ethernet is connected.
4. Check the router, access point, and relevant cables.
5. Restart affected equipment when appropriate.
6. Review the computer's network settings.
7. Test another trusted website or network service.
8. Check whether the internet provider reports an outage.
9. Record any error messages.
10. Contact the network administrator or service provider if necessary.

If other devices can access the internet but one computer cannot, the issue may be specific to that computer or its configuration.
If several devices lose connectivity simultaneously, the router, network infrastructure, or internet service may be responsible.
Change advanced network settings only when you understand the purpose of the change or have received appropriate instructions.

Chapter 21: Email and Digital Communication

Topic 356: Introduction to Email
Email, short for electronic mail, is a digital communication service that allows users to exchange messages through computer networks.
Email is widely used in education, business, customer support, personal communication, and official correspondence.
An email message can contain text, attachments, links, and other supported content.
Main Components of Email Communication

- **Sender:** The person or system sending the message.
- **Recipient:** The intended person or group receiving it.
- **Subject:** A short description of the message's purpose.
- **Message Body:** The main content.
- **Attachment:** A file included with the message.
- **Timestamp:** Information indicating when the message was sent or received.

Email services may be accessed through a web browser, desktop application, or mobile application.
Email is asynchronous, meaning the sender and recipient do not need to be online at the same time. The recipient can read and respond when convenient.

Topic 357: Understanding Email Addresses
An email address identifies a mailbox or destination used by an email service.
A typical email address contains a local part, the @ symbol, and a domain name.
Example:
student@example.com
In this example:

- student is the local part.
- @ separates the local part from the domain.
- example.com is the domain.

The domain is associated with the email service or organization responsible for receiving messages for that address.

Email addresses can use personal domains, educational domains, or business domains.
Users should enter addresses carefully because a small spelling error may send a message to the wrong destination or cause delivery failure.
Before sending confidential information, verify the recipient's complete address rather than relying only on a display name.

Topic 358: Creating and Managing an Email Account
An email account allows a user to send, receive, and organize messages through an email service.
The registration procedure depends on the provider, but it commonly involves choosing an available address, setting authentication credentials, and completing any required verification.
General Account Creation Process
1. Open the official website or application of the email provider.
2. Select the account creation option.
3. Enter the requested information.
4. Choose an available email address.
5. Create a strong, unique password.
6. Complete the provider's required verification steps.
7. Review recovery and security settings.

After creating the account, users should keep recovery information current and enable multifactor authentication when available.
Account Management Tasks
- Update account security settings.
- Review recovery methods.
- Organize messages into folders or labels.
- Check storage usage.
- Remove unnecessary messages and attachments.
- Sign out of shared devices when finished.

Users should never share their passwords or verification codes with someone who requests them unexpectedly.

Topic 359: Understanding the Email Inbox
The inbox is the main area in an email service where incoming messages are displayed. Its layout varies between providers and applications.
Common inbox elements include the sender's name, subject line, message date, and an indicator showing whether a message has been read.
Common Email Folders and Categories
- **Inbox:** Contains incoming messages that have not been moved elsewhere.
- **Sent:** Stores copies of messages sent by the account.
- **Drafts:** Contains messages saved before sending.
- **Spam or Junk:** Holds messages classified as potentially unwanted.
- **Trash or Deleted Items:** Holds messages marked for deletion, often temporarily.
- **Archive:** Removes messages from the main inbox without necessarily deleting them.

Some services use labels or categories rather than traditional folders.
Users can search for messages by sender, subject, date, or keywords where supported.
Organizing email regularly helps users locate important correspondence and distinguish current tasks from completed conversations.

Topic 360: Composing and Sending an Email
Composing an email means preparing a message for delivery to one or more recipients.
A well-written email should have a clear purpose, an appropriate tone, and enough information for the recipient to understand the request or message.
Main Fields in an Email
- **To:** The primary recipient or recipients.
- **CC:** People who receive a copy and whose addresses are generally visible to other recipients.
- **BCC:** Recipients whose addresses are hidden from other recipients in the usual message display.

- **Subject:** A concise summary of the message.
- **Body:** The main message.
- **Attachment:** A file included when needed.

Steps for Sending an Email

1. Select Compose or New Message.
2. Enter and verify the recipient's address.
3. Add a clear subject.
4. Write the message in the body.
5. Attach files if required.
6. Review the message and recipients.
7. Select Send.

Before sending, check that the correct files are attached and that confidential information is being sent only to authorized recipients.
Use BCC when appropriate for protecting a group of recipients' email addresses, while following organizational policies and the context of the communication.

Topic 361: Email Etiquette and Professional Communication

Email etiquette refers to the guidelines for communicating clearly, respectfully, and professionally through email.
Professional emails are commonly used for job applications, academic communication, business inquiries, customer support, and official requests.

Essential Email Etiquette Guidelines

- Use a clear and meaningful subject line.
- Begin with an appropriate greeting.
- State the purpose of the email early.
- Use polite and professional language.
- Organize longer messages into short paragraphs.
- Check spelling and grammar before sending.
- Avoid unnecessary capital letters, which may appear aggressive.
- Use an appropriate closing.
- Review the recipient's address before sending.

For example, an email requesting information from a teacher should clearly explain what information is needed and use a respectful tone.
Users should avoid sending confidential information to unintended recipients. They should also remember that professional messages may be retained, forwarded, or used as records.
Good email etiquette helps prevent misunderstandings and creates a positive impression.

Topic 362: Reply, Reply All, and Forward

Email services commonly provide three options for responding to messages: Reply, Reply All, and Forward.

Reply

Reply sends a response to the original sender. Other recipients are generally not included unless the user adds them.

Reply All

Reply All sends a response to the original sender and the other recipients included in the relevant message fields, subject to the email service's behavior.
Use this option only when the response is relevant to everyone receiving it.

Forward

Forward sends an existing message to another recipient. The forwarded content may include the original message and its attachments, depending on the service and the user's choices.

Responsible Use

- Check the recipient list before responding.
- Avoid Reply All when a private reply is sufficient.
- Remove unnecessary confidential information before forwarding.
- Verify attachments and links in forwarded messages.
- Do not assume that an email is safe simply because it came from a familiar contact.

Choosing the correct option helps protect privacy and keeps communication organized.

Topic 363: Sending and Receiving Email Attachments
An email attachment is a file sent with an email message. Attachments may include documents, spreadsheets, presentations, images, PDFs, and other supported file types.
How to Attach a File
1. Compose a new email or reply.
2. Select the attachment or paperclip option.
3. Choose the required file.
4. Wait for the file to finish attaching.
5. Confirm that the correct file is listed.
6. Complete and send the email.

Receiving Attachments
Recipients can usually select an attachment to preview or download it. The available options depend on the email service and file type.
Attachment Safety
- Open attachments only when their source and purpose are reasonably verified.
- Be especially cautious with unexpected executable files or documents requesting unusual actions.
- Scan suspicious downloads using suitable security software.
- Verify unexpected attachments with the sender through a separate trusted communication method.
- Avoid uploading confidential attachments to untrusted online services.

Email providers may limit attachment sizes. Large files may need to be shared through a trusted file-sharing service with suitable access permissions.

Topic 364: Understanding Email Spam and Junk Mail
Spam is unwanted or unsolicited electronic messaging, commonly delivered through email. It may contain advertisements, misleading offers, suspicious links, or malicious attachments.
Not every unwanted email is dangerous, but some spam messages attempt to steal information or persuade recipients to take unsafe actions.
Common Signs of Spam
- Unexpected promotional messages.
- Repeated messages from unfamiliar senders.
- Misleading subject lines.
- Unusual requests for personal information.
- Suspicious attachments or links.
- Offers that seem unrealistic.

Email services use filtering systems to identify and separate suspected spam from ordinary messages. These systems are not perfect, so legitimate emails may sometimes enter the spam folder, while some unwanted messages may reach the inbox.
Recommended Practices
1. Avoid opening suspicious attachments.
2. Do not respond to obvious spam.
3. Report unwanted messages using the service's spam controls.
4. Unsubscribe only from legitimate mailing lists when the sender is trustworthy.
5. Review the spam folder occasionally for incorrectly filtered messages.

Spam filters reduce unwanted messages but cannot guarantee that every suspicious email will be detected.

Topic 365: Recognizing Email Phishing
Phishing is a form of social engineering in which attackers impersonate trusted organizations or individuals to trick people into revealing information, opening malicious files, or performing unauthorized actions.
Phishing messages may imitate banks, educational institutions, employers, delivery services, or popular online platforms.
Common Warning Signs
- Unexpected requests to confirm passwords.
- Urgent threats involving account suspension.

- Requests for authentication codes.
- Links leading to unfamiliar domains.
- Unexpected payment requests.
- Attachments that require unusual actions.
- Messages that do not match the sender's normal communication style.

Some phishing messages are carefully written and may appear authentic. Spelling mistakes are not the only indicator.

How to Respond Safely

- Do not click suspicious links.
- Visit the official website through a known address.
- Never share passwords or verification codes in response to unexpected requests.
- Confirm unusual payment or account requests through a separate trusted channel.
- Report suspected phishing to the relevant service or organization.

If credentials have already been entered on a suspected phishing website, use a trusted device to change the affected password and review account security. Contact the relevant organization if financial or sensitive information may have been exposed.

Topic 366: Email Security and Account Protection

Email accounts often contain personal conversations, account recovery messages, business documents, and other sensitive information. Protecting an email account is therefore an important part of digital security.

Recommended Security Measures

1. Use a strong, unique password.
2. Enable multifactor authentication where available.
3. Keep recovery information accurate and secure.
4. Never share verification codes.
5. Review active sessions and signed-in devices.
6. Check forwarding rules and filters for unexpected changes.
7. Keep devices and email applications updated.
8. Avoid signing in through suspicious links.
9. Sign out of shared computers.
10. Review security alerts from the email provider.

Multifactor authentication adds another verification step beyond the password. Depending on the method, this may involve an authentication application, security key, or other supported factor.

Users should also protect the devices used to access email. A secure password alone cannot protect an account if an attacker gains control of an already authenticated device.

Topic 367: Managing Email Folders, Labels, and Filters

Email organization features help users manage large numbers of messages and locate important information efficiently.

Folders and Labels

Folders allow messages to be grouped by purpose. Labels provide a way to categorize messages, and some email services allow a message to have multiple labels.

Examples of useful categories include:

- Education
- Work
- Clients
- Receipts
- Important Documents
- Personal Messages

Email Filters and Rules

Filters automatically perform actions when messages match specified conditions. Depending on the service, a filter may apply a label, move a message, mark it as read, or direct it to a particular folder.

For example, a user may create a filter to organize messages from a specific educational institution.

Filters should be configured carefully. Incorrect rules may hide important messages, delete information, or redirect messages unexpectedly.

Review filters periodically and avoid creating rules that automatically forward confidential email to unknown addresses.

Topic 368: Email Signatures and Professional Identity
An email signature is a block of text automatically added to the end of an email. It commonly contains the sender's name and relevant contact or professional information.
A professional signature helps recipients identify the sender and understand how to make contact.
Common Signature Elements
- Full name or professional name.
- Job title or area of work.
- Organization, where appropriate.
- Professional email address.
- Website or portfolio link, when relevant.
- Other necessary contact details.

For example, a computer teacher may include a professional title and the educational institution's name when writing official correspondence.
A signature should be concise, accurate, and suitable for the audience. Decorative images, excessive quotations, and large blocks of promotional content may make messages harder to read.
Users should avoid including unnecessary personal details, private addresses, or information that could create security or privacy concerns.

Topic 369: Email Storage and Message Archiving
Email services allocate storage for messages, attachments, and sometimes other account data. The available storage depends on the provider and account plan.
Large attachments and accumulated messages can gradually consume available space.
Ways to Manage Email Storage
- Remove unnecessary messages and attachments.
- Review large emails when the service provides size-based search.
- Empty the trash when appropriate.
- Delete unwanted promotional messages.
- Download important files only when permitted and necessary.
- Archive messages that should be retained but do not need to remain in the inbox.

Archiving generally removes a message from the main inbox without permanently deleting it. The message may remain searchable and accessible through the archive or all-mail view.
Deleting a message may move it to a trash folder before permanent removal. Exact retention periods and recovery options vary by provider.
Important records should be retained according to applicable organizational, legal, and personal requirements. Users should not assume that an archived message is a separate backup.

Topic 370: Digital Communication Ethics
Digital communication ethics concerns the responsible and respectful use of email, messaging, forums, video meetings, and other online communication systems.
People should communicate honestly, respect privacy, and consider how their messages may affect others.
Key Principles
- Respect other people's privacy.
- Obtain permission before sharing private messages or photographs.
- Avoid harassment, threats, and abusive language.
- Do not impersonate another person.
- Respect copyright and intellectual property.
- Verify information before forwarding it.
- Use professional language in official settings.
- Avoid spreading rumors or unverified claims.
- Follow the rules of the relevant platform or organization.

Users should remember that digital messages can be copied, stored, or forwarded beyond their original audience.

Responsible communication protects relationships, supports collaboration, and contributes to a safer online environment.

Chapter 22: Messaging, Video Conferencing, and Digital Collaboration

Topic 371: Introduction to Instant Messaging

Instant messaging is a form of digital communication that allows users to exchange messages through an application or online service.

Unlike traditional email, instant messaging is designed primarily for rapid exchanges. However, messages may still be delivered or read later if a recipient is offline.

Common features include:

- One-to-one conversations.
- Group chats.
- Text messages.
- Image and document sharing.
- Voice messages.
- Audio and video calls.
- Message reactions and replies.
- Notifications and contact management.

Messaging tools are used in education, business, customer service, and personal communication.

Users should check the recipient before sending sensitive information. They should also review privacy settings and understand how the service stores messages and manages account data.

Topic 372: Managing Messaging Applications

Messaging applications provide interfaces for organizing conversations, managing contacts, and controlling communication settings.

Although individual applications differ, many include similar features.

Common Management Tasks

- Add or remove contacts.
- Start a new conversation.
- Create or leave a group.
- Mute notifications.
- Archive conversations.

- Block unwanted contacts.
- Report abusive or suspicious accounts.
- Review privacy and security settings.

Some applications synchronize messages across multiple devices. Others may provide different features depending on the device, account type, or subscription.
Before connecting a messaging account to another computer, review the authorization process and make sure the device is trusted.
On a shared device, sign out when finished and remove access if the application provides a device-management feature.
Users should not install unofficial modified versions of messaging applications because they may introduce security, privacy, or account risks.

Topic 373: Group Chats and Online Communities
A group chat allows multiple participants to communicate within a shared conversation. Group chats are commonly used for school classes, project teams, families, clubs, and workplaces.
Online communities may use group messaging, discussion boards, or dedicated collaboration platforms.
Good Group Communication Practices

- Use a clear and relevant group name.
- Keep messages related to the group's purpose.
- Respect members' time and privacy.
- Avoid repeatedly sending irrelevant messages.
- Share files only when appropriate.
- Confirm important instructions before acting on them.
- Use suitable administrative controls.
- Remove access when someone no longer has permission to participate.

Group membership does not mean that every participant should receive confidential information.
Administrators should understand the platform's controls for inviting members, assigning roles, moderating content, and removing access.
Users should also be cautious about joining unfamiliar groups that request personal information, payments, or unusual account permissions.

Topic 374: Voice and Video Calling
Voice calling transmits audio between participants, while video calling transmits video and usually audio over a communication network.
These technologies support remote meetings, online education, customer support, interviews, and personal communication.
Requirements for Good Call Quality

- A stable network connection.
- A working microphone.
- Suitable speakers or headphones.
- A compatible camera for video calls.
- Appropriate application permissions.
- A reasonably quiet environment.

Poor call quality may result from weak connectivity, network congestion, hardware problems, background applications, or incorrect audio settings.
Before an important meeting, test the microphone, speakers, and camera. Close unnecessary applications if the device is struggling to maintain performance.
Users should also check their surroundings and avoid showing private documents, personal information, or other sensitive content unintentionally.

Topic 375: Introduction to Video Conferencing
Video conferencing allows participants in different locations to communicate through live audio and video. It is widely used for online classes, business meetings, interviews, training sessions, and remote collaboration.
A video-conferencing application may provide:

- Meeting links.
- Participant lists.
- Microphone and camera controls.
- Screen sharing.
- Chat.
- Virtual backgrounds.
- Recording, where supported and authorized.
- Host and participant permissions.

The available features depend on the application, account type, and meeting settings.

Preparing for a Meeting

1. Check the meeting time and time zone.
2. Open the correct meeting link.
3. Test the microphone and camera.
4. Use a stable internet connection.
5. Join from a suitable environment.
6. Mute the microphone when not speaking, if appropriate.
7. Follow the host's instructions.

Meeting participants should respect organizational rules and avoid recording or distributing meeting content without the required permission.

Topic 376: Screen Sharing and Remote Presentations

Screen sharing allows a participant to show selected content from a computer or mobile device to other people during an online meeting.

It is useful for demonstrations, teaching, presentations, technical support, and collaborative problem-solving.

Depending on the application, users may be able to share an entire screen, a specific window, or a browser tab.

Safe Screen-Sharing Practices

- Close unnecessary windows.
- Turn off private notifications where appropriate.
- Check the displayed content before sharing.
- Share only the required screen or application.
- Avoid exposing passwords or confidential documents.
- Stop sharing when the demonstration is complete.

Sharing an entire screen may reveal information outside the intended presentation, including messages and personal files.

If a presenter needs to demonstrate software, sharing only the relevant application window can reduce accidental exposure.

Users should grant remote-control permissions only when necessary and only to trusted participants.

Topic 377: Online Collaboration Tools

Online collaboration tools allow multiple people to work together using internet-connected services.

These tools may support shared documents, spreadsheets, presentations, task lists, project boards, and team communication.

Common Collaboration Features

- Shared access to files.
- Real-time editing.
- Comments and suggestions.
- Version history.
- Task assignment.
- Notifications.
- Role-based permissions.
- Activity tracking.

For example, a group of students can work on a shared presentation while each member contributes content from a different location.

Collaboration tools can improve coordination, but users must manage permissions carefully. A link that allows unrestricted access may expose information to unintended viewers.
Before sharing a document, confirm who needs access and whether they should be allowed to view, comment, or edit it.
Important collaborative files should be protected against accidental deletion and retained according to the relevant project requirements.

Topic 378: Cloud File Sharing and Permissions
Cloud file sharing allows users to store files on a remote service and provide access to other people through invitations or links.
The available permissions vary by provider, but common options include viewing, commenting, and editing.
Common Permission Types
- **Viewer:** Can view the content but generally cannot change it.
- **Commenter:** Can add comments where supported.
- **Editor:** Can make changes to the content.

Some services also support download restrictions, expiration dates, or additional access controls.
Safer Sharing Practices
1. Select the correct file or folder.
2. Identify the intended recipients.
3. Choose the minimum permissions necessary.
4. Check whether the link is restricted or publicly accessible.
5. Send the invitation through an appropriate channel.
6. Review and remove access when it is no longer needed.

A link-based sharing setting may allow anyone possessing the link to access the content, depending on the service configuration.
Users should avoid sharing confidential documents through unrestricted links and should periodically review access to important files.

Topic 379: Managing Digital Calendars and Meeting Invitations
A digital calendar is an application or service used to organize events, appointments, deadlines, and reminders.
Calendars help individuals and teams coordinate schedules and reduce the likelihood of missing important activities.
Common Calendar Features
- Creating events.
- Setting start and end times.
- Adding locations or meeting links.
- Inviting participants.
- Setting reminders.
- Scheduling recurring events.
- Viewing daily, weekly, or monthly schedules.

When creating an event, verify the date, time, and time zone. Incorrect time-zone settings can cause meetings to appear at the wrong time for participants.
Meeting invitations should contain enough information for recipients to understand the purpose and prepare appropriately.
Users should also check calendar-sharing permissions. A shared calendar may reveal event titles, locations, or other information depending on its access settings.

Topic 380: Digital Collaboration Etiquette and Troubleshooting
Successful online collaboration requires clear communication, reliable technology, appropriate permissions, and respectful behavior.
Technical problems can interrupt work, while unclear instructions may cause confusion even when the software operates correctly.
Good Collaboration Practices
- Define the task and expected outcome.

- Use clear file names and organized folders.
- Assign responsibilities where appropriate.
- Respect deadlines and meeting schedules.
- Keep discussions relevant.
- Review changes before publishing shared work.
- Protect confidential information.
- Communicate technical problems promptly.

Common Problems and Solutions

- **Cannot Access a Shared File:** Verify the account being used and request the correct permission.
- **Changes Are Missing:** Check synchronization status, connection quality, and version history.
- **Meeting Audio Does Not Work:** Review microphone selection, application permissions, and system sound settings.
- **A Meeting Link Fails:** Confirm the link, meeting time, and required sign-in method.
- **Notifications Are Missing:** Review application and operating-system notification settings.

Before changing settings or deleting files, identify the problem carefully. For important collaborative work, preserve a backup or version history when available.

Effective collaboration combines technical knowledge with communication, accountability, and respect for other participants.

Chapter 23: Audio, Video, and Digital Media

Topic 381: Introduction to Digital Audio

Digital audio is the representation of sound in a digital format that computers and electronic devices can store, process, and reproduce.

Computers convert sound into digital data through a process called analog-to-digital conversion. During this process, an analog audio signal is sampled and represented numerically.

Two important concepts in digital audio are:

- **Sampling Rate:** The number of audio samples recorded per second, measured in hertz (Hz).
- **Bit Depth:** The number of bits used to represent each audio sample in supported audio formats.

For example, a sampling rate of 44.1 kHz means that 44,100 samples are taken per second for each channel in a standard audio recording.

Higher sampling rates and bit depths can increase the amount of data required, although the resulting quality also depends on the recording equipment, processing, and listening conditions.

Digital audio is used in music, voice recordings, online meetings, educational videos, podcasts, and multimedia presentations.

Topic 382: Common Audio File Formats

Audio file formats define how sound data is stored and encoded. Different formats offer different combinations of compatibility, file size, and audio quality.

Common Audio Formats

- **MP3:** A widely supported lossy compressed audio format.
- **WAV:** A container format often used for uncompressed PCM audio, although other encodings are possible.
- **AAC:** An audio coding format commonly used in digital media and streaming.
- **FLAC:** A lossless audio compression format that preserves the original encoded audio data.
- **OGG:** A multimedia container commonly associated with formats such as Vorbis and Opus.
- **M4A:** A filename extension commonly used for audio stored in an MPEG-4 container.

Lossy Compression removes some audio information to reduce file size.

Lossless Compression reduces file size without losing information from the encoded source data.

The best format depends on the intended use. MP3 may be suitable for convenient distribution, while FLAC may be preferred when preserving lossless audio is important.
File extensions help identify formats, but the actual encoding and compatibility should also be considered.

Topic 383: Microphones, Speakers, and Headphones
Microphones, speakers, and headphones are common audio devices used with computers.
Microphone
A microphone captures sound and converts it into an electrical signal that can be processed by a computer or recording device.
Microphones are used for voice calls, lectures, podcasts, voice commands, and audio recording.
Speakers
Speakers convert electrical audio signals into sound that people can hear. They may be built into a laptop, monitor, or external audio system.
Headphones
Headphones provide personal audio output. Headsets combine headphones with a microphone and are commonly used for online meetings and gaming.
Common Troubleshooting Steps
- Confirm that the correct input or output device is selected.
- Check volume and mute settings.
- Review application permissions.
- Inspect cables and wireless connections.
- Test the device in another compatible application.
- Update or troubleshoot audio drivers when necessary.

For online meetings, users should test their microphone and speakers before the session begins. Selecting the wrong audio device is a common reason for sound problems.
Topic 384: Understanding Digital Video
Digital video represents moving images as a sequence of frames stored and processed electronically. Video files may also contain audio, subtitles, and other information.
Important video concepts include:
- **Resolution:** The dimensions of the video image, commonly expressed in pixels.
- **Frame Rate:** The number of frames displayed or recorded per second, measured in frames per second (fps).
- **Aspect Ratio:** The proportional relationship between image width and height.
- **Bitrate:** The amount of data used per unit of time.
- **Codec:** A technology used to encode and decode audio or video data.

For example, a video with a resolution of 1920 × 1080 contains 1920 horizontal pixels and 1080 vertical pixels per frame.
A higher resolution can provide more image detail, but it may also require more storage and processing power.
Video quality depends on several factors, including resolution, frame rate, compression, source quality, and display capabilities.

Topic 385: Common Video File Formats
Video file formats determine how video, audio, subtitles, and related information are packaged and stored.
Common Video Formats
- **MP4:** A widely supported container commonly used for online videos and general-purpose playback.
- **MKV:** A flexible container that can hold multiple audio tracks, subtitles, and video streams.
- **AVI:** An older multimedia container still supported by some applications.
- **MOV:** A multimedia container associated with Apple's QuickTime technology.
- **WebM:** A media container designed primarily for web use.

A video container is not the same as a video codec. For example, an MP4 file may contain video encoded using different supported codecs.
A media player must support the relevant container and its included audio and video formats to play a file correctly.

When a video does not open, users should check the file extension, file integrity, and player compatibility before assuming that the video is damaged.

Topic 386: Media Players and Video Playback

A media player is an application used to play audio and video files. Some media players also support streaming, subtitles, playlists, and playback controls.

Common Playback Controls

- Play and pause.
- Stop.
- Seek forward or backward.
- Adjust volume.
- Change playback speed.
- Enable subtitles.
- Select an audio track.
- Switch between windowed and full-screen modes.

Different players support different formats and features.

If a media file does not play correctly, the problem may be caused by an unsupported codec, an incomplete download, a damaged file, or insufficient system resources.

Basic Troubleshooting

1. Confirm that the file has downloaded completely.
2. Try opening it with a trusted compatible media player.
3. Check whether the player supports the file's format.
4. Update the player through its official distribution channel.
5. Check system performance if playback stutters.
6. Obtain another copy from a trusted source if the file appears damaged.

Avoid downloading unknown codec packages from unfamiliar websites, because they may contain unwanted or malicious software.

Topic 387: Basic Video Editing Concepts

Video editing is the process of arranging and modifying video clips, audio, images, text, and other media to produce a finished video.

It is used in education, advertising, entertainment, business presentations, and online content creation.

Common Editing Operations

- **Trimming:** Removing unwanted material from the beginning or end of a clip.
- **Splitting:** Dividing a clip into separate sections.
- **Joining:** Combining multiple clips into one sequence.
- **Transitions:** Creating visual changes between clips.
- **Titles and Captions:** Adding text to provide information.
- **Audio Adjustment:** Changing volume or arranging sound.
- **Color Correction:** Adjusting image characteristics for a more consistent appearance.
- **Exporting:** Producing the final video file.

A video editor commonly arranges clips on a timeline. Separate tracks may be used for video, audio, text, and other elements.

Before exporting, users should select an appropriate resolution, frame rate, and file format for the intended platform.

They should also respect copyright rules when using music, images, video clips, and other people's creative work.

Topic 388: Screen Recording and Capturing Media

Screen recording captures activity displayed on a computer or mobile device. It is commonly used for tutorials, software demonstrations, online lessons, presentations, and technical support.

Screen-recording tools may capture the entire screen, a selected window, or a defined area. Some tools can also record microphone input and system audio.

Preparing a Screen Recording

1. Decide what needs to be demonstrated.

2. Close unnecessary applications and private windows.
3. Check the screen-recording settings.
4. Select the required screen or window.
5. Confirm microphone and audio settings.
6. Start recording and perform the demonstration.
7. Stop recording when finished.
8. Review the saved file.

Screen recordings may reveal notifications, account details, private messages, or confidential information.
Users should check the visible content before recording and sharing it.
Recording another person's meeting or conversation may require consent and must follow applicable rules and policies.
A short, clearly organized screen recording is often easier to understand than a long recording containing unnecessary activity.

Topic 389: Image Formats and Basic Image Handling

Digital images store visual information in a form that computers can display, edit, and share.
Different image formats are designed for different purposes.

Common Image Formats

- **JPEG or JPG:** Commonly used for photographs with lossy compression.
- **PNG:** Supports lossless compression and transparency.
- **GIF:** Supports simple animated images and a limited color palette.
- **SVG:** A vector format that describes graphics using shapes, paths, and other scalable elements.
- **WebP:** Supports lossy and lossless image compression and transparency.
- **TIFF:** Commonly used in professional imaging, scanning, and publishing workflows.

Raster images consist of pixels, while vector graphics represent shapes and paths mathematically.
Resizing a raster image to a larger size can reduce its apparent sharpness. Vector graphics can often be scaled without the same pixelation, provided the software supports the format correctly.

Basic Image Management

- Use descriptive filenames.
- Organize images into folders.
- Choose a suitable format for the intended purpose.
- Preserve original files when making significant edits.
- Check image dimensions and file size.
- Respect copyright and licensing requirements.
-

Topic 390: Understanding Digital Media Storage and Compression

Digital media files can require substantial storage space, particularly high-resolution video and uncompressed audio.
Compression reduces the amount of data required to store or transmit a file.

Two Main Compression Types

Lossless Compression: Reduces file size while preserving all information represented by the source data.
Lossy Compression: Reduces file size by discarding some information. Depending on the settings, the changes may or may not be noticeable to the viewer or listener.
Compression can be applied to images, audio, video, and other types of data.
For example, a high-resolution video may be compressed to make it easier to upload or share online.
Excessive lossy compression, however, may reduce visual detail or create noticeable artifacts.
Users should choose compression settings according to the purpose of the file. Archival work may require preserving original files, while online distribution may prioritize smaller file sizes.
Important media files should be backed up to a suitable storage location rather than relying on a single copy.

Chapter 24: Printing, Scanning, PDF Documents, and Projectors

Topic 391: Understanding Printers and Printing

A printer is an output device that produces a physical copy of digital content on paper or another supported medium.

Printers are commonly used in homes, schools, offices, laboratories, and businesses.

Common Printer Types

- **Inkjet Printer:** Uses liquid ink and is often used for documents and photographs.
- **Laser Printer:** Uses toner and an electrostatic printing process.
- **Thermal Printer:** Uses heat-based technology and is common in receipts and labels.
- **Dot Matrix Printer:** Uses pins striking an ink ribbon and is still used in certain specialized environments.

Basic Printing Process

1. Open the document.
2. Select Print.
3. Choose the required printer.
4. Set the page range and number of copies.
5. Select paper size and orientation.
6. Review color and quality settings.
7. Confirm the print job.

Before printing, use Print Preview where available to check margins, page breaks, and layout.

Correct printer selection and suitable settings help prevent wasted paper and ink or toner.

Topic 392: Understanding Printer Settings and Troubleshooting

Printer settings determine how a document is transferred to paper. Available settings depend on the printer model, driver, application, and operating system.

Common Printing Settings

- **Paper Size:** Selects the required paper dimensions, such as A4.
- **Orientation:** Sets portrait or landscape layout.
- **Page Range:** Specifies which pages to print.

- **Copies:** Determines how many copies are produced.
- **Color Mode:** Selects color or grayscale printing where supported.
- **Print Quality:** Controls available quality settings.
- **Duplex Printing:** Prints on both sides of the paper when supported.

Common Printer Problems

- Printer is offline.
- Paper is jammed.
- Ink or toner is low.
- Printed text is faint.
- Pages are misaligned.
- The wrong printer is selected.
- The print queue is stuck.

Basic Troubleshooting

1. Check power and connection status.
2. Confirm that paper is loaded correctly.
3. Check for error messages.
4. Review ink or toner status.
5. Verify the selected printer.
6. Check the print queue.
7. Restart the printer if appropriate.
8. Consult the manufacturer's instructions for persistent errors.

Do not pull jammed paper forcefully or touch hot internal components. Follow the printer's safety instructions when clearing a jam.

Topic 393: Introduction to Scanners and Digital Scanning

A scanner is an input device that converts physical documents, photographs, or other supported materials into digital images.

Scanners are used to digitize certificates, forms, books, photographs, receipts, and business records.

Common Scanner Types

- **Flatbed Scanner:** Uses a flat glass surface to scan documents or images.
- **Sheet-Fed Scanner:** Feeds individual pages through a scanning mechanism.
- **Handheld Scanner:** Moves across a document manually.
- **Multifunction Printer Scanner:** Combines scanning with printing and often copying.

Basic Scanning Process

1. Connect or select the scanner.
2. Place the document correctly.
3. Open the scanning application.
4. Select the required resolution and color settings.
5. Preview the scan when available.
6. Adjust the scan area if needed.
7. Scan the document.
8. Save the result in a suitable format.

For ordinary text documents, moderate resolution may be sufficient. Photographs and detailed images may require higher resolution.

Scanned files should be named and organized clearly. Users should protect confidential documents and avoid uploading sensitive scans to untrusted services.

Topic 394: Understanding OCR Technology

Optical Character Recognition (OCR) is a technology that identifies printed or supported handwritten characters in an image and converts them into machine-readable text.

A standard image scan records the appearance of a document. OCR adds a text-recognition layer that allows supported software to search, select, or process recognized words.

Common Uses of OCR

- Making scanned documents searchable.

- Extracting text from printed pages.
- Digitizing paper records.
- Processing forms and invoices.
- Supporting document accessibility.
- Reducing manual retyping.

OCR accuracy depends on factors such as image quality, font style, language, page layout, and handwriting clarity.
Poor lighting, skewed pages, low resolution, and damaged documents can reduce recognition quality.
OCR results should be reviewed before they are used in official or important documents. Names, dates, numbers, and technical terms may be misrecognized.
OCR is particularly useful when large collections of paper documents need to become searchable digital records.

Topic 395: Understanding PDF Documents
PDF stands for Portable Document Format. It is widely used to distribute documents while preserving their layout across compatible devices and applications.
PDF files may contain text, images, links, forms, annotations, and other supported content.
Common Uses of PDF

- Reports and assignments.
- Official forms.
- Manuals and user guides.
- Digital invoices and receipts.
- Scanned documents.
- Books and publications.
- Documents intended for printing.

PDF readers allow users to open, search, zoom, print, and sometimes annotate documents.
A PDF is not necessarily editable in the same way as a word-processing document. Editing options depend on the document structure and the software being used.
Some PDFs contain selectable text, while others consist mainly of scanned page images. OCR may be needed to make scanned text searchable.
Users should obtain important PDF documents from trusted sources and use appropriate security precautions before opening unexpected attachments.

Topic 396: Creating and Managing PDF Files
PDF creation converts or exports digital content into a PDF document. Many word processors, presentation applications, browsers, and scanning tools provide this feature.
Creating a PDF

1. Open the document in the appropriate application.
2. Review the layout and formatting.
3. Select Export, Save As, or Print to PDF, depending on the application.
4. Choose PDF as the output format.
5. Select available quality or page options.
6. Save the file.
7. Open the resulting PDF to verify the output.

Some applications allow users to combine pages, add bookmarks, create forms, or apply security settings.
PDF Management Practices

- Use descriptive filenames.
- Keep related documents in organized folders.
- Review page orientation and margins.
- Check that fonts and images display correctly.
- Retain an editable source document when future revisions are expected.
- Protect confidential PDFs with suitable access controls.

Password protection may reduce unauthorized access, but its effectiveness depends on the method used and how the password is managed.

Before distributing a PDF, confirm that it does not contain unintended personal information or hidden content.

Topic 397: Understanding Digital Signatures and Document Security
Digital document security includes methods used to protect files, control access, and verify information about a document's origin or integrity.
A digital signature is a cryptographic mechanism that can help verify whether signed data has changed and whether the signature corresponds to a particular signing key.
A digital signature is different from a picture of a handwritten signature placed in a document.
Important Concepts
- **Integrity:** Helps determine whether signed content has been altered.
- **Authentication:** Helps establish a relationship between a signature and a signing identity, depending on the trust system.
- **Certificate:** May associate a public key with an identity through a certificate authority or another trust arrangement.
- **Access Control:** Restricts who can view or modify a document.
- **Encryption:** Protects data from being read by unauthorized parties when properly implemented.

A valid digital signature does not automatically mean that every statement in the document is truthful. It provides specific security assurances that depend on the signing method and trust arrangements.
Users should verify signatures through appropriate software and trusted certificate information, particularly when handling important business or official documents.

Topic 398: Understanding Projectors and Presentation Equipment
A projector is an output device that displays computer content on a larger surface, such as a screen or wall. Projectors are commonly used in classrooms, training sessions, conferences, meetings, and public presentations.
Common Projector Connections
- HDMI.
- VGA on compatible older equipment.
- USB-C with supported video output or an appropriate adapter.
- Wireless projection technologies supported by the devices.

The connection method depends on the projector and the computer.
Important Display Settings
- Resolution.
- Display mode.
- Screen orientation.
- Brightness and focus.
- Aspect ratio.
- Audio output, where supported.

Before a presentation, confirm that the computer detects the projector and that the displayed image is readable.
Room lighting, projection distance, screen quality, and projector brightness can affect visibility.
Users should also prepare a backup copy of important presentation files and check that fonts, videos, and other embedded media work correctly.

Topic 399: Connecting a Computer to an External Display
External displays include monitors, televisions, projectors, and other compatible screens connected to a computer.
An external display can extend the workspace, duplicate the main screen, or serve as the primary display.
Common Display Modes
- **Duplicate:** Shows the same content on both displays.
- **Extend:** Treats the displays as a larger combined desktop.
- **Second Screen Only:** Displays content on the external screen while disabling the computer's built-in display where supported.
- **PC Screen Only:** Uses the computer's primary display.

The available modes and shortcuts depend on the operating system.

Basic Connection Procedure

1. Check the available ports on the computer and display.
2. Select a compatible cable or adapter.
3. Connect the devices securely.
4. Turn on the external display.
5. Select the correct input source on the display.
6. Open the computer's display settings.
7. Choose the required display mode and resolution.
8. Test the arrangement.

If the display does not work, check the cable, input source, adapter compatibility, and display settings.
Disconnect equipment carefully and follow the manufacturer's instructions.

Topic 400: Essential Computer Maintenance and Digital Best Practices

Computer maintenance involves activities that help keep a computer reliable, secure, organized, and suitable for its intended tasks.

Maintenance includes hardware care, software updates, storage management, security protection, and data backup.

Hardware Maintenance

- Keep ventilation openings clear.
- Place the computer in a suitable environment.
- Protect devices from liquids and excessive dust.
- Use compatible power equipment.
- Inspect cables and accessories for damage.
- Follow manufacturer instructions when cleaning equipment.

Software Maintenance

- Install updates from trusted sources.
- Remove applications that are no longer needed.
- Monitor storage capacity.
- Review startup applications.
- Use suitable security software.
- Avoid untrusted downloads.

Data Maintenance

- Back up important files regularly.
- Verify that backups can be restored.
- Organize documents into meaningful folders.
- Remove unnecessary duplicate files.
- Protect sensitive information.
- Retain important records according to applicable requirements.

Digital Best Practices

- Use strong, unique passwords.
- Enable multifactor authentication where available.
- Verify websites and email senders.
- Respect privacy and copyright.
- Use appropriate security settings.
- Seek qualified assistance when a technical problem could damage equipment or cause data loss.

Effective maintenance is a continuous process. Regular, sensible care can reduce avoidable problems and help users work more confidently with computers.

GLOSSARY

Computer Fundamentals

This glossary provides simple definitions of important computer and information technology terms used throughout this book. The terms are arranged alphabetically to help readers find their meanings quickly.

A

Access Point: A networking device that allows wireless devices to connect to a network.

Algorithm: A step-by-step procedure for solving a problem or completing a task.

Application (App): A software program designed to perform a particular task.

Artificial Intelligence (AI): Technology that enables computer systems to perform tasks that normally require human-like intelligence.

Attachment: A file sent along with an email or digital message.

Audio: Sound that is recorded, stored, processed, or played using a digital device.
Authentication: The process of verifying the identity of a user or device.

B

Backup: An additional copy of data kept for recovery if the original data is lost or damaged.
Bandwidth: The data-carrying capacity of a network connection.
Binary: A number system that uses only two digits: 0 and 1.
BIOS: Basic Input/Output System, firmware used to initialize hardware and begin the computer startup process on many systems.
Bit: The smallest basic unit of digital information, represented by 0 or 1.
Bluetooth: A wireless technology used for short-range communication between compatible devices.
Booting: The process of starting a computer and loading the software required for operation.
Browser: Software used to access and view websites.
Byte: A unit of digital information that usually consists of eight bits.

C

Cache Memory: High-speed memory that stores frequently or recently needed data and instructions to improve processing performance.
Central Processing Unit (CPU): The main processing component of a computer that executes instructions and performs calculations.
Cloud Storage: A service that stores files on remote servers and allows access through a network, commonly the internet.
Compression: The process of reducing the amount of space required to store or transfer data.
Computer: An electronic device that accepts input, processes data according to instructions, produces output, and can store information.
Computer Network: A group of connected devices that communicate and share resources.
Control Panel: A Windows utility used to access and manage various system settings.
Cursor: A visual indicator showing the current position or location for an action, such as text entry or pointer movement.
Cybersecurity: The practice of protecting computers, networks, accounts, and digital information from threats and unauthorized access.

D

Data: Raw facts, figures, symbols, or observations that can be processed into meaningful information.
Database: An organized collection of data that can be stored, accessed, managed, and updated.
Desktop: The main workspace displayed by a graphical operating system, commonly containing icons, a taskbar, and windows.
Device Driver: Software that enables an operating system to communicate with and control a hardware device.
Digital: Information represented and processed using discrete values, commonly binary digits in computers.
Domain Name: A human-readable name used to identify an internet resource, such as a website.
Download: The process of receiving data or files from another device or online service.

E

Email (Electronic Mail): A digital system for sending and receiving messages over a network.
Encryption: The process of converting information into a protected form so that it cannot be understood without the appropriate key or method.
Ethernet: A widely used technology for wired computer networking.
Extension (File Extension): The part of a filename, usually following a dot, that often indicates the file format, such as .pdf or .jpg.

F

File: A named collection of digital information stored on a computer or another storage system.
File Explorer: A Windows application used to browse and manage files, folders, and storage locations.
File Format: The structure and encoding used to store information in a file.
Firewall: A security system that monitors and controls network traffic according to defined rules.
Firmware: Software stored in a device that provides low-level instructions for operating or initializing its hardware.
Folder: A container used to organize files and, in many systems, other folders.

G

Gigabyte (GB): A unit of digital storage commonly equal to 1,000 megabytes in decimal notation.
Graphical User Interface (GUI): A user interface that allows people to interact with a computer through visual elements such as icons, buttons, menus, and windows.
H
Hardware: The physical components of a computer system, such as the motherboard, keyboard, monitor, and storage drive.
Hard Disk Drive (HDD): A storage device that records data magnetically on rotating platters.
HDMI: High-Definition Multimedia Interface, a digital interface commonly used to transmit video and audio.
Home Network: A network that connects devices within a home to share internet access and other resources.
Hyperlink: A clickable element that takes a user to another webpage, file, document location, or online resource.
I
Icon: A small graphical symbol representing an application, file, folder, command, or system function.
Input Device: Hardware used to enter data or instructions into a computer.
Input: Data or instructions supplied to a computer for processing.
Installation: The process of setting up software or hardware so that it can be used.
Internet: A worldwide network of interconnected computer networks that communicate using common protocols.
IP Address: A numerical address used to identify an interface on an Internet Protocol network.
J
JPEG: A commonly used image format that typically uses lossy compression.
Junk Email: Unwanted email, often including advertisements, scams, or other unsolicited messages.
K
Keyboard: An input device containing keys used to enter text, numbers, commands, and other instructions.
Kilobyte (KB): A unit of digital information equal to 1,000 bytes in decimal notation.
L
LAN (Local Area Network): A network connecting devices within a limited area, such as a home, school, or office.
Link: A connection that allows a user to navigate to another digital resource.
Login: The process of accessing an account or system by providing the required credentials.
M
Malware: Software designed to damage systems, disrupt operations, steal information, or perform other unauthorized actions.
Megabyte (MB): A unit of digital information equal to 1,000 kilobytes in decimal notation.
Memory: Computer components used to hold data and instructions for processing. The term often refers to primary memory such as RAM.
Microphone: An input device that captures sound and converts it into a signal that a computer can process.
Monitor: An output device that displays text, images, video, and other visual information.
Motherboard: The main circuit board that connects and allows communication between many computer components.
Mouse: A pointing device used to move the pointer, select items, click buttons, and interact with a graphical interface.
Multitasking: The ability of an operating system to manage multiple tasks or applications so that they can make progress during overlapping periods.
N
Network-Attached Storage (NAS): A storage system connected to a network that allows authorized users and devices to access shared files.
Network Interface Card (NIC): A hardware component that enables a computer or device to connect to a network.
Operating System (OS): System software that manages computer hardware and resources and provides services for applications.
Output: Information produced by a computer after processing data.
Output Device: Hardware that presents processed information, such as a monitor, printer, or speaker.
P

Password: A secret string of characters used to authenticate access to an account or system.
PDF (Portable Document Format): A file format designed to preserve document layout across supported devices and software.
Peripheral: A hardware device connected to a computer to provide additional input, output, storage, or other functions.
Phishing: A deceptive attempt to obtain sensitive information by pretending to be a trustworthy person or organization.
Pixel: The smallest addressable picture element in a digital image or display.
Platform: The hardware and software environment on which an application or service operates.
Port: A physical connector or logical communication endpoint used to connect devices or exchange data.
Power Supply Unit (PSU): A component that converts electrical power into the forms required by computer components.
Printer: An output device that produces physical copies of digital documents, images, or other content.
Processor: An electronic component that executes instructions and processes data.
Program: A set of instructions that tells a computer how to perform a task.
Q
QR Code: A two-dimensional barcode that stores information which can be read by a compatible camera or scanning device.
Query: A request to retrieve, search, or manipulate information in a database or another system.
R
RAM (Random Access Memory): Volatile main memory that temporarily holds data and instructions needed by active programs.
Read-Only Memory (ROM): A category of non-volatile memory traditionally associated with data or instructions that are not changed during ordinary operation.
Reboot: The process of restarting a computer or device.
Resolution: The amount of detail represented by a digital image or display, commonly described by its pixel dimensions.
Restore: To return data, files, settings, or a system to an earlier or saved state.
Router: A networking device that forwards data between networks.
S
Scanner: An input device that converts physical documents or images into digital form.
Search Engine: An online service that helps users find information on the web.
Secondary Storage: Non-volatile storage used to retain data over time, such as SSDs, HDDs, and USB flash drives.
Security Update: A software update that fixes or reduces security vulnerabilities.
Server: A computer or software system that provides services or resources to other devices or programs.
Shortcut: A quick method of performing an action, such as a keyboard combination or a link to a file or application.
Software: Programs and related instructions that tell computer hardware what tasks to perform.
Solid-State Drive (SSD): A non-volatile storage device that uses flash memory and has no moving mechanical parts.
Spam: Unwanted or unsolicited digital messages, especially email.
Storage: The retention of digital data for later use.
Synchronization (Sync): The process of keeping data consistent across devices, applications, or storage locations.
T
Taskbar: A Windows interface element commonly used to access open applications, pinned apps, and system controls.
Terabyte (TB): A unit of digital storage equal to 1,000 gigabytes in decimal notation.
Touchpad: A touch-sensitive pointing device commonly built into laptops.
Troubleshooting: The process of identifying, diagnosing, and resolving problems with a computer, device, or software.
U
Upload: The process of sending data or files from a local device to another system or online service.

URL (Uniform Resource Locator): The address used to identify a resource on the web or another supported network.
USB (Universal Serial Bus): A standard used to connect devices and transfer data and, where supported, electrical power.
User Account: A collection of settings and permissions associated with a person or service using a computer system.
V
Video Conferencing: Real-time communication that allows participants to see and hear one another through compatible devices and software.
Virtual Memory: A memory-management technique that uses storage space to extend the memory address space available to programs. It is generally slower than physical RAM.
Virus: A type of malware that attaches itself to files or programs and can spread when infected content is executed or shared.
W
Web Page: A document or resource that can be accessed through a web browser.
Webcam: A camera used to capture live images or video for a computer or network application.
Website: A collection of related web pages and resources commonly available under a domain name.
Wi-Fi: A family of wireless networking technologies based on IEEE 802.11 standards.
Window: A rectangular area of a graphical interface used to display an application, document, or system content.
Wireless Network: A network that uses wireless communication technologies to connect devices.
Word Processor: Software used to create, edit, format, and print text documents.
World Wide Web (WWW): A system of interlinked resources accessed over the internet, commonly using web browsers and web protocols.
X
XML (Extensible Markup Language): A markup language used to structure, store, and exchange data.
Y
YouTube: An online video-sharing platform used to publish, watch, and share video content.
Z
ZIP File: An archive file format that can combine multiple files and folders and may compress their contents.

Note to Readers
Computer technology continues to evolve. Some terms, interfaces, and examples may change as hardware, software, and operating systems develop. Readers should consult current product documentation when working with specific devices or software versions.